RESEARCH SYNTHESIS / PHYSICAL SYSTEMS + LIFECYCLES

Physical Systems Across the Lifecycle

Tom Klootwijk's cross-domain model suite

Royal Philips as a healthcare and technology facet of cradle-to-grave and circular applications

MATTER + ENERGY

Physical laws and coupled systems

PEOPLE + SERVICE

Care journeys and useful function

INFORMATION + TIME

Models, observations and histories

10 repositories / 56 domain profiles / 787 Philips vacancies

Prepared for Tom Klootwijk
2 November 2026

Independent analytical synthesis with primary-source references, physical derivations, lifecycle applications and a complete recruitment snapshot.

Reading guide

This report brings together the physical interpretation of the complete model suite and the companywide Philips opportunity review. It treats Royal Philips as one detailed healthcare and technology facet within a wider cradle-to-grave framework. Product and material history, human care needs, and the life of data and models are carried as three connected lifecycles.

If you want to...	Start with...
Understand the central proposition	Part I: the lifecycle thesis, common accounts and cross-sector application space.
Inspect the mathematics and all ten repositories	Part II: physical foundations, equations, worked examples and representation relationships.
See where the framework meets Royal Philips	Part III: current business facets, public circular programmes and care-journey connections.
Build a concrete demonstration	Part IV: integrated applications and a sequence from physical model to lifecycle decision.
Use the work professionally	Part V: companywide role connections, followed by 26 detailed comparisons in the appendices.
Inspect complete coverage and sources	Part VI: glossary, method, 56 domain profiles, all 787 requisitions and the primary-source register.

The equations retain their physical units and mathematical conditions. Published constructions, report-derived applications and documented Philips programmes are identified in the narrative. Source codes link to the final register; job IDs in the vacancy catalogue open the exact official record retained in the research snapshot.

Recruitment snapshot: 2 November 2026, 08:37 CET. Model revisions are pinned in the source register. Corporate results retain their reporting year.

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Part I | The lifecycle thesis

Matter, people and information provide the organising axes. Royal Philips is one case within the larger physical application space.

01 Executive thesis: a physical system persists through its lifecycle

The central object of this report is an evolving physical system. It has an identity, occupies a geometry, carries material and energy, interacts with its surroundings, is observed by instruments, and changes through use. A product is one instance of that object. A production line, a laboratory experiment, a circulation model, a building, a power network and a material recovery process are others. Tom Klootwijk's model suite offers a common way to describe these systems while preserving the equations and units that give each application its physical meaning. [M1, M3, M6, M9]

Royal Philips is treated here as a detailed healthcare and technology facet of that wider application space. Its devices, software, services, supply chains and circular programmes provide practical settings in which physical modelling, model selection, traceable decisions and lifecycle accounting meet. The report carries the earlier companywide vacancy review into this broader setting, while retaining the physical depth of the subsequent model analysis. The resulting document is both a technical account of the suite and a map of where its components could contribute across an organisation.

The core proposition is that the same product should remain intelligible as it moves through different disciplines. A component begins as a material and geometric design, becomes a manufactured specimen, enters an assembled system, is calibrated and commissioned, performs a service, accumulates usage history, undergoes repair or upgrade, and eventually supplies reusable components or recovered material. Each transition changes some properties while preserving others. Stable identity, explicit conversions, physical balances and versioned observations make those changes traceable.

Cradle-to-grave is the baseline accounting boundary used in this report: extraction and material production through final treatment. Circular routes introduce additional loops for maintenance, repair, upgrading, refurbishment, component reuse and recycling. Those loops can be represented within the same framework. Their environmental and economic consequences depend on the actual materials, energy, logistics, service delivered and remaining useful life attached to each route.

The physical analysis identifies several especially useful structures. AURUMTOMATON carries inventories through reciprocal exchange networks. Port-Hamiltonian components connect electrical, mechanical and other energy stores through power-conjugate interfaces. XD-TAU connects geometry to fields, material behaviour and transport. Maktub makes changes to an effective response law part of the evolving state. Twin Viper carries a system between coordinate descriptions. TOMK makes competing explanations and informative measurements explicit. TNOMTOM retains complete state and intervention histories. TOM-GROK supplies a broad application catalogue, and TOMElinedT provides a local interaction layer. [M1, M2, M3, M4, M5, M6, M7, M8, M9, M10]

The result is a framework with broad architectural reach: a domain enters through its state, laws, interfaces and observation maps. Its physical contribution can then be composed with the others.

The strongest company application is therefore a connected lifecycle capability, where design decisions, measured operation and circular recovery can refer to the same modelled object.

What this report contains

The technical chapters examine all ten repositories and retain the equations that made the previous physical analysis substantive. They include thermal exchange and entropy, electromechanical conversion, thermomechanics, moving fluid domains, radiation, catalytic boundary formation, adaptive response, coordinate transport, action-based dynamics, graph-state constructions and spacetime models. The Philips chapters identify actual public business and circularity programmes. The application chapters propose specific ways to connect the suite to those settings. The career chapters retain the earlier snapshot of 787 vacancies, including 26 selected comparisons and a complete compact catalogue. [S1, S4]

Three labels keep the narrative precise without interrupting it. A **published suite construction** is found in the retained model sources. A **derived application** follows from those structures and is developed in this report. A **Philips public programme** is grounded in the company's own publications. Proposed Philips applications join the first two to the third and are described as designs with measurable outputs.

02 Three connected lifecycles

The product and material lifecycle

The physical lifecycle begins with resources, materials and components. It continues through design, manufacture, distribution, installation, use, maintenance, upgrade and end-of-life treatment. The unit of analysis can be a complete system, a replaceable assembly, a consumable, a material batch or an individual component. Each has its own geometry, mass, composition, ownership, condition and transformation history.

The suite's identity and state structure allows a product to retain continuity across those levels. An assembly can contain several individually identified modules. A module can retain its own calibration and service history when moved into a refurbished system. A material lot can be related to the components produced from it. Recovered material can be linked to its treatment record and measured properties when it re-enters production. These are proposed lifecycle applications of the source identity, composition and history machinery. [M1, M3, M9]

At each stage the physical balance is explicit. Material received is partitioned into finished products, intermediate inventory, reusable offcuts, waste streams and releases. Energy supplied is partitioned into stored energy, useful work, process heat and transfers to the surroundings. A measured balance residual has an operational meaning: it can indicate an unrecorded stream, a sensor offset, an inconsistent boundary or a model term that needs to be included.

The person and care lifecycle

The second lifecycle concerns human needs over time: early life, prevention, daily wellbeing, investigation, diagnosis, treatment, recovery and management of long-term conditions. Philips participates in selected parts of this landscape through personal health, diagnosis and treatment, monitoring, informatics and related services. The company is a facet of the broader care system, alongside patients, families, clinical teams, laboratories, public services and other organisations. [P1]

Within this facet, a physical model can support instrument performance, signal interpretation, resource planning or the consistency of a measurement chain. A physiological model can describe a compartment balance or a waveform. A service model can describe the availability of the equipment required for a care pathway. The same framework can link these representations while keeping their quantities distinct. A pressure waveform, an appointment duration, a maintenance interval and a cost are separate observations with different units and decision roles.

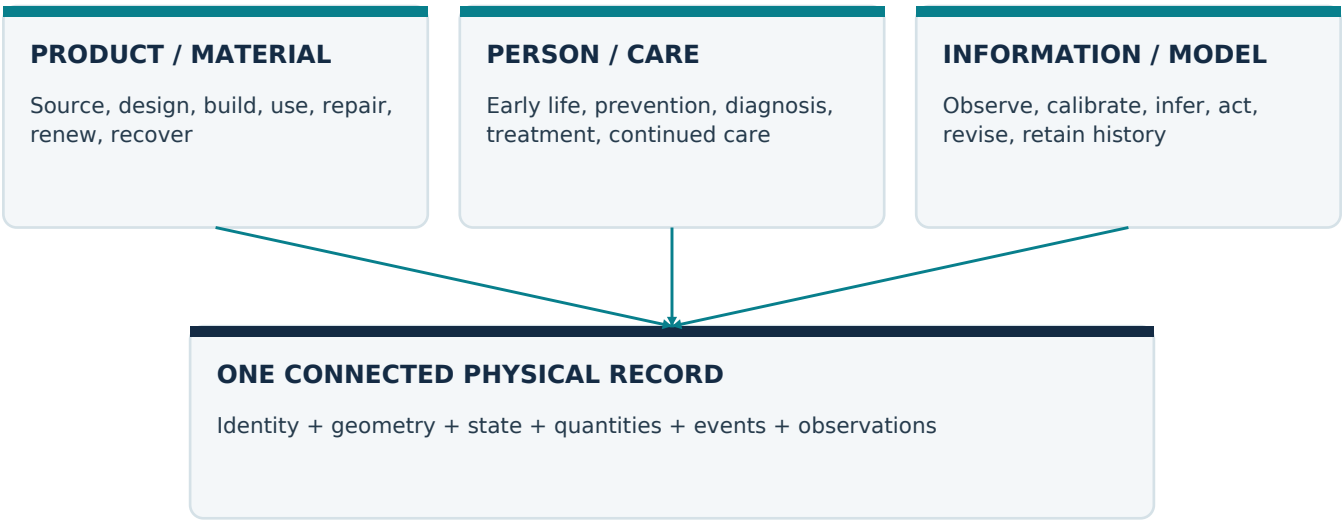
The care lifecycle also changes the meaning of the functional unit. For a diagnostic instrument, a useful unit may be one completed examination with a declared protocol and accepted image-quality measure. For a monitoring installation, it may be a defined period of usable observation coverage. For a consumer device, it may be a specified number of completed use cycles at the intended performance. These units tie lifecycle expenditure to the service actually delivered.

The information and model lifecycle

The third lifecycle follows the descriptions used to understand and operate the physical system. A model begins with a hypothesis and a parameterisation. Measurements identify some parameters, discriminate among alternative models and establish a calibration state. A model version is then used for prediction, design comparison, monitoring or control. New observations can lead to revised parameters, a different representation or a different response policy. The previous version remains part of the historical explanation of earlier decisions.

This is where Maktub, TOMK and TNOMTOM connect most directly. Current action uses the currently active model or policy. New evidence changes a subsequent version. The history records which observation informed which change and which action followed. Twin Viper adds transformations between representations of that same state; XD-TAU supplies physical geometry, units and couplings. [M3, M5, M7, M9]

These lifecycles overlap at real events. Replacing a sensor changes a physical component, affects measurement continuity in a care pathway, and starts a new calibration epoch. Refurbishing an instrument changes its component composition, remaining service capacity and model parameter set. Recycling an assembly closes one product history and opens a material-recovery record. The common state should make each of these relationships explicit.



Circular routes preserve useful function and material; each account keeps its own quantities and boundary.

Figure. Three connected lifecycles share physical identity and event history.

03 A common account of matter, energy, information and value

Material continuity

For a declared system boundary, a material balance can be written as:

$$m_{in} = m_{product} + \Delta m_{inventory} + m_{recovered} + m_{waste} + m_{release} \quad (1)$$

The terms refer to changes or transfers over the same interval, with mutually exclusive output streams at the chosen boundary. Inventory is a signed stock change. Composition can be tracked by element, alloy, polymer family or another declared material class. The stoichiometric left-nullspace idea used in the chemistry profiles is the corresponding local statement for conserved chemical quantities. A production model can aggregate those local transfers into a facility or product account. [M1, M6]

At end of life, the object changes representation. A functioning device may be described by performance and configuration; a disassembled device by component condition; a recycling stream by mass, composition and recoverability. A typed conversion preserves the relationship among these records. Recovery yield is an observed or specified process quantity, not an automatic property of declaring a material recyclable.

Energy continuity

For an operating product, the central balance relates supplied energy, exported work, environmental heat and the change in stored energy. Open fluid and manufacturing systems also carry energy across the boundary with matter, including enthalpy and the relevant kinetic or potential terms. The suite's port structures are valuable because they allow internal transfers to cancel when subsystems are composed. Electrical losses can become a thermal source; hydraulic resistance can become fluid heating; mechanical damping can become heat in a housing. [M1, M3]

Over a lifecycle, operating modes matter. Active operation, standby, cooling, charging, idle time, service procedures and transport each contribute differently. A useful energy model integrates power over time and retains the mode that generated each interval. This gives the same history both engineering and operational meaning: a thermal transient can be linked to an operating state, and its energy can be attributed to the corresponding service period.

$$E_{use} = \int_0^{t_{end}} P(t, mode(t), state(t))\, dt \tag{2}$$

Environmental accounting as a typed transformation

Physical inventory and environmental impact are related through an explicit characterisation model. Let an activity amount be a quantity such as kWh, kg of a material, or tonne-kilometres. Let its factor convert that amount into an impact category. The proposed lifecycle application calculates:

$$I_k = \sum_p \sum_j a_{p,j} f_{k,j} \tag{3}$$

Here p identifies a lifecycle phase, j an activity or flow, and k an impact category. Factors retain their source, geography, time basis, units and version. The account can therefore be recomputed when a factor set changes while preserving the original result used for an earlier decision. This is a natural application of the suite's separation between historical execution and later alternative projections.

Product comparisons use the same functional unit, boundary and service horizon. One illustrative performance measure is:

$$I_{per\ service} = \frac{I_{manufacture} + I_{logistics} + I_{use} + I_{maintenance} + I_{end}}{N_{qualified}} \tag{4}$$

The denominator is a measured or explicitly modelled service quantity. It connects resource use to the actual purpose of the product. Multiple impact categories remain visible rather than being silently compressed into one score.

Value and operational accounts

Financial records provide another view of the same events. A replacement part changes the physical configuration, incurs a cost, affects equipment availability and may extend service capacity. An energy event contributes physical joules and a tariff-based expense. A refurbished system can have acquisition, service and residual-value histories that differ from a new system. TNOMTOM's exact minor-unit and reconciliation examples show how these accounts can retain deterministic identities and balance conditions. [M3]

The shared event is the bridge among the accounts. It does not make euros, joules, kilograms and clinical measures interchangeable. It makes their relationships inspectable. An analyst can ask how a service intervention changed energy use, availability, maintenance expenditure and remaining useful life, with each answer retaining its own measurement and model.

Refurbishment and replacement: a worked lifecycle comparison

Consider a proposed comparison of two ways to deliver the same service over a common future horizon. One route refurbishes an existing system; another manufactures a new system with lower operating power. Let the new manufacture account be M, refurbishment be R, the operating-power saving be delta P, and the electricity characterisation factor be g. Hold other route-specific impacts equal, or include their difference in M minus R. For constant assumptions, the operating-hour crossover is:

$$h^{*} = \frac{M - R}{\Delta P \, g} \tag{5}$$

For an illustrative calculation, use M = 6,000 kg CO2e, R = 1,200 kg CO2e, delta P = 0.8 kW and g = 0.25 kg CO2e/kWh. The crossover is 24,000 operating hours. At 4,000 hours per year, that corresponds to six years. These are example inputs chosen for the derivation, not Philips product measurements.

The useful result is the decision structure. The comparison becomes sensitive to remaining useful life, actual duty cycle, refurbishment work, service differences, electricity mix and performance equivalence. A state-based lifecycle model can replace each illustrative term with measured records and can represent time-varying power and electricity factors. Circularity decisions then become explicit comparisons of service and resources.

04 Lifecycle applications beyond one company

Royal Philips makes the discussion concrete, but the model structures carry into other sectors through their physical adapters. The table gives a set of derived application routes. Each begins with a physical quantity or law and follows it across the lifecycle.

Sector	Physical object and law	Lifecycle question	Suite connection
Buildings	Thermal mass, airflow and moisture transport	How do material, control and retrofit choices change comfort and lifetime resource use?	AURUMTOMATON exchange, XD-TAU geometry, adaptive policy and history
Energy systems	Charge, electrical power and thermal storage	How should operating modes, maintenance and second-life use be compared?	Port coupling, battery profiles, model selection and event records
Water systems	Hydraulic head, volume and dissolved species	Where are losses, quality changes and recovery opportunities introduced?	Conserved inventories, fluid graphs and observation maps
Manufacturing	Geometry, stress, heat and process state	How does a design or process change affect yield, energy and repairability?	CSG, thermoelasticity, hybrid events and causal histories
Transport	Mechanical energy, drag and load cycles	How do operating conditions alter performance, wear and component replacement?	Mechanical models, effective response updates and measured identification

Sector	Physical object and law	Lifecycle question	Suite connection
Agriculture	Water, heat, nutrients and biomass compartments	How do local exchanges affect resource efficiency through a season?	Transport, reaction networks and spatial fields
Materials	Structure, constitutive response and degradation	How do processing and use histories change reuse or recovery options?	Typed material state, Maktub adaptation and geometry
Space systems	Orbital state, power, communication and thermal fields	How do mission modes and component condition shape useful service?	Orbital adapters, graph interfaces and clocks
Environmental recovery	Material streams, reaction and separation	Which recovered fractions retain useful properties and where do losses occur?	Stoichiometry, balance ledgers and conversion records
Healthcare technology	Instrument physics, physiological signals and service capacity	How can reliable measurement and resource use be maintained across product life?	Complete suite with Philips as the detailed organisational facet

The cross-domain advantage is strongest when a decision spans several rows. A battery-powered medical device combines electrochemistry, power electronics, thermal design, software, user interaction, logistics and material recovery. A refurbished imaging component combines mechanical precision, electrical performance, calibration, service planning and circular accounting. The framework becomes useful at those interfaces.

Part II | The complete physical framework

All ten repositories are examined through their states, equations, interactions, observations and lifecycle roles.

05 A common description of organised physical systems

Tom Klootwijk's suite can be read as a framework for connecting physical organisation, evolution and observation across disciplines. Its common objects are entities that retain identity, occupy geometry, carry state, exchange quantities, respond to inputs and change over time. The practical question is how those objects remain intelligible as an application grows from a component into a product, an installed system, a service network and finally a stream of recovered materials. Royal Philips supplies a particularly rich case study because a health-technology product brings physical hardware, software, measurements, people, service processes and resource use into the same lifecycle.

The suite's breadth is best understood through three forms of generality. First, the same mathematical structure can describe different physical phenomena: a weighted exchange graph can carry temperature, concentration, voltage or momentum. Second, unlike laws can be composed through interfaces that account for the quantities crossing them: electrical work can become motion and heat. Third, one physical evolution can be represented through several coordinate systems, resolutions and clocks while preserving selected relationships. Together, these capabilities support broad application across science and engineering. [M1, M2, M9]

The shared XD-TAU state is:

$$S = (E, H, P, \xi, b, F, O, C) \tag{6}$$

The entity table E identifies the objects. H carries hierarchy and addressing. P is the geometry program. Continuous parameters are stored in xi, discrete configurations in b, physical and graph fields in F, operations and events in O, and declared couplings in C. A component's identity therefore remains available while its shape, location, temperature, material state, configuration or ownership changes. This is directly relevant to the lifecycle of an engineered object: a repaired or relocated component can retain its identity while its configuration and condition acquire new values. [M1, M2]

For a temperature-controlled actuator, the entities could include a coil, moving mass, support, sensor and drive circuit. Its fields would contain current, position, momentum and temperature. For a chemical device, the entities could instead be compartments and membranes, with concentration, pressure and species inventory as fields. The architecture supplies the relationship between these descriptions; the chosen physical laws give their evolution.

How the ten repositories contribute

Repository	Principal contribution to the physical framework	Lifecycle connection
XD-TAU-JKT	Entities, geometry, fields, time, physical laws and declared coupling	Product definition, configuration and system modelling
AURUMTOMATON v1.5.1	Symmetry, physical adapters, conserved exchange, representation maps and realization records	Reusable component models, reduced models, calibration and comparison
JTK-Twin-Viper / CATHIROGLYPHDRALE	Coordinate transport and a typed grammar of composition	Consistent views across design tools, operators and model resolutions
TK-MK-ULTRA	A complete action-based oscillator linking motion, energy, geometry and clocks	Reference physics and consistency of alternative descriptions
Maktub	Evolution of state together with its effective response description	Adaptation, changing material condition, calibration and policy updates
TOM-MAKTUBIII	Typed fields, interfaces, modal transforms and time-coordinate relationships	Sensing, signal analysis, physical interfaces and multiscale representations
TOMK	Explicit candidate-model selection and ambiguity	Calibration, parameter identification and informative experiments
TOM-GROK	Evolving operational descriptions and a 56-domain application catalogue	Coordinated experiments, operational scenarios and cross-domain reuse
TNOMTOM	Integrated physical applications and complete state/action histories	Traceable commissioning, operation, maintenance and requalification
TOMElinedT	Local execution and interaction with the framework	An edge workbench for model inspection, decisions and observations

The lifecycle connections in this table are analytical applications developed for this report. The source constructions and implementations are described in the pinned repositories. [M1, M2, M3, M4, M5, M6, M7, M8, M9, M10]

06 Storage, transport and conservation: the AURUMTOMATON core

One exchange law, several physical realizations

The newest extension makes a reusable physical pattern particularly explicit. Let x contain local intensive quantities, M convert those quantities into stored inventories, L represent reciprocal exchanges, and s represent external supply or removal:

$$M\dot{x} = -Lx + s \tag{7}$$

For a closed system, define the extensive inventory q = Mx. The graph Laplacian has zero column sums, so internal transfers cancel:

$$\frac{d}{dt}(\mathbf{1}^T M x) = 0 \tag{8}$$

This is a physical balance statement. Every amount leaving one connected subsystem enters the other with the opposite sign. It also makes a lifecycle interpretation possible: materials, charge, energy or another conserved quantity can be traced across boundaries without being counted twice. [M1]

The thermal and chemical adapters are explicit source constructions. The electrical, hydraulic and viscous-mechanical rows below are derived realizations of the same balance equation.

System	Intensive state	Storage coefficient	Exchange coefficient	Closed-system inventory
Lumped heat exchange	Temperature, K	Heat capacity, J/K	Thermal conductance, W/K	Internal energy, J
Conserved tracer in compartments	Concentration, mol/m3	Volume, m3	Reciprocal exchange, m3/s	Tracer amount, mol
Capacitors connected through resistors	Voltage, V	Capacitance, F	Electrical conductance, S	Charge, C
Linear hydraulic storage	Pressure, Pa	Compliance, m3/Pa	Hydraulic conductance, m3/(Pa s)	Stored volume relative to a reference, m3
Masses with viscous coupling	Velocity component, m/s	Mass, kg	Viscous coupling, kg/s	Momentum component, kg m/s

The reusable object is the relationship between storage and exchange. Each physical adapter supplies its units, observations and constitutive meaning. This supports applications ranging from instrument cooling and battery systems to fluid compartments and vibration damping. For a Philips-oriented demonstrator, the thermal adapter could describe a set of instrument subassemblies with measured capacities and conductances, while the hydraulic adapter could describe a separate cooling circuit. Their coupling would explicitly account for the heat carried by the fluid.

A complete three-body thermal example

Use three heat capacities of 1, 2 and 1 J/K, connected as a chain by conductances of 1 W/K. For a one-second implicit update, the source transition has the following coefficients; every row is divided by 12.

Output temperature	Contribution from body 1	Contribution from body 2	Contribution from body 3
Body 1	7	4	1
Body 2	2	8	2
Body 3	1	4	7

Starting at 306, 300 and 300 K gives the next temperatures 303.5, 301 and 300.5 K. The capacity-weighted total is 1206 J both before and after the step. The equilibrium temperature is 301.5 K. The same numerical table describes conserved tracer exchange after assigning concentration, volume and exchange-rate units. [M1]

The general discrete map is:

$$B_h = (M + hL)^{-1} M \tag{9}$$

For positive capacities, symmetric nonnegative conductances and a nonnegative step, its entries are nonnegative and each row sums to one. Every output is consequently a weighted average of the inputs. The transition preserves the total inventory and remains within the initial range. This gives an implementation-level expression of physically useful structure: the chosen exchange step does not manufacture an interior temperature beyond the supplied range.

The continuous model uses the exponential evolution exp(-t M^-1 L). AURUMTOMATON also gives a particular three-node spectral construction with rational states at times m ln(2) in the chosen time unit. The discrete implicit transition and those exact continuous samples are separately defined evolution maps. Their shared invariants allow implementations to be compared without confusing a numerical step with a physical duration. [M1]

Entropy and equilibration

For positive absolute temperatures and constant heat capacities, the thermal graph has entropy relative to a reference temperature T*:

$$S = \sum_i C_i \ln(T_i / T_*) \tag{10}$$

Differentiating under the reciprocal heat-exchange law gives:

$$\dot{S} = \sum_{i<j} g_{ij} \frac{(T_i - T_j)^2}{T_i T_j} \geq 0 \tag{11}$$

This is a derived thermodynamic reading of the source graph. Total thermal energy is conserved while temperature differences diminish and entropy increases. The framework can therefore retain both the inventory being transferred and a measure of the direction of physical evolution.

The quadratic disagreement function has units determined by its adapter. For capacitors it can represent electrical energy relative to common voltage; for viscously coupled masses it can represent relative kinetic energy. With temperature as the state, its units are J K. Assigning these different meanings explicitly is important when the same equation is reused across a lifecycle model. An energy account and a quality-of-energy or equilibration measure answer different design questions.

For manufacturing and service, this distinction is useful. A thermal model may conserve the measured heat input while still revealing excessive temperature gradients that threaten dimensional stability or slow readiness. A designer can then optimize the conductance pattern or warm-up schedule against both energy consumption and allowable temperature spread.

Symmetry, reduction and equivalent representations

AURUMTOMATON writes a transformation law that carries state, input and model parameters together:

$$F_{g\cdot p,h}(g\cdot x, g\cdot u) = g\cdot F_{p,h}(x,u) \tag{12}$$

For a physical system, this can represent relabelling equivalent components, changing reference frames or rotating a configuration together with its material tensors and loads. The observation map transforms as well. A temperature sensor remains attached to the same physical component even when the network's index order changes. [M1]

The related representation condition is:

$$HF_{\alpha} = F_{\beta}H \tag{13}$$

It states that evolving and then translating produces the same represented result as translating and then evolving. In the three-reservoir example, averaging the two endpoints produces a two-state description with capacities 2 and 2 and conductance 2. That reduced model retains the chosen group temperatures and total inventory exactly.

This is valuable across product generations and operating environments. Detailed engineering models can be reduced to the quantities needed by an installed controller; service dashboards can retain aggregates while linking back to component states. The physical meaning of the retained quantities is a design decision that the model makes explicit.

07 Energy-based interconnection across disciplines

Storage, reversible transfer and dissipation

The port-Hamiltonian form in XD-TAU supplies an especially useful connection between unlike physical domains:

$$\dot{x} = (J-R)\nabla H + Gu \tag{14}$$

H is stored energy, J is skew-symmetric and describes reversible internal exchange, R is symmetric nonnegative and describes dissipation, and Gu describes supplied interaction. With y = G^T grad H, the energy balance is:

$$\dot{H} = -(\nabla H)^{T} R \nabla H + y^{T}u \tag{15}$$

The skew contribution cancels because a vector has zero quadratic product with a skew-symmetric matrix. This algebra expresses a physical fact: exchange between internal stores does not create a net source. The remaining terms describe dissipation and power crossing the system boundary. [M1]

Compatible ports make this useful across electrical, mechanical, hydraulic and chemical components. Force times velocity, torque times angular velocity, voltage times current, pressure

times volumetric flow, and chemical potential times molar flow all have power units. Absolute temperature times entropy flow is the corresponding thermal power pairing. A coupling law supplies equal-and-opposite transfer terms at the participating ports.

That composition principle is relevant throughout a lifecycle. During design it checks subsystem interfaces. During manufacturing it connects test inputs to measured responses. During operation it distinguishes useful work from losses. During refurbishment it provides a physical basis for comparing the same assembly before and after a replacement.

The published capacitor, coil and moving-mass profile

TNOMTOM contains an implemented physical-network profile with four coordinates: capacitor charge q, coil flux linkage phi, position z and momentum p. The energy is:

$$H = \frac{q^2}{2C} + \frac{\phi^2}{2L} + \frac{p^2}{2m} \tag{16}$$

Its dynamics are:

$$\dot{q} = \phi / L \tag{17}$$

$$\dot{\phi} = -q/C - R\phi/L - Kp/m + V_{\text{in}} \tag{18}$$

$$\dot{z} = p/m, \qquad \dot{p} = K\phi/L \tag{19}$$

Charge produces capacitor voltage. Current stores magnetic energy. The coil produces mechanical force, and motion generates back electromotive force. The same reciprocal conversion constant K appears in both directions. Substitution into the energy derivative leaves:

$$\dot{H} = V_{\text{in}} I - RI^2 \tag{20}$$

The published nominal coil profile uses 4.4 mH inductance, 22 ohm resistance, 0.154 kg unloaded moving mass and a conversion constant of 22.5 N/A, with its equivalent back-EMF units. A declared ideal 1 microfarad capacitor adds electrical storage. These values define a concrete physical component model whose variables have measurable meanings. [M13]

The source midpoint step satisfies a discrete work-and-loss identity:

$$H_{n+1} - H_n = hV_{\text{in}} I_{\text{mid}} - hRI_{\text{mid}}^2 \tag{21}$$

The exact arithmetic implementation evaluates this declared model while retaining the physical work ledger. The same application package includes series and parallel RLC profiles, a 16-coordinate mechanical chain and a three-node thermal-storage profile. These examples show how a common energy kernel accepts distinct physical storage laws. [M14]

Derived application: an electromechanical thermal instrument

A useful extension adds a spring, mechanical damping, a load and a thermal state. With fixed positive L, m, k and thermal capacity, positive resistance, nonnegative damping and conductance, a reciprocal K and positive temperatures, use:

$$L\dot{I} = V - R(T)I - Kv \tag{22}$$

$$\dot{z} = v, \qquad m\dot{v} = KI - kz - bv - F_{\text{load}} \tag{23}$$

$$C_{\text{th}}\dot{T} = R(T)I^2 + bv^2 - G(T - T_a) \tag{24}$$

Electrical resistance and mechanical damping produce heat. The same temperature can change resistance, so the thermal state feeds back into electrical dynamics. For a fixed ambient reference, the combined energy is:

$$E = \frac{LI^2}{2} + \frac{mv^2}{2} + \frac{kz^2}{2} + C_{\text{th}}(T - T_a) \tag{25}$$

Its balance reduces to:

$$\dot{E} = VI - F_{\text{load}}v - G(T - T_a) \tag{26}$$

Every internal transfer cancels. A proposed Philips-oriented demonstrator could use this pattern for a moving instrument subassembly, then add its geometry, position observations, drive decisions and service history. Across the lifecycle, the same balances would support sizing, commissioning, thermal readiness, maintenance diagnosis and comparison after component replacement.

A nonnegative thermal availability relative to ambient adds a second useful measure:

$$B_{\text{th}} = C_{\text{th}}[(T - T_a) - T_a \ln(T/T_a)] \tag{27}$$

Its combination with electrical, kinetic and spring energy gives a dissipation account relative to the environment. This allows resource efficiency and physical stability to be considered alongside total energy conservation.

08 Geometry, mechanics, radiation and moving material

Geometry as part of the governing system

XD-TAU treats a geometric region as an authoritative object with associated fields. A region can supply a boundary for stress, a surface for optical absorption, a volume for inventory, or a path for transport. Its constructive geometry is therefore active in the physical model. A changed thickness can alter stiffness; a changed area can alter heat transfer; a changed channel can alter flow. [M1, M2]

An implicit interface can evolve through a level-set equation:

$$\partial_t \Phi + v \cdot \nabla \Phi + V_n |\nabla \Phi| = 0 \tag{28}$$

The velocity v transports the interface, while Vn advances it normally. A physical adapter may obtain the normal speed from deposition, erosion, curvature, pressure or a material law. The same geometric structure can consequently represent a manufactured surface, an evolving membrane or a moving fluid boundary.

For lifecycle analysis, this makes geometry a historical quantity. A component begins with a designed shape, acquires manufacturing deviations, deforms under operation and may lose or gain material. Retaining a geometric state and its observations creates a coherent link between those stages. A proposed service application could compare measured dimensions with the configuration and loads recorded for that component, then evaluate a repair or replacement using the same physical model.

Thermomechanics from one free energy

The physical model uses a material motion x = chi(X,t), deformation gradient F and positive volume ratio J = det F. Its reference-volume Helmholtz energy has elastic, thermal and coupling contributions. The mechanical part includes the neo-Hookean expression and a volumetric penalty; the coupling term is proportional to the product of temperature change and log volume change. [M1]

$$W_{\mathrm{mech}} = \frac{\mu}{2}[\mathrm{tr}(F^T F) - 3 - 2\ln J] + \frac{\lambda}{2}(\ln J)^2 \tag{29}$$

$$W = W_{\mathrm{mech}} - b(T - T_0)\ln J + \rho_0 c_v [T - T_0 - T\ln(T/T_0)] \tag{30}$$

Stress and entropy follow from that same constitutive object:

$$P = \frac{\partial W}{\partial F}, \qquad \eta = -\frac{\partial W}{\partial T} \tag{31}$$

Heating changes stress and deformation. Deformation also changes the thermal state through a reciprocal term involving volumetric deformation rate:

$$-bT\,\mathrm{tr}(F^{-1}\dot{F}) \tag{32}$$

This relationship is useful for optical mounts, heated housings, pressure-bearing structures and assemblies whose alignment changes with temperature. The proposed lifecycle use is to retain the free-energy parameters and material configuration as the object moves from initial design into production and service. A replacement material or changed mounting condition then enters as an explicit change in the constitutive model.

The scalar energy is invariant under a rigid rotation of the material configuration, while stress transforms with the rotation. This supplies a concrete physical example of the newer equivariance structure: the representation changes consistently with the quantity being represented.

Fluids and moving control volumes

The fluid state groups mass density, momentum density and total-energy density:

$$U=(\rho,\rho v,\rho E) \tag{33}$$

A shared face exchanges the same quantity with opposite orientation for its two adjacent cells. For moving boundaries, the relative flux includes the boundary velocity:

$$F_n(U)-w_nU \tag{34}$$

The finite-volume update uses the cell volume and face geometry:

$$V_i^+U_i^+=V_iU_i-h\sum_f A_f\widehat{F}_f \tag{35}$$

The same swept face geometry updates both physical volume and transported inventory. This is the geometric conservation law: consistent mesh motion preserves a uniform physical state. The source includes a moving one-dimensional Euler profile with positive density and internal energy and explicit mass, momentum and energy checks. [M1]

This pattern supports deforming chambers, pistons, fluid passages and fluid-structure constructions. A Philips-oriented research application could connect a cooling circuit's flow model to the geometry of an enclosure and its heat sources. Manufacturing measurements would define the realized passages; operation would provide boundary conditions; service observations would reveal changes in conductance or blockage. The mathematics links those lifecycle stages through physical quantities rather than isolated status labels.

Radiation, heat and deformation

The radiation model stores material thermal energy and photon-bin energies. Absorption transfers photon energy to the material, emission transfers material energy into radiation, and scattering redistributes photon energy among directions. Summing the equations gives the closed balance:

$$\frac{d}{dt}\left(E_{\text{material}} + \sum_j A_j\right) = 0 \tag{36}$$

The associated composition joins geometry, optical paths, thermal response and mechanical deformation. A geometric change alters absorption. Absorption heats a material. Temperature changes deformation, and deformation changes the next optical geometry. This is a useful template for optomechanical assemblies and laser-heated materials. [M1]

A lifecycle study can then follow the same optical assembly through design tolerances, manufacturing alignment, warm-up, repeated use, contamination, replacement and requalification. The proposed model carries the state and cause of a change. It also gives the measurement plan a clear purpose: observations should discriminate between geometry, absorption, thermal conductance and material-response explanations for the measured behaviour.

09 Chemistry that produces and modifies its own boundary

The catalytic membrane construction

The membrane model is one of the suite's clearest cross-domain constructions. It combines species inventories, reaction rates, deposition, spherical geometry, precursor transport and thermal coupling. The idealized reactions transfer a precursor F into catalysts A and B and a boundary-forming product M. Their rates are:

$$v_1 = k_1 f b, \qquad v_2 = k_2 f a, \qquad v_3 = k_3 f a b \tag{37}$$

Dissolved product deposits into a shell at rate vd = kd m. If Ns is shell material in moles and the molar volume is fixed, its outer radius is:

$$R_o = \left(R_i^3 + \frac{3\bar{v}_M N_s}{4\pi}\right)^{1/3} \tag{38}$$

Transport through the shell depends on that evolving geometry:

$$G = 4\pi D_s K_{\text{part}} \frac{R_i R_o}{R_o - R_i} \tag{39}$$

$$J_{\text{in}} = G(f_b - f) \tag{40}$$

The core chemistry produces boundary material; deposition changes shell thickness; shell thickness changes precursor supply; altered supply changes chemistry. This is an explicitly physical feedback loop with concentrations, moles, radii, diffusivity, volumes and flow rates. [M1]

The global material inventory includes dissolved species, deposited material and the surrounding finite reservoir:

$$\mathcal{M} = V(f + a + b + m) + N_s + V_b f_b \tag{41}$$

Its derivative is zero. All reaction and transfer terms cancel in the combined account. With nonnegative initial amounts and the stated positive geometry, the source construction also gives a nonnegative, bounded evolution for its finite model.

A derived consequence clarifies the feedback:

$$\frac{dG}{dR_o} = -4\pi D_s K_{\text{part}} \frac{R_i^2}{(R_o - R_i)^2} < 0 \tag{42}$$

Growth of the shell reduces conductance in this geometry. The boundary produced by the system changes the conditions of its future growth. This supplies a concrete example of self-organisation and boundary production without requiring an abstract metaphor.

Thermal and lifecycle extensions

Reaction enthalpies supply a volumetric heat source:

$$Q_{\text{rxn}} = -\sum_r \Delta H_r \nu_r \tag{43}$$

Temperature then affects the kinetic coefficients, adding an energy-mediated feedback to the material-mediated one. The same construction pattern can inform models of deposition, protective layers, fouling, corrosion products, coatings and growing interfaces when their own reaction and transport laws are supplied.

For a cradle-to-grave study, this is useful because a boundary is not always static over an object's life. A coating can protect a component, wear can remove it, and deposits can change a passage's transport properties. The lifecycle account should retain both material inventory and the functional consequence of the changed surface. A proposed Philips-related engineering study could use this pattern for surface-condition effects in a component, with the relevant materials and observations defining that particular application.

The broad discipline connection is between chemistry and morphology. Material conversion creates geometry; geometry constrains subsequent conversion. Joining the two permits questions about growth history, resource availability, transport time and the conditions under which a target shape or state can be reached.

10 Adaptation, sensing and selecting physical explanations

Maktub: changing effective response

Maktub evolves a state together with the description that governs its effective response:

$$x_{n+1} = \mathrm{Act}(x_n, d_n, e_n) \tag{44}$$

$$d_{n+1} = \mathrm{Rewrite}(x_n, d_n, e_n) \tag{45}$$

Both successors read the same pre-event state. A current observation or input is processed using the current response description; the revised description applies to later events. This supplies a clear causal structure for adaptation. [M7]

A physical interpretation of d is an internal condition. Material damage changes effective stiffness; catalyst condition changes activity; battery condition changes resistance; sensor calibration changes how a voltage is interpreted; and a controller changes its future gains. In each case, the response description belongs to the evolving state of the system.

For a material, a derived application is sigma = E(d) epsilon, with d evolving under a specified damage or recovery law. The material's mechanical response and internal condition evolve together. For a catalyst, a possible adapter multiplies an Arrhenius rate by an activity fraction d, then gives that fraction its own deactivation and regeneration law. These are concrete ways to use the paired-state structure in physical modelling.

The native adaptive threshold profile uses:

$$\tau_{n+1} = (1-\lambda)\tau_n + \lambda e_n \tag{46}$$

For constant input, its deviation contracts by a factor of 1-lambda per step. Choosing lambda = 1 - exp(-h/Tr) gives the exact sampled form of a first-order relaxation with time constant Tr. The connection is useful for baseline tracking, sensor adaptation and control.

Across a lifecycle, the effective law matters as much as the current reading. A repaired sensor can have a new calibration; an aged component can have new parameters; a revised controller can change later behaviour. Maktub's event timing makes those changes attributable to an identifiable point in the history.

TOM-MAKTUBIII: interfaces, fields and modes

TOM-MAKTUBIII's shell description carries a carrier, metric, units, representation, identity and field sampler. This is a natural contract for a material surface, membrane, wavefront or sensing interface. Its Hadamard field product combines values at the same coordinates and retains their resulting units and sign semantics. That is useful for constructing interaction fields and joint observations. [M8]

The normalized two-channel Hadamard transform separates common and differential modes:

$$(a,b) \mapsto ((a+b)/\sqrt{2}, (a-b)/\sqrt{2}) \tag{47}$$

It preserves the sum of squared magnitudes. For physically normalized wave amplitudes, this yields a power-preserving representation change. Paired sensors, acoustic channels, optical modes and electrical waves can all use this pattern after their physical normalization is specified.

Fourier transformations provide another view of a physical signal: a time record can be expressed through its frequency components. This is useful for vibration, acoustic response, periodic interference and phase relationships. The lifecycle application is to compare repeated observations in a representation that exposes the component behaviour of interest, while retaining the original samples and their acquisition configuration.

The source also transports kinematics through a time-coordinate change:

$$v_t = v_T \dot{T} \tag{48}$$

$$a_t = a_T \dot{T}^2 + v_T \ddot{T} \tag{49}$$

A changing clock rate contributes an acceleration term. This is useful when slow degradation, fast vibration and event-driven decisions share a model but use different time resolutions. Physical seconds, simulation steps and transformed time coordinates remain related by explicit maps.

TOMK: candidate models and informative measurements

TOMK's finite candidate families correspond to recognizable modelling tasks. Rational polynomials can express calibration curves or low-order response surfaces. Piecewise affine maps can represent operating regimes. Transition tables can describe instrument modes. Bounded expression trees can compose candidate relationships among measured quantities. [M5]

The physical value of explicit ambiguity is experiment selection. Suppose two normalized candidate responses are y = 2x and y = x squared. Measurements at x = 0 and x = 2 agree with both. At x = 1, their predictions differ: one gives 2 and the other 1. The surviving alternatives identify an input that can separate the explanations.

A proposed lifecycle application would retain the candidate family, physical normalization, measurement conditions and model-selection result for each component. Design experiments could identify material or device parameters. Production measurements could compare an individual unit with the reference family. Service measurements could test whether the current parameterization still explains its behaviour. Requalification after refurbishment could repeat the same discriminating observations.

This creates a coherent sequence from scientific model choice to operational decision. The important output is not only a selected parameter but the relationship between that parameter, the measurements that determine it and the alternatives that remain possible.

A measured thermal-identification example

TNOMTOM contains a thermal identification application using a published TCLab temperature trace. Its model is:

$$\dot{T} = au - \lambda(T - T_a) \tag{50}$$

The input u is heater command in percent. The retained fit gives a heating gain of approximately 0.00244497 degrees Celsius per second per command-percent and a decay rate of 0.00477496 per second. These imply a thermal time constant of 209.43 seconds and a steady command gain of 0.51204 degrees Celsius per percent. [M12]

The source uses 400 training rows and 201 later observations for free-running prediction. Its recorded held-out RMSE is about 1.447 degrees Celsius, compared with 3.834 degrees Celsius for the specified fixed-temperature baseline. The physical interpretation is the effective heating response and cooling timescale of that measurement trace.

Writing the model as C Tdot = alpha u - UA(T-Ta) identifies a = alpha/C and lambda = UA/C. Temperature response therefore identifies ratios and timescales; an independent power measurement supplies the extra scale for absolute capacity and conductance. This is a constructive link between experiment design and physical identification. For lifecycle use, the specimen, fixture, calibration epoch and command interpretation become part of the parameter record.

11 Coordinate transport, action-based dynamics and clocks

Twin Viper: the same evolution in another representation

Twin Viper transports a transition through an invertible coordinate map W:

$$F_W = WFW^{-1} \tag{51}$$

$$F_W W = WF \tag{52}$$

One can evolve the physical state and then transform its description, or transform the description and evolve it under the transported rule. The two routes give the same represented successor. [M9]

For a positive coordinate, the signed-log construction contains the branch u = ln(1+x/ell), with inverse x = ell(exp(u)-1). The physical metric transforms accordingly:

$$ds^2 = \ell^2 e^{2u} du^2 \tag{53}$$

Coordinate compression therefore carries the physical scale needed to recover lengths and gradient norms. This supports multiscale views in which near and distant structures share one representation. A lifecycle record can offer different engineering or operator views while retaining the identity and physical meaning of the underlying state.

A derived mechanical extension transports momenta as covectors. If Q = W(q), then P = (DW)^-T p preserves the canonical pairing p dot dq = P dot dQ. The same motion can be written in transformed coordinates with the corresponding kinetic-energy metric. This is useful for connecting geometry, dynamics and alternative computational descriptions.

The source's inverse-driver exponential response also admits a direct chemical adapter. With G = T/Tref and alpha = Ea/(R Tref), exp(-alpha/G) becomes exp(-Ea/(RT)). This gives the same dimensionless response form an Arrhenius interpretation. The adapter illustrates how domain changes work: the physical quantities assigned to the argument define the meaning of the reusable operator. [M9]

TK-MK-ULTRA: a complete physical action and orbit

TK-MK-ULTRA embeds a selected Twin Viper physical evolution in XD-TAU. Its action uses a position-dependent rest-energy profile:

$$\mathcal{A} = -\int M(x)c^2\sqrt{1-\dot{x}^2/c^2}\,dt \tag{54}$$

$$M(x) = M_0\sqrt{1+(x/L)^2} \tag{55}$$

The Hamiltonian and equations are:

$$H = \sqrt{M(x)^2c^4 + c^2p^2} \tag{56}$$

$$\dot{x} = c^2p/H, \qquad \dot{p} = -M_0^2c^4x/(HL^2) \tag{57}$$

The published selected orbit has amplitude 3L/4, momentum amplitude 3M0c/4, energy 5M0c squared/4 and angular frequency 4c/(5L). Its physical motion is carried into geometric views, an energy allocation and a configured clock through explicit maps. [M10]

A broader amplitude family can be derived from the same action. For amplitude aL:

$$H_a = M_0c^2\sqrt{1+a^2}, \qquad \omega_a = \frac{c}{L\sqrt{1+a^2}} \tag{58}$$

$$x(t) = aL\sin(\omega_a t), \qquad p(t) = M_0ca\cos(\omega_a t) \tag{59}$$

Consequently:

$$H_a\omega_a = M_0c^3/L \tag{60}$$

Increasing amplitude increases energy and lowers frequency. The low-amplitude expansion reveals a harmonic limit with angular frequency c/L. This is a useful analytical reference for checking the consistency of different evolution and representation routes.

The configured conformal clock for the selected orbit is:

$$\tilde{\tau}(t) = \frac{41t}{80} - \frac{9\sin(2\omega t)}{160\omega} \tag{61}$$

Its rate stays between 2/5 and 5/8, so chronology remains strictly increasing. Physical time, phase, log-time and configured clock readings are stored as related quantities. The lifecycle connection is a disciplined treatment of time: event chronology, physical duration and alternative model coordinates are each available without being conflated.

12 Broader physical domains and composition

Quantum graph structures and spacetime fields

XD-TAU supplies an explicit graph-to-quantum description. A symmetric binary adjacency defines stabilizers:

$$K_i = X_i \prod_{j \in N(i)} Z_j \tag{62}$$

Their commutation supplies the graph-state construction. This connects the suite's finite graph representation to Pauli-operator descriptions and quantum-information structures. The physical profile determines preparation, evolution and observation. Within a lifecycle framework, the useful pattern is that configuration, measurement protocol and recorded outcomes remain attached to the same declared experiment. [M1]

The relativistic chapters use an ADM spacetime description with spatial metric, lapse, shift, extrinsic curvature and matter fields:

$$ds^2 = -N^2 dT^2 + h_{ij}(dx^i + \beta^i dT)(dx^j + \beta^j dT) \tag{63}$$

The log-time coordinate is related to physical time by t = t*(exp(T)-1). A supplied massless-scalar expanding profile has scale factor proportional to (t+t*) raised to the one-third power. Lapse, field evolution and constraints transform consistently. This is a concrete example of the framework carrying geometry and time into a physical field theory. [M1]

These domains demonstrate the breadth of the representational framework. The same organisational questions remain useful: which state is being described, which evolution law applies, how quantities are observed, and how different descriptions are connected. A company's immediate engineering needs form one region within that larger space of applications.

TOM-GROK's cross-domain catalogue

TOM-GROK's catalogue contains 56 profiles: 16 computation/formal profiles, 16 physical profiles, 16 biological or biochemical profiles and eight human-system profiles. Each describes state, inputs, operators, rewrite timing, outputs, equations and application meaning. The catalogue makes the suite's disciplinary range explicit. [M16]

Examples include programmable optical transfer matrices with heater dynamics; power converters with current and voltage states; batteries with energy and temperature; reaction networks with stoichiometric balances; neuronal membrane voltage with threshold events; cardiovascular compartments with volume conservation; and reaction-diffusion pattern formation.

Several physical domains share a recognizable structure. In a reaction network, a left-null vector of the stoichiometric matrix identifies a conserved pool. In circulation, internal flows cancel in the total volume balance. In a neuron model, continuous voltage evolution is interrupted by a defined threshold/reset event. In photonics, a configured transfer matrix acts on field amplitudes while thermal state alters phase settings. Each can be coupled to observations and future rule updates.

The cradle-to-grave value is reuse of the modelling discipline. Materials and components can be developed through experiments; manufacturing can record the executed process; installed systems can compare expected and observed behaviour; service can revise parameters; and recovery can retain material identity and quantities. The governing domain changes, while the discipline of state, transfer and history remains consistent.

Composition as the source of architectural reach

CATHIROGLYPHDRALE's constructor grammar includes identity, typed leaves, routing, sequence, parallel product, tagged choice, guards, finite repetition, events and stored feedback. Its composition result assigns a defined meaning to finite well-typed expressions whose supplied contracts establish their domains, invariants and transfer rules. The grammar has no fixed maximum expression-tree size. [M9]

A separate finite-state result represents every total finite deterministic operational system through a finite lookup-table descriptor. Continuum descriptions retain the meaning of their differential equations and boundary conditions. The common architecture can therefore carry finite state machines, fields, hybrid systems and observational interfaces in appropriately defined mathematical roles.

For a physical composition, dimensional compatibility and equal-and-opposite transfer laws work together. A thermal-to-chemical coupling uses enthalpy in J/mol to translate a material conversion into an energy transfer. A motor coupling uses reciprocal force and back-EMF constants. A change in a stiffness parameter has an associated energy consequence. These connections make a composed model physically interpretable.

The broad application claim can consequently be stated constructively: an exceptionally wide range of systems can be expressed when their states, governing laws, interfaces and observation maps can be specified. Adaptation extends this description when the effective parameters, policies or model structure also evolve. This architectural reach is the wider context within which Royal Philips can be examined as one detailed industrial facet.

13 Integrated physical applications across the lifecycle

An existing coupled plant in TNOMTOM

TNOMTOM already executes a chain from geometry through mechanics, pumping, thermal exchange and reaction kinetics. Compiled support geometry determines beam stiffness. Pump pressure supplies a load. Support qualification permits or disables pumping. Flow changes heat extraction, temperature changes reaction rate, and the observed state informs a later decision. [M15]

The mechanical relationships include:

$$I_{\mathrm{beam}} = bh^3/12 \tag{64}$$

$$\delta = FL^3/(3EI_{\mathrm{beam}}) \tag{65}$$

Fluid power is pressure difference times volume flow. Reaction kinetics use a temperature-dependent rate. A shared ledger accounts for reactor energy, receiving-reservoir energy, exported work and external electrical input.

The published modelled alternatives illustrate the physical consequences of design changes. Increasing support thickness from 20 mm to 30 mm reduces deflection by a factor of 3.375. Reducing pump drive changes the load and can permit operation with the thinner support. Disconnecting thermal coupling lets pumping consume power while removing no heat from the reacting vessel. The model therefore distinguishes component activity from effective system-level function.

This is a useful cradle-to-grave pattern. Geometry and material selection belong to design and manufacture. Measured loading and temperature belong to commissioning and operation. Changes in coupling or parameters belong to service analysis. Resource use can then be connected to physical work and output under the selected scenario. The model provides the causal chain linking them.

TNOMTOM and TOMElinedT as an experimental workbench

The history layer has a physical purpose: it retains which specimen, component, configuration and calibration were active when an observation or action occurred. It can connect measured data, inferred state, selected model, admitted command and subsequent response. A repaired component receives a new configuration record while retaining its identity. A model revision can be evaluated on the recorded history with its own version and assumptions. [M3, M11]

TOMElinedT supplies a local application direction for working with these objects. Its architectural goal includes typed models, programs, ports, couplings, physical clocks and digital-twin composition. A proposed instrument workbench would present the current state, expected trajectory, observations and candidate action together. The current 0.17 delivery and 0.18 source checkpoint define the application versions reviewed for this report. [M4]

For a Philips-oriented demonstration, one coherent object could pass through all stages: an actuator or thermal subassembly with defined geometry, material parameters, sensors and control inputs. The design model becomes the manufacturing test reference. Commissioning

identifies the realized parameters. Operation records input and response. Service compares candidate explanations for a deviation. Refurbishment repeats discriminating measurements and updates the assembly record. Retirement preserves recoverable materials and configuration history.

A lifecycle state is more than a current snapshot

A cradle-to-grave application benefits from three connected histories. The physical history records loads, temperatures, material changes and operating exposure. The informational history records observations, calibration, model versions and inferences. The intervention history records assembly, control, maintenance, repair, replacement and recovery actions. The suite's shared state and event structures can relate all three.

This creates a useful basis for decisions about readiness, efficiency, maintenance timing and reuse. A temperature reading becomes more informative when connected to the current material and cooling configuration. A vibration feature becomes more informative when connected to loading and prior interventions. A recovered component becomes more informative when its identity and exposure history remain available.

The physical framework and the enterprise view then meet naturally. Energy and material accounts can be connected to test completion, installed-base records, service actions and circular flows. Royal Philips is one detailed setting in which those connections matter. The same modelling discipline can also apply to manufacturing equipment, transport systems, energy devices, scientific instruments and other long-lived engineered objects.

Part III | The Royal Philips facet

Public company programmes and product families provide the setting. The proposed model connections remain explicitly identified.

14 Three lifecycles that meet inside one industrial case

Royal Philips is a useful bounded case for a much broader physical framework because healthcare technology connects durable equipment, consumable materials, software, clinical workflows, service organisations and people. The product lifecycle begins with material choices and component manufacture, passes through installation and repeated use, and continues through maintenance, upgrades and recovery. The care journey describes changing human needs and transitions between home, community and hospital. The information lifecycle follows observations, calibration, inference, decisions, versions and eventual retirement of records or models.

These lifecycles overlap without becoming interchangeable. A person is not a product; a clinical observation is not a spare part; an environmental impact estimate is not a physical energy store. The suite's typed quantities and persistent identities give each object its own meaning while making their relationships explicit. An imaging system can retain one asset identity through several hardware revisions, serve many care episodes, produce multiple data products, and later contribute components to a refurbished system.

Philips practice. The company's 2025 environmental accounting methodology explicitly uses a cradle-to-grave value-chain boundary: raw materials, components and packaging; its operations and logistics; product use; and disposal. Software, medical-equipment hardware upgrades and parts replacement are excluded from the EP&L calculation. Service-engineer business travel is included, while repair parts are included separately in the material-flow account. These are different reporting views of the lifecycle. [P2]

Application proposed in this report. A common lifecycle model can retain those accounting boundaries as named views. Physical history, service history and software history can coexist even when a published environmental account includes only selected activities. This supports comparison without silently changing the meaning of an existing metric.

15 The Philips facet: actual portfolio and current company boundary

Philips practice. Annual Report 2025 organises the company around Diagnosis & Treatment, Connected Care, and Personal Health. The first combines diagnostic and image-guided treatment businesses; the second links monitoring, informatics and care delivery across settings; the third addresses healthy living and preventive care. The report also states that the Emergency Care sale completed by the end of 2025, with that business operating independently as Heartstream. Philips reported 27.9% circular revenue for 2025. [P1]

This portfolio creates a practical collection of physical domains: electromagnetism and cryogenics in MR; acoustics and transducers in ultrasound; radiation, geometry and motion in image-guided

systems; physiological sensing and communication in monitoring; actuation, batteries, polymers and interaction design in personal-health devices; and data integration across the enterprise.

Philips practice. Its current diagnostic-informatics portfolio names radiology, cardiology, digital pathology, enterprise imaging and integrated diagnostics. [P9] The image-guided therapy portfolio includes interventional systems, devices, software and services, with Azurion, FlexArm and Zenition among the named system families. [P10]

Application proposed in this report. Philips becomes one industrial slice through the suite's larger domain space. The same state, coupling and observation structures could support energy systems, manufacturing, transport or environmental instrumentation elsewhere. Within Philips, the most informative unit of analysis is often a complete physical service: producing an image, keeping a monitored care area available, sustaining a device fleet, or returning a component to useful service.

16 Materials, suppliers and environmental design

Philips practice. EcoDesign is embedded in the company's innovation process. Its stated design concerns include energy efficiency, material and packaging reduction, recycled or renewable inputs, disassembly, recyclability, substance management and longer useful life. Philips uses lifecycle analysis and its Environmental Profit & Loss account to identify impact hotspots and guide design choices. [P3] Its supplier-sustainability programmes include responsible mineral sourcing, substances management, supplier assessments and supply-chain decarbonisation. [P4]

Application proposed in this report. Begin each component with a material identity, geometry, material properties and provenance. XD-TAU's entity and region structure can connect the bill of materials to the physical design; TNOMTOM can retain the source and revision of each coefficient; TOMK can compare candidate material or process relationships against measurements. Relevant quantities include mass, stiffness, thermal conductivity, electrical resistance, expected duty and recovery route.

A lightweight component changes more than purchased mass. Its altered geometry may affect heat transfer, vibration, acoustic response, mechanical strength and ease of disassembly. A useful comparison therefore carries all affected properties through the model. For example, making a support thinner reduces mass but changes bending stiffness with the cube of thickness in the selected beam model. The resulting lifecycle decision should retain both the physical performance requirement and the resource consequence.

The information object should also retain evidence dates. A supplier declaration, measured material property and assumed transport distance are different kinds of input. Keeping them distinct makes a later supplier substitution or design revision traceable and allows the same physical model to feed several environmental or procurement views.

17 Manufacturing, assembly and commissioning

Manufacturing turns a nominal design into a particular specimen. Tolerances, material batches, assembly settings and calibration turn a generic equation into the parameterisation of an actual component or system. This is where the suite's distinction between identity, geometry, coefficients and observations is useful: two units can share a design while carrying different measured parameters and histories.

Application proposed in this report. A manufacturing record can contain the intended geometry, measured geometry, assembly configuration, test conditions and accepted parameter range. AURUMTOMATON supplies conservation and exchange structure; the port-Hamiltonian component model can describe electrical and mechanical energy transfer; Maktub can represent an explicitly versioned change in the subsequent test or adjustment policy. The comparison is between predicted and observed response under the same identified test conditions.

Philips practice. Its MRI lifecycle account explicitly includes development, installation and integration, operation and maintenance, upgrades/refurbishment, and replacement. Installation includes adaptation to local conditions and integration into clinical workflows. [P5]

Application proposed in this report. Commissioning binds the product model to a site: power and cooling connections, room geometry, environmental conditions, connected systems, calibration state and service responsibilities. The installed system is a new configuration of the same underlying asset. Its history should show which boundaries and coefficients changed during transport, assembly and local acceptance.

This is also an opportunity for model reduction. Detailed development models can be projected onto the smaller set of temperatures, positions, currents, vibration measures or throughput variables needed during installation and service. The projection should identify the observables it preserves, making a compact operational model understandable to the engineer using it.

18 Operation: physical resources become a diagnostic or care service

Philips practice. BlueSeal MRI uses a sealed, small helium inventory and micro-cooling to support helium-free operation without refills. Philips also describes siting and lifecycle benefits of that design. [P6] This is an unusually clear example of an upstream material choice changing installation and operating requirements.

Application proposed in this report. Model operation as a coupled flow of energy, material, computation and human work. For an imaging subsystem, a state might contain temperatures, coil or actuator states, geometric calibration, operational mode, acquisition state and the observations used to assess performance. Electrical input, heat removal, stored energy and delivered mechanical work then fit a common balance. A schedule adds workload and time; an observation map connects physical state to the engineering measurements.

The useful lifecycle question is not simply whether one operating mode consumes fewer watts. It is how a change affects a stated service over a stated interval: delivered acquisitions, available operating hours, repeat work, thermal recovery and component duty. A system that draws less instantaneous power but requires more time or additional repetitions can produce a different total result. The report can therefore compare both instantaneous power and integrated energy, with a specified service denominator.

In image-guided equipment, geometry and motion provide another strong mapping. Spatial transformations, sensor registration and mechanical state must refer to a common frame and time. Twin Viper's coordinate transport offers a mathematical lens for retaining the same physical relationship across alternative views. XD-TAU's geometry and coupling structure connects that relationship to the material system and its observations.

19 Monitoring, connected care and enterprise information

Philips practice. Its patient-monitoring portfolio includes bedside and transport monitors, fetal and maternal monitoring, central monitoring systems, clinical measurements, mobile applications and remote monitoring. The company describes a flow of device data across care settings and facilities for device operational management. [P7] Integrated Diagnostics is Philips' stated approach to connecting data and workflows across radiology, cardiology and pathology. [P8]

Application proposed in this report. The suite can represent observations as typed events with physical units, sensor identity, acquisition time, calibration context and processing history. That creates a precise separation among a measurement, a reconstructed quantity, an inferred state and an action proposed from that state. A new model version can change future interpretations while the original observation and the historical decision retain their identities.

The same approach connects equipment state to enterprise analytics. An installed-base record should be linked to the corresponding device and configuration; a service event should identify the component actually changed; an operational metric should declare its time window and denominator. These links connect the original Data and AI Engineer vacancy's commercial and operational questions to the physical fleet that produces the business data.

For example, an apparent mismatch between asset counts, service contracts and utilisation may arise because those records describe different entities, periods or definitions of active status. A typed semantic layer can expose the difference. TNOMTOM's historical-state and durable-event patterns supply a useful engineering example for preserving how a metric was computed at a particular time. TOMK's candidate comparison can support an explicit choice among explanations, while Maktub records when a revised decision rule begins to apply.

20 Service, repair and upgrades as continued physical evolution

Philips practice. Philips offers remote services involving proactive monitoring, predictive maintenance and alert response. [P11] Its MRI lifecycle offering includes tailored upgrades and renovation, while its circular offering identifies Technology Maximizer and SmartPath as routes to continued system capability and lifetime extension. [P5, P12]

Application proposed in this report. A service intervention is a state transition of a physical asset. Replace a component, recalibrate a sensor, change a software version or adjust a mechanical assembly, and the operative configuration changes. The history should retain the old state, the intervention, the newly accepted state and the observations used to judge the result.

AURUMTOMATON's observation and representation maps are useful here: service engineers often observe only a small portion of the internal state. A model can identify which additional measurement would separate plausible fault or parameter explanations. TOMK supplies a small, inspectable example of this principle. The aim is to make the next measurement or intervention informative and attributable.

Maintenance also changes resource flows. A repair consumes a spare part, technician time, transport and test energy, but may retain the useful body of the system. Replacing the whole unit changes a different set of inventories. Both alternatives should be compared over a common

service horizon. A decision about lifetime extension therefore belongs to the same physical account as the original design, rather than appearing only as a financial transaction.

21 Refurbishment, second use and recovery

Philips practice. Circular Edition and Select Edition are two Philips equipment portfolios. The former comprises refurbished imaging systems; the latter comprises serviced systems inspected and tested against Philips specifications. Philips identifies MR, CT, ultrasound, image-guided therapy and mobile C-arm equipment within its circular offering. Its Close the Loop route makes returned equipment available for refurbishment, parts recovery or recycling. [P13]

Application proposed in this report. A refurbished system is a new accepted configuration with retained physical ancestry. Component identities can persist while their system membership changes. A lifecycle model can record what was reused, replaced, repaired, calibrated or retired. That allows a recovery process to produce both a bill of recovered materials and a technically meaningful history of the new assembly.

This is an especially good fit for XD-TAU's separation of identity from hierarchy. A component's identity need not disappear when its parent system changes. Likewise, the amount of recovered material and the amount embodied in the refurbished product must remain connected through a material balance. Losses, testing consumables and rejected parts belong in the same accounting boundary.

Philips practice. Its recycling programme provides recycling passports for professional healthcare equipment and information for consumer-product recovery. Philips also describes collection and treatment arrangements and product take-back in relevant markets. [P14]

Application proposed in this report. At end of use, the model can expose disassembly dependencies, substance information, remaining inventories and intended recovery routes. A terminal record closes the old operational asset while its materials or components begin another lifecycle. The resulting loop is naturally cradle-to-cradle where recovery succeeds, with disposal recorded as an explicit terminal flow where it occurs.

22 The care journey: a second axis through the same framework

The human life-course lens concerns changing needs and care settings. It should be represented as its own axis, with devices, services and information supporting particular episodes and transitions. A product lifecycle can span many people; a person's care journey can involve many products and organisations.

Philips practice. In pregnancy and childbirth, the Avalon portfolio combines fetal and maternal measurements with information systems. [P15] Philips Avent's Pregnancy+ has introduced community features around pregnancy and continuing connection after birth. [P16] A September 2026 launch adds a wearable Avent breast-pump product, illustrating the connection between electromechanical design and daily use. [P17]

Application proposed in this report. These facets connect sensing, geometry, user interaction, pressure/flow and information timing. For engineering work, a prototype can study the physical interface, signal quality, energy use or interaction sequence while preserving what

was measured and what was inferred. The suite's model structure supports that traceability across different scales of device and use context.

Philips practice. In oral healthcare, Philips announced a July 2026 Sonicare product with on-device AI, spatially aware brushing guidance and pressure sensing. [P18] The broader SenseIQ explanation identifies movement, pressure, duration and frequency as observed aspects of use. [P19]

Application proposed in this report. This is a direct meeting point for native edge computation, physical sensing, actuation, human feedback and adaptive behaviour. TOMElinedT's local-interaction direction, Maktub's timed adaptation, TOMK's explicit hypothesis space and XD-TAU's state/geometry description can be arranged around such a device-engineering problem. Later-life diagnosis, treatment and monitoring are represented by the imaging, intervention and connected-care portfolios already described. Continuity is achieved by connecting observations and services across episodes, while keeping the person, device and model identities distinct.

23 Measurement framework for a cradle-to-grave application

Application proposed in this report. The lifecycle application should produce several linked accounts. A material account retains mass and component movement. An energy account retains supplied, stored, transferred and dissipated energy. A service account retains the specified work delivered and system availability. A financial account retains prices, expenditures and revenues. An environmental account applies named impact factors and accounting rules to identified activities.

For material stock, a simple balance is:

M_next = M_current + M_in - M_out - M_loss.

For an impact category k, an activity-based view can use:

Impact_k = sum_j Activity_j * Factor_jk.

The activity, factor version, geography, time and boundary must travel with the result. Impact weights transform an inventory into an assessment; they are not conserved physical quantities. Likewise, an environmental valuation in euros and an operating cash amount are different accounts even though they use the same currency symbol. Philips publishes distinct environmental and emissions-accounting methodologies, giving a concrete source structure for retaining those meanings. [P20]

A useful common comparison is one specified service delivered over the same horizon by competing designs or lifecycle decisions. Track energy per completed service, materials retained, maintenance interventions, downtime, recovered component yield and the evidence behind the model parameters. The suite can then connect a physical change to a lifecycle consequence and to the observation that would decide whether the predicted benefit occurred.

24 Portfolio narrative for Royal Philips

The resulting narrative is a physically grounded systems-engineering proposition: preserve the relationships among a product's geometry, material state, energy flows, observations, interventions and resource history from its first design through successive uses and eventual recovery. At Philips, this can be illustrated through MR thermal/resource behaviour, image-guided motion and registration, connected-device observations, service decisions, consumer edge devices and circular equipment.

The employment lens then becomes a set of entry points into that larger system. Scientific modelling roles address governing equations and observables; software roles implement dependable transitions and interfaces; data and AI roles reconcile the fleet's operational and commercial information; test and systems-engineering roles define what must remain true across changes; service and circularity teams use the resulting histories to decide how assets continue to provide value. Each role touches a different facet of the same lifecycle, and each proposed application should be demonstrated with a concrete physical model, data product or reproducible experiment.

Part IV | Integrated lifecycle applications

Five connected application routes take the framework from a physical model to design, operation, service, recovery and business decisions.

25 A persistent lifecycle state for a Philips application

Identity, condition and observation

A proposed lifecycle record should identify the product family, individual specimen, assembly revision, component relationships, material declarations, geometry, active configuration, calibration epoch and relevant model versions. Measurements retain physical time, sensor identity, units, location, observation function and the model state to which they were compared.

This structure builds on XD-TAU's stable entity identity and explicit distinction among organisational, spatial and interaction relations. A component can be owned by one service record, occupy another location, and interact with several physical subsystems. Those are different relationships. Keeping them explicit allows refurbishment and component reuse to preserve a coherent account. [M1]

Couplings as named physical relationships

Each coupling states what crosses an interface. A coil transfers force and back EMF. A thermal contact transfers heat. A fluid connector carries volume, mass, enthalpy or species. A sensor maps an underlying state to a measured quantity. A business-system connector relates a serialised component or service event to an order, contract, cost or installed-base record.

The original Data and AI Engineer vacancy is relevant at this interface. Its commercial, financial and operational datasets could refer to the same installed assets that the physical models describe. A model of remaining service capacity can inform a service forecast; actual service events can update the asset's history; the resulting costs and revenues can be reconciled to enterprise records. This is a proposed bridge between the physical suite and the role's requested enterprise analytics. [J2]

Changes have an effective time

Repair, calibration, software update, configuration change and controller adjustment should each have an explicit effective event. Maktub's paired update expresses this clearly: the current response uses the current description, and the revised description governs a later response. TNOMTOM's historical-state mechanisms preserve the version actually active at an event. [M3, M7]

This matters in an instrument's lifecycle. A drift observed before a sensor replacement belongs to the previous observation system. A calibration fitted after the replacement belongs to the new one. A revised thermal model may explain both periods, but the retrospective analysis is a separate view from the historical decisions made at the time. The distinction creates an inspectable connection between learning and operation.

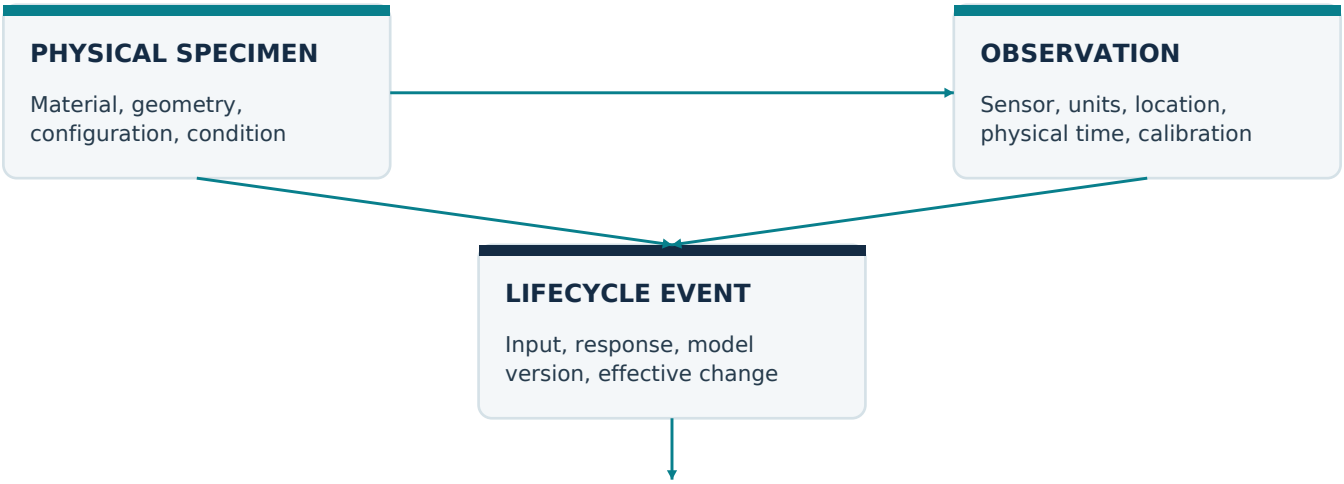


Figure. A proposed lifecycle record connects the specimen, observations and effective changes.

26 Proposed application: an electromechanical module from design to recovery

Design and material selection

The first proposed Philips application uses the suite's published electrical-mechanical profile as a starting point for an instrument module. A coil, moving assembly, support, housing and sensors are given stable identities, physical dimensions and material properties. The design model carries inductance, resistance, force constant, mass, stiffness, damping and thermal capacity. Geometry determines clearances, support dimensions and thermal contact areas. [M1, M3]

The port-Hamiltonian description connects electrical input to motion and heat. The design question is expressed in measurable terms: required movement, permitted temperature range, settling time, electrical consumption and repeated-cycle performance. A design search compares candidate geometries or parameters while retaining the balance equations. TOMK can represent a declared candidate family; TOM-GROK's deterministic operational descriptions can preserve the search and selection history.

Manufacture and commissioning

The manufactured specimen receives measured dimensions and calibration values. An observation model relates voltage, current, position and temperature readings to the physical state. Tests identify parameters such as resistance, force constant and thermal time constants under stated conditions. The fitted parameter set is attached to the specimen and its instrument configuration.

The commissioning record includes the input waveform, sensor configuration, model version, measured response and acceptance quantities. A measured discrepancy can lead to an updated parameter estimate, a new observation model or a physical adjustment. The history retains the relationship among those choices. This is an application of the same experiment-design logic that distinguishes competing candidate laws in TOMK.

Operation and service

During use, current and temperature observations can be compared with the expected response under the recorded command. A change in effective resistance, friction or damping becomes a candidate explanation for a changed trajectory. An informative next diagnostic interval can be selected from the candidate models. The resulting recommendation is tied to the observations and parameter versions that generated it.

A service intervention records the actual component replaced or adjusted. The system is then re-characterised and the new parameterisation becomes effective. Physical energy, operational availability, replacement cost and material movement are connected through the same event. This supplies a concrete bridge among engineering, service, finance and installed-base analytics.

Refurbishment and recovery

A returned module can be evaluated against the requirements of its next intended use. Its measured condition and component history determine the repair or requalification work. The lifecycle model compares reuse, replacement of selected components, full refurbishment and recovery routes on a common service basis.

The final material account records which masses entered reuse, recycling or another treatment stream. The original design's material and disassembly information can therefore be compared with the actual recovery result. That closes the loop from design intent to physical outcome.

27 Proposed application: an optical, thermal and mechanical measurement chain

The second application follows an optical assembly whose performance depends on heat and deformation. XD-TAU's published thermal-geometric coupling supplies the organising structure: geometry determines optical paths; absorption deposits heat; temperature changes material response; deformation changes geometry; observations measure the resulting optical behaviour. [M1]

At design time, the model carries surface geometry, optical properties, heat capacities, conductances, expansion coefficients, support stiffness and observation locations. A candidate material or mounting change affects several of these quantities at once. The physical model makes those consequences explicit rather than treating optical, thermal and mechanical design as separate scorecards.

At manufacture, actual dimensions and material lots enter the specimen record. At commissioning, reference measurements identify alignment and calibration. During operation, the model can relate temperature history to predicted drift. During service, a replacement mount, source or sensor changes the geometry or observation map and starts a new calibration epoch.

A useful application output is a paired plot of predicted and observed measurement drift over an operating cycle, with thermal and mechanical contributions identified. Another is a design comparison that shows resource use and performance over the same declared duty cycle. A third is a refurbishment assessment relating retained components to the calibration work needed for their next service interval.

The physical question is not limited to one instrument category. The same structure applies to optical sensing, inspection systems, laser processing, photonic circuits and imaging assemblies.

Philips supplies a healthcare technology setting in which the relationship among image quality, stability, energy and service is particularly concrete.

28 Proposed application: a production process with geometry, cooling and reaction

TNOMTOM already includes a modelled plant where a support's geometry affects mechanical qualification, qualification permits or disables a pump, fluid flow determines cooling, and temperature changes an abstract reaction rate. That source construction is a useful lifecycle demonstration because each discipline changes a downstream physical condition. [M15]

The geometric relation is especially instructive. For a rectangular support, the second moment of area varies with thickness cubed. A thickness change therefore affects deflection, permitted operating load, material use and potential service life together. Pump drive also changes the load. A design choice and an operating choice can achieve related physical outcomes through different resource paths.

In a proposed industrial application, each component of that model is attached to a measured equipment or material record. The output is a common comparison of delivered function, energy input, material usage, thermal response and resulting product quality. The business account can then associate those physical results with actual cost records using the same event identifiers.

Across the lifecycle, this application has several roles. Design selects geometry and equipment capacity. Manufacturing determines actual component properties. Commissioning identifies the operating model. Use reveals changing efficiency or fouling. Maintenance restores or changes the physical pathway. End-of-life decisions compare recovered component condition with the requirements of another process. The same state description remains useful throughout.

29 Proposed application: a connected observation and service system

An instrument's useful output depends on the entire observation chain: the physical phenomenon, transducer, electronics, signal processing, calibration, model interpretation and presentation. The suite can represent each stage as a typed map with units and an effective version. That creates a common account of how a physical state became a displayed quantity.

For a monitoring or imaging fleet, this observation history can be connected to installation, service and configuration records. A model can compare observed behaviour across a declared population of comparable devices. A proposed change can be evaluated against retained traces before becoming effective for later processing. Each comparison preserves the source data and model version.

The application should measure quantities meaningful to both technical and operational teams: usable observation coverage, signal-quality indicators, calibration stability, unnecessary repeat measurements, resource consumption, downtime and time to restore the intended function. The selected quantities can be aggregated without erasing the underlying device and event identities.

This is also a useful setting for agentic AI. An agent can retrieve a trusted model or record, call a deterministic calculation, inspect the resulting balance or evaluation output, and explain the relationship to an operational question. TOMK and the physical models contribute explicit calculations; TNOMTOM contributes the state and provenance; the interaction layer presents the

answer and its evidence. The agent's role is part of the workflow rather than the source of the underlying physical law.

30 Proposed application: circular installed-base intelligence

A circular installed-base model connects engineering condition to the commercial and service records requested in the Data and AI Engineer vacancy. The central entities are product, component, installation, contract, service event, observation, model version and recovery route. Each event can change several linked accounts. [J2]

A refurbishment decision can use a remaining-service model, measured component condition, expected requalification work, logistics and resource inventories. A service forecast can distinguish planned maintenance, model-supported diagnostic work and actual failure history. A commercial view can reconcile the same installed asset with contract coverage, backlog, order intake and recognised service activity.

The state model makes common disagreements diagnosable. An installed-base count may differ because one source counts complete systems while another counts modules. A service revenue measure may refer to billing dates while an engineering measure refers to the intervention date. A refurbished product may retain a source serial history while receiving a new configuration revision. The correct response is to make those definitions and mappings explicit.

The proposed output is a semantic and metric layer in which each metric has an entity grain, event time, unit, inclusion rule and source lineage. Physical models become callable computations over curated observations. Agent tools retrieve those definitions, perform approved calculations and return the supporting state. This joins the user's physical portfolio to the enterprise-data facet of Philips without collapsing either into the other.

31 A practical sequence of demonstrators

Demonstrator A: conserve and explain

Begin with a small coupled electrical, mechanical and thermal system. Publish the state and units, the governing equations, one complete input sequence, the resulting trajectory and the energy balance. Show both an expected operating sequence and a changed physical parameter. The purpose is to make the physical semantics readable and reproducible.

Demonstrator B: measure and identify

Connect a declared measurement dataset to the same state structure. Fit a bounded candidate family, show which parameter combinations are identified, and select a measurement that separates remaining candidates. Retain training and evaluation intervals as distinct records. The purpose is to connect mathematical prediction to observed behaviour.

Demonstrator C: preserve history through a change

Introduce a calibration, component or policy change at an explicit event. Reproduce the observations and actions before and after it. Show a historical query using the version active at the time, then a separate retrospective analysis using a revised model. The purpose is to demonstrate causal continuity across a product's life.

Demonstrator D: compare lifecycle routes

Use the measured and modelled state to compare continued use, repair, refurbishment and replacement over a common future service horizon. Include actual material, energy, service and cost inputs where available. Retain a sensitivity analysis for uncertain future use. The purpose is to make circular choices concrete and reviewable.

Demonstrator E: answer a business question through the physical record

Connect the asset and event history to an enterprise-style dataset. Ask a question such as which service interventions restored the most usable capacity per unit of resource, or which configurations have the strongest evidence for an upgrade route. Show the metric definition, query, physical calculation and final explanation. The purpose is to demonstrate the full bridge from physical reality to organisational decision-making.

Each demonstrator has a distinct output and can stand on its own. Together they form a cradle-to-grave application: specify, build, observe, adapt, maintain and recover. The same modules can then be transferred to another domain by supplying its state, constitutive laws, interfaces and observation functions.

32 A professional contribution across the lifecycle

The broadest professional account of this portfolio is the ability to connect physical models to executable calculations, experiments, operational records and human decisions. Within Philips, that can support several teams: scientific software, product engineering, systems engineering, test and verification, design technology, services, sustainability analytics and enterprise informatics.

The companywide vacancy review provides entry points into those functions. The physical chapters explain why the connection is substantive. A candidate narrative can lead with one real system, show the governing quantities, demonstrate how its components interact, and explain how observations changed the next decision. This is more useful to a technical reviewer than a catalogue of model names without a physical example.

The report's central application thesis is therefore a persistent, interpretable account of a system through its lifecycle. The system's material, geometry, behaviour, observations, service history and recovery route remain connected. Royal Philips supplies a detailed case study; the mathematical structures remain available for other physical and organisational settings.

Part V | Professional contribution

The earlier global recruitment review is retained as a map of entry points and collaborators across the lifecycle.

33 From physical architecture to work across the Philips lifecycle

The companywide search as a map of contribution

The physical interpretation of the suite changes how the Philips opportunity should be approached. Its contribution can begin with geometry, material response and model construction; continue through software, verification, deployment and observation; and extend into service histories, operational decisions and recovery of assets. Employment opportunities are points of entry into that larger lifecycle. A vacancy title identifies one responsibility within it, while the portfolio can explain relationships across responsibilities.

The completed recruitment census captured 787 unique public requisitions on 2 November 2026 at 08:37 CET. Full descriptions were retrieved for 786, and those records were marked posted and accepting applications. Thirty-five had the Netherlands as their primary country; 41 primary countries were represented. All available descriptions were screened, 45 received additional detailed requirements review, and 26 were retained for structured comparison. The complete catalogue appears in the appendix. Requisition 591655 remains an unresolved entry because the live feed supplied its identifier without a usable current description. [S1]

The original vacancy, Data and AI Engineer 590735 in Amsterdam, remained open in the live recruitment record. It belongs to one important layer: connecting enterprise information to business decisions. The wider search exposes additional routes through scientific modelling, native software, systems engineering, product experience and test development. Together they give the suite several possible organisational homes. [J2]

The appendix preserves the original screening labels as research metadata. Those labels describe the evidence available for a direct employment match. The lifecycle reading adds a second question: which teams could use or collaborate around a particular application? A procurement role, for example, may be a stakeholder in material provenance even when the model portfolio is chiefly evidence for an engineering role. Keeping these questions separate creates a useful companywide map.

Lifecycle phases and relevant working relationships

Lifecycle phase	Contribution from the suite	Philips role families to connect
Need, concept and exploratory design	Typed problem definition, geometry, candidate hypotheses, physical demonstrators and observable targets	Research, design technology, function development, systems engineering
Component and system engineering	Coupled physical models, native software, energy and material balances, coordinate consistency	Scientific modelling, C++ software, algorithms, hardware engineering
Verification and industrial delivery	Repeatable experiments, test oracles, failure handling, configuration identity and traceable results	Test development, systems integration, design quality, industrial engineering
Deployment and use	Interfaces, observations, state estimation, interpretable actions and instrument software	Product software, clinical applications, deployment and service teams
Operation, maintenance and renewal	Historical state, component replacement, performance drift, calibration and intervention records	Services, reliability, installed-base operations and lifecycle engineering
Commercial and resource decisions	Reconciled asset, contract, cost and usage data linked to physical systems	Enterprise Informatics, sales operations, finance, supply chain and business operations
Recovery and next use	Persistent component identity, condition evidence, material inventories and disposition decisions	Service, refurbishment, sourcing, logistics and sustainability teams

This mapping is an application design proposed in this report. It uses the responsibilities visible in the vacancy research and the physical capabilities documented in the model chapters. It makes Philips a concrete organisational facet of the wider cradle-to-grave framework.

Netherlands: three distinct application narratives

Native software in a physical product

Software Designer (C++) - Image Guided Therapy, 588449 in Best, is the clearest Netherlands software route in the reviewed set. Its display software, hardware interaction, model-based design and code-generation responsibilities connect to the portfolio's geometry, native integration and testable execution. TOMElinedT provides a local application example; TOM-GROK supplies native/GPU implementation; TNOMTOM supplies reproducible state and verification examples. [J1]

The published requirements include approximately five years of relevant development, strong C/C++, Windows and legacy software, TDD/BDD, and regulated-industry experience. The application should therefore pair a concise technical case study with a precise professional record of C++ delivery. A useful demonstration would follow one physical or user-visible requirement through an interface, a native implementation, a test and an observed outcome. This locates the broader architecture within a component that a product team can assess.

Enterprise information across the installed lifecycle

Data and AI Engineer, 590735 in Amsterdam, connects the enterprise side of the lifecycle: pipeline, orders, revenue, backlog, contracts, service activity and installed base. The role explicitly names Salesforce, SAP and ServiceNow, as well as Databricks, Python, SQL, Spark/PySpark, Delta Lake and cloud platforms. Its experience requirement is five years with a bachelor's degree or three with a master's. [J2]

TNOMTOM's durable store and reconciliation work provide a useful foundation for discussing consistent records and repeatable decisions. The thermal-identification example supplies an analytical case with a defined model and observed measurements. A lifecycle application could connect component identity, configuration, service history and operating observations to business datasets, with definitions for measures such as utilisation, service cost or renewal opportunity. The CV should identify the actual enterprise platforms and commercial analyses used professionally, so those experiences accompany the published model work.

Systems methods across product generations

AI Senior Systems Engineer, 591185 in Eindhoven, connects most directly to traceability, modelling, verification and reusable engineering methods. Its remit includes adoption and transformation across complex product organisations. The required background is a master's or PhD and extensive systems engineering, architecture, RAMS or digital-engineering experience. A strong presentation would show how one physical requirement, model assumption and verification result remain connected as a product changes. [J11]

Senior Software Architect - Azurion Eye, 580273 in Best, is another systems-level route. It asks for typically seven or more years in complex regulated software, distributed cloud/edge architecture, containers/Kubernetes, Go/Python and architecture leadership. Typed interfaces, historical state and edge execution are relevant portfolio themes. The application record should make architecture ownership and delivered systems visible. [J12]

Two further Best roles belong to more established leadership paths: Product Owner/Technical Lead 588477 specifies more than ten years of development and more than three years as a Product Owner; Senior Software Design Quality Engineer - AI 570549 specifies fifteen-plus years with design-quality and regulatory responsibilities. These roles can be revisited when the corresponding career evidence is explicit. [J13, J14]

International: matching physical disciplines to particular roles

Scientific software and physical model construction

Senior Methods HW Modelling Engineer, 589928 in Suzhou, is the closest scientific-software connection to XD-TAU and AURUMTOMATON. It involves translating MR hardware and physical behaviour into reusable software models, expressing assumptions, and comparing predictions with bench, scanner and phantom observations. A bachelor's is accepted; MR knowledge is preferred. The central experience requirement is substantial professional modern C++ and scientific-software delivery, with Windows, testing and deployment practices. [J3]

A suitable portfolio discussion would combine three layers: the governing physical relationship, its numerical or executable realisation, and the observation used to judge it. The thermal graph, port-Hamiltonian voice coil, and measured thermal-response fitting offer distinct examples. Their value is the ability to explain how geometry, material parameters and observed response belong to the same engineering question.

The Hamburg research roles have narrower disciplinary requirements. Research Scientist - AI in CT, 586913, asks for a master's, actual deep-learning training and PyTorch competence, and CT or medical-imaging understanding. Research Scientist - Physics of CT, 586911, asks for a master's and expertise in X-ray generation, interactions, detection and CT. Both connect to scientific modelling and evaluation; German is optional in their retrieved descriptions. [J16, J17]

Verification, automation and native implementation

GenAI Solutions & Prompt Engineer, 584997 in Bangalore, connects to model evaluation, regression, traceability and AI-assisted engineering within MR systems integration and verification. It asks for five-plus years of software/test automation, C#/.NET, and practical agents, RAG and LLM evaluation. The current title is authoritative even though the URL retains an older integration-test title. The most relevant application package would show a repeatable engineering workflow, its evaluation set, and how a change is accepted or rejected. [J4]

Software Technologist C++, 589859 in Bangalore, offers a broader native-software path and lists a bachelor's qualification. Its separate four-to-eight-year experience wording is attached to vocational education. Senior Software Engineer, 593840 in Shenyang, concentrates on component integration, tests, performance and code quality; its minimum is two years with a bachelor's or no prior experience with a master's. Software Technologist I, 593980 in Bangalore, states a broad minimum but prefers six-plus years and C#/WPF/.NET experience. Each can use a readable implementation case rather than requiring the whole theoretical suite to be understood at once. [J6, J7, J8]

The US image-reconstruction roles 580470 and 586928 connect to C++/CUDA, scientific computation and performance engineering. They ask for five-plus years of professional C++; the latter also highlights image/signal processing and accepts either GPU experience or readiness to move into GPU development. Both require existing US work authorisation and state no sponsorship or relocation. TOM-GROK's kernel and conformance work are appropriate technical exhibits, accompanied by the candidate's actual imaging or signal-processing experience. [J9, J10]

Test Development Engineer, 588485 in Böblingen, connects model evaluation to instruments and measured signals. It values Python/test software and AI coding assistants, alongside electronics, sensor interfaces, hardware measurement and both German and English. The appointment is fixed term through September 2028. This role can connect the suite's precise test histories to physical test benches and manufacturing verification. [J15]

Prototyping, human interaction and function development

Design Technologist, 585168 in Murrysville, combines product/design judgement with coded prototypes and explicitly welcomes AI-assisted coding examples. It asks for four-plus years, a relevant degree and a portfolio linking user need, prototype, feedback and impact. Existing US work authorisation and commuting proximity are required; sponsorship and relocation are not offered. TOMElinedT and earlier creative-technology work form the natural narrative. [J5]

Assistant Function Development Engineer, 590246 in Shenzhen, connects particularly well to physical demonstrators and exploratory engineering. It accepts relevant engineering degrees including computer science and specifies at least three years of relevant systems, design or hardware work for the bachelor's route. Its scope includes physical principles, dynamic mechanics, fluid dynamics, electromagnetics and measurement. A case study should show the path from a physical principle to a built or measured demonstrator. [J18]

Additional comparisons retain useful specialisations: UI/UX Designer - Advanced Imaging 592685 asks for shipped-interface design and a required portfolio; Full Stack C# Software Technologist

587348 emphasises Angular, Node.js, C#/.NET and web delivery; Embedded C++ Software Technologist 591151 requires ThreadX and four to nine years of embedded systems. E-Commerce Data, Automation & AI Manager 585855 connects commercial analytics to Amazon Vendor/Seller/Ads or digital-shelf experience. These are distinct professional routes within the wider lifecycle. [J22, J23, J24, J25]

Study routes and specialist research paths

Three internships were retained because their topics connect strongly to the suite. Avent Product Innovation & AI Prototyping 586187 explicitly requires current study. Robot-Assisted Toothbrush IMU Validation 593360 asks for an MSc student or final-year BSc, with Python/Linux, rotations and signal-processing knowledge; robotics, IMU and motion-capture experience strengthen the application. Dental Synthetic Data Generation 593359 is an MSc-level internship involving Python, PyTorch/computer vision and 3D rendering; its retrieved text does not explicitly state active enrolment. Their eligibility should be handled according to each posting. [J19, J20, J21]

The selected comparison set also retains Senior/Advanced Algorithm Engineer 593837 in Shenyang, whose master's/PhD, ten-plus years and medical-imaging algorithm requirements make it a specialised experienced-hire route. Keeping these roles in the catalogue is useful for identifying research collaborators and long-term capability directions as well as immediate applications. [J26]

A portfolio package organised around the lifecycle

The public professional record supplies a coherent starting narrative. The supplied LinkedIn profile describes Amsterdam-based creative technology, invention and game development associated with Dawnwood Interactive. It lists Amsterdam University of Applied Sciences study from 2014 to 2018, cum laude, and an emergent-behaviour game thesis. The exact degree title was not exposed. A separate credential records MIT Professional Education's Machine Learning: From Data to Decisions, issued 27 November 2020. [S2, S3]

Three case studies can connect this background to the lifecycle argument.

A physical model connected to observation. Start with the thermal exchange or electromechanical model. State the quantities, units and coupling; show the expected balance; then explain the measurement or test that answers the engineering question. This serves modelling, research, systems and test roles.

A state history connected to an intervention. Use TNOMTOM to show which configuration was active, what was observed, which action occurred and how later behaviour is reconstructed. Tie the example to a component replacement, calibration revision, service event or repeated experiment. This serves product software, operations and enterprise-data roles.

A usable prototype connected to a person. Use TOMElinedT or a creative-technology project to show the task, interface, coded behaviour and feedback. For compute-focused roles, substitute the TOM-GROK kernel with a clearly defined workload, correctness comparison and performance result. This serves design technology, native software and computational engineering.

The broad suite then becomes the explanatory layer around those concrete examples. It shows why an implementation can connect to physics, why a measurement has a particular meaning, and how results can remain traceable across product generations. The application record should supply the exact degree, employers, dates, responsibilities, professional technologies, languages and geographic eligibility that determine the appropriate entry point. No applications were submitted as part of this research.

Part VI | Reference and complete catalogues

Technical terms, research methods, exact sources and complete coverage are collected here for detailed review.

Appendix	Purpose	Coverage
A. Glossary	Physical and lifecycle terms with their application meaning.	40 terms
B. Method and evidence	Retained source revisions, earlier execution results and research method.	Six fresh reproduction groups
C. Philips matrix	Documented company facets connected to proposed model outputs.	10 lifecycle phases
D. Domain catalogue	The full cross-domain atlas, indexed by governing relationships.	56 profiles
E. Role comparisons	Portfolio connections, published role conditions and application emphasis.	26 comparisons
F. Vacancy census	Every requisition retained in the companywide snapshot, with official links.	787 records
G. Sources	Pinned model sources, official company publications and job records.	66 source records

The appendices preserve detail while keeping the main argument readable. They can be used independently: follow a model source, inspect a lifecycle mapping, compare a role, or search the complete catalogue by requisition ID.

A. Physical and lifecycle glossary

Term	Meaning and use
Automorphism	A transformation that preserves the complete structure and coefficients of a fixed model. Swapping identical endpoints of a symmetric exchange network can reveal equivalent configurations and reduce repeated analysis.
Availability or exergy	Useful-work potential defined relative to a specified environment and a stated set of allowed exchanges. It helps distinguish conserving total energy from retaining the ability to perform useful work as heat approaches ambient temperature.
Calibration	Establishing a relationship between instrument indications and reference quantity values under stated conditions. A calibration epoch identifies which measurement relationship was active before and after sensor replacement or adjustment.
Characterisation factor	A factor that converts an inventory flow into a selected impact-category indicator under a specified assessment method. Its units, geography, time basis and version connect physical activity records to a reproducible environmental result.
Circularity	Organisation of material and product flows to retain useful function or material value through maintenance, repair, reuse, refurbishment and recovery. The model tracks each loop, its resource requirements, delivered service and actual recovered quantities.
Conjugacy	An invertible change of representation that carries one evolution rule into another while preserving the represented dynamics. Twin Viper can compare evolving a state before a chart change with evolving it under the transported chart rule.
Conservation law	A statement that a specified quantity remains constant in a closed system or changes only through accounted sources and boundary transfers. Internal exchanges cancel when connected components are considered together, allowing mass, charge or energy to be traced.
Constitutive law	A relationship that specifies how a material or component responds, supplementing general conservation equations. Elastic stiffness, electrical resistance, heat conduction and chemical rate laws supply the particular behaviour of an application.
Coordinate chart	A chosen coordinate description of a region of a state space or physical geometry, with a defined relation to other descriptions. Cartesian and signed-log views can display the same component or trajectory at different spatial scales.
Coupled model	A model in which outputs or state quantities from one subsystem enter the equations of another through declared relationships. Electrical resistance can supply a thermal source, while the resulting temperature changes the resistance used in later evolution.
Cradle-to-grave boundary	A lifecycle boundary covering resource extraction and material production through the final treatment of the product and its outputs. It makes upstream manufacture, use-stage resource consumption and end-of-life processes part of one declared account.
Digital twin	A digital representation linked to an identified physical asset or process through defined observations and update relationships. It can connect design assumptions, measured operation, maintenance events and the changing condition of an installed product.
Dissipation	Conversion that reduces recoverable mechanical, electrical or other organised energy, commonly transferring it into heat and producing entropy. Resistive and damping losses enter both performance analysis and the thermal model of an instrument.
Entity identity	A persistent identifier for an object, kept distinct from its location, organisation, physical condition and current representation. A component can retain its identity through installation, repair, transfer to a refurbished system and final material recovery.

Term	Meaning and use
Entropy	A thermodynamic state quantity used to express the direction of physical processes and the second-law balance. A closed thermal exchange network conserves energy while its entropy increases as temperatures equalise.
Equivariance	Consistency between a transformation of the complete model inputs and the corresponding transformation of its outcome. Relabelling components or changing frames requires parameters, sensors and inputs to move consistently with the state.
Extensive quantity	An accounted amount associated with the size or contents of a system, such as mass, chemical amount or internal energy. It supports balances that add compatible inventories and account explicitly for transfers between subsystems.
Functional unit	A quantified service or function used as the common basis for a lifecycle comparison. Examples include a defined examination, a period of usable observation coverage or a specified number of qualified use cycles.
Graph Laplacian	A matrix encoding network exchange: diagonal entries sum connected weights and off-diagonal entries carry the corresponding negative weights. It describes reciprocal heat, tracer, charge or other exchanges and exposes the cancellation of internal transfers.
Hybrid model	A model combining continuous evolution with discrete modes, events or resets. An instrument can evolve thermally between mode changes, while a neuron model combines voltage dynamics with threshold events.
Implicit geometry	A representation of a region or boundary through a field, predicate or constructive expression rather than only an explicit surface mesh. It can define material domains, contact surfaces, moving fronts and the geometry used by physical equations.
Intensive quantity	A local property that does not scale with the amount of material at an otherwise fixed state, such as temperature, pressure or concentration. AURUMTOMATON converts intensive states into inventories using capacities, volumes or other storage coefficients.
Invariant	A property retained by every admitted evolution step of a model on its declared domain. Examples include nonnegative inventories, a conserved total or a geometric relationship that must survive each update.
Inventory balance	An equation relating the change in stored quantity to external inputs, outputs and any defined production or conversion terms. Material received can be partitioned into inventory change, products, recovered outputs, waste and releases over one interval.
Lifecycle inventory	A structured account of resource inputs, energy use, emissions and other flows associated with a declared lifecycle boundary. It provides the physical activity data for comparing manufacture, operation, service, refurbishment and end-of-life routes.
Metric	The mathematical structure that assigns lengths, angles or spacetime intervals in a chosen description. Transporting the metric with a nonlinear coordinate change preserves the physical interpretation of distance and gradients.
Model reduction	Construction of a smaller state description designed to retain selected behaviour or observables of a larger model. A detailed thermal network can be reduced to the group temperatures needed by a controller or service dashboard.
Model validation	Evaluation of how a model represents the quantities and behaviour relevant to its intended application, using defined comparison evidence. Predicted trajectories are compared with observations over a stated operating range and with meaningful performance measures.
Model version	An identified specification of equations, parameters, numerical choices and interpretation rules used for a particular calculation or decision. Historical analysis retains the version active at an event; a later reinterpretation receives its own version.

Term	Meaning and use
Numerical time step	The finite interval used by a numerical evolution rule, together with the physical-time interpretation of that rule. Recording the step and integration scheme makes trajectories comparable across model versions and implementations.
Observation map	A specified relationship from model state to the quantity represented by a sensor or reported measurement. It connects temperature, position or a physical field to the actual readings used for calibration and model comparison.
Parameter identifiability	The extent to which specified observations and inputs can distinguish the parameters of a selected model. A thermal relaxation trace may determine conductance-to-capacity ratios, while a power measurement supplies an additional absolute scale.
Physical port	An interface through which a model exchanges a specified physical quantity with another model or with its surroundings. A connector can carry electrical current and voltage, fluid flow and enthalpy, or heat between an instrument and a cooling circuit.
Port-Hamiltonian model	An energy-based model that separates storage, reversible interconnection, dissipation and external ports. It connects electrical, mechanical and other components while making the combined power balance explicit.
Power-conjugate variables	A pair of physical variables whose product is power when their units and orientation are consistent. Voltage and current, force and velocity, and pressure and volumetric flow connect otherwise different engineering domains.
Provenance	The recorded origin and transformation history of a quantity, dataset, model or result. It links an operational conclusion to specimen identity, measurements, calibration, calculation and effective configuration.
Refurbishment	A defined set of inspection, repair, replacement, configuration and requalification activities that prepares an existing product for further use. Its comparison with replacement uses measured condition, remaining service capacity and a common future functional basis.
Sensitivity analysis	Evaluation of how changing selected inputs or assumptions changes a model result or decision. Duty cycle, electricity factors, remaining life and refurbishment effort can be varied to examine a lifecycle crossover.
State	The quantities needed to describe a system at a chosen instant and to evaluate its future evolution under the selected model. An actuator state can include current, position, velocity, temperature and the active controller configuration.
System boundary	The declared separation between the system being accounted for and its surroundings, including the interfaces across which quantities are exchanged. It determines which heat, work, material and service flows belong inside a lifecycle account and which cross its boundary.

B. Research method and reproducible evidence

This report extends the preceding portfolio and vacancy review. Public repository revisions were retained by commit identity. The physical reading examined the 87-page AURUMTOMATON publication, embedded source attachments, selected code and mathematical proofs, the TNOMTOM physical applications, and the TOM-GROK domain catalogue. New lifecycle mappings and algebraic extensions are labelled as derived applications.

The earlier execution review reran the following components. The results describe the specified calculation or test population, with each physical interpretation supplied by its model.

Component	Fresh result	Meaning and scope
XD-TAU AURUMTOMATON v1.5.1 extension	19/19 checks passed; complete result JSON matched the publication.	Conserved inventory, positive exchange, unit transformations, symmetry and retained-observable dynamics. 256 three-node graph/capacity/step profiles; 1,024 single-step state cases; six permutations of one profile; one exact sampled spectral family.
Eight inherited XD-TAU verification programs	8/8 programs completed; outputs matched the latest embedded reproduction record.	Geometry, energy/material transfer, mechanics, thermal/fluid evolution, radiation, shells and spacetime constraints. Published finite examples and algebraic/numerical checks. Older standalone floating results differ in roundoff; the latest embedded record is the matching baseline.
TNOMTOM durable event store	29/29 tests passed.	Reliable retention of observations, configurations, actions and histories for a physical-system workflow. SQLite transaction, idempotency, revision conflicts, replay, outbox integrity and actual multiprocess/crash-before-commit recovery tests.
Maktub reference	13/13 tests passed.	Causal changes to effective response, reproducible adaptation, coordinate and channel transformations. Old-state action/rewrite semantics, adaptive threshold, repeatability, log-polar transforms, Hadamard norm, field combination and save/restore behaviour.
TOM-MAKTUBIII reference	43/43 tests passed.	Typed interfaces, geometry fields, modal/spectral representations, time transformations and ordered change. Shell units/domains, transform identities, Fourier samples, kinematic chain, event ordering, failure behaviour, provenance and reconstruction.
Twin Viper numerical reference	10/10 tests passed.	Consistent coordinate descriptions, conjugate evolution, energy accounting and configured oscillator/clock observations. Chart round trips, independent derivative comparison, six exact transition-table rows, energy shares, oscillator/clock, domain checks and command-line fixture.

GPU benchmark records, larger aggregate verification records and Android delivery sources were inspected as published evidence. The present lifecycle report adds source research and mathematical synthesis; it does not recategorise those earlier inspections as new execution runs.

Recruitment method: 40 catalogue pages yielded 787 unique requisitions. Full details were retrieved for 786. All available descriptions were screened, 45 roles received additional requirements review, and 26 were selected for comparison. The catalogue includes one unresolved entry, 591655. Company figures remain attached to their original reporting period. [S1]

C. Philips lifecycle application matrix

Proposed mappings to documented Philips lifecycle facets. The company activities are sourced; the module combinations and proposed outputs are the report's application design.

Lifecycle phase	Philips facet / quantities	Suite / proposed output
01. Need and research	Diagnostic, treatment, monitoring and personal-health needs. Required observables, spatial resolution, timing, energy and service capacity. [P1, P7, P9, P10]	XD-TAU, TOMK, JTK-Twin-Viper. A requirement-to-observation map that links each engineering question to a model, measurable quantity and comparison condition.

Lifecycle phase	Philips facet / quantities	Suite / proposed output
02. Materials and sourcing	EcoDesign, supplier sustainability and material selection. Mass, composition, dimensions, conductivity, stiffness and transport inventory. [P3, P4]	XD-TAU, TNOMTOM, TOMK. A typed material and supplier record connecting physical properties, evidence dates, geometry and proposed recovery routes.
03. Design and simulation	Imaging, actuation, sensing and consumer-device development. Charge, magnetic flux, force, displacement, temperature, stress and optical state. [P3, P5, P10, P18]	AURUMTOMATON, XD-TAU, TK-MK-ULTRA, TOM-MAKTUBIII. A coupled physical design model with energy balances, conserved quantities, geometry-dependent response and retained alternatives.
04. Manufacture and calibration	Product realisation and supplier/component interfaces. As-built geometry, parameter tolerances, test response, sensor gain and offsets. [P3, P4, P5]	XD-TAU, TOMK, TNOMTOM, Maktub. A specimen-specific record linking nominal design, measured properties, calibration state and the effective time of adjustments.
05. Logistics and commissioning	MRI installation, local integration and site requirements. Equipment mass and volume, transport activity, power, cooling and installed geometry. [P5, P6]	XD-TAU, JTK-Twin-Viper, TNOMTOM. An installed configuration binding asset identity to its site, reference frames, boundary connections and acceptance observations.
06. Use and service delivery	Imaging, image-guided intervention, monitoring and personal health. Power, integrated energy, temperature, motion, sensor signals and qualified service time. [P6, P7, P10, P18]	AURUMTOMATON, TOM-GROK, TOMElinedT, XD-TAU. A mode-resolved operational account relating physical resource use to a declared examination, observation interval or device-use cycle.
07. Observation and model revision	Integrated diagnostics, enterprise informatics and remote operational monitoring. Timestamped observations, calibration epochs, inferred states and response residuals. [P7, P8, P9, P11]	TOMK, Maktub, TNOMTOM, JTK-Twin-Viper. A versioned measurement-to-model history showing which observations informed each revised parameter, interpretation or future policy.
08. Maintain, repair and upgrade	Remote services, SmartPath and Technology Maximizer. Component condition, replaced material, service energy, downtime and remaining service capacity. [P5, P11, P12]	TNOMTOM, TOMK, Maktub, XD-TAU. An intervention record connecting the prior state, work performed, new configuration and measured response after service.
09. Refurbish and redeploy	Circular Edition, Select Edition and component recovery. Retained mass, component condition, restored performance and projected service horizon. [P12, P13]	XD-TAU, TNOMTOM, AURUMTOMATON, TOMK. A new accepted configuration with retained component ancestry and a common-horizon comparison against replacement.
10. Recover and close the loop	Take-back, disassembly, recycling passports and material recovery. Material fractions, recovery yield, treatment energy, waste and final inventories. [P13, P14]	XD-TAU, TNOMTOM, TOM-GROK. A closed asset history plus material/component successor records, preserving balance through reuse, recycling and final treatment.

D. All 56 application domains

The catalogue below preserves every domain title from TOM-GROK's source atlas. The governing relation is a compact index into its longer state, input, operation, observation and reference definitions. The catalogue contains 16 formal/computational, 16 physical, 16 biological/biochemical and eight human/creative profiles. [M16]

Profile	Domain	Governing relation / organising object
comp-01	Interpreters, staging and specialization	Eval(d,x,u) = y; Eval(d,x,u) = Run(Specialize(Eval,d),x,u)

Profile	Domain	Governing relation / organising object
comp-02	Parameterized refinement and inductive invariants	I(s) and Admit(s,u) => I(Step(s,u)); R(s,b) => R(Step(s,u),K(b,u))
comp-03	Cellular automata and evolving local rule tables	x_(t+1)(i) = f_(d_t)(x_t restricted to N(i),u_t); d_(t+1) = R(d_t,x_t,u_t,x_(t+1))
comp-04	Distributed replication and agreement on event order	s_n = Fold(Step,s_0,[e_0,...,e_(n-1)]); L_A[0:n] = L_B[0:n] and s_A0 = s_B0 => s_An = s_Bn
comp-05	Graph rewriting and structural growth	rule = (L <- K -> R); G = Dctx +_K L; H = Dctx +_K R
comp-06	Scheduling as a mutable deterministic policy	Ready_t = {v not in Done_t : Pred(v) subseteq Done_t}; v_t = argmin_(v in Ready_t) (Score(d_t,v,u_t),ID(v))
comp-07	Databases, transactions and event-sourced projections	View_n = Fold(Project,View_0,[e_0,...,e_(n-1)]); available = received - shipped - reserved
comp-08	Optimization and reproducible populations of candidates	x_trial = Propose(d_t,P_t,u_t); x_next = argmin_(x in {x_old,x_trial}) (f(x),ID(x))
comp-09	Adaptive machine-learning pipelines as explicit descriptions	yhat_t = Predict(d_t,x_t); ell_t = Loss(yhat_t,y_t)
comp-10	Recommendation and retrieval as typed ranking transitions	C_t(q) = Generate(d_t,index_t,q); score_t(q,j) = sum_l w_(t,l) * feature_l(q,j)
comp-11	Compression, descriptions and explicit reconstruction cost	Decode_m(Encode_m(x)) = x; L_total = L(m) + L(x|m) + L(metadata)
comp-12	Information theory of deterministic evolving trajectories	H(X) = -sum_x p(x) * log2(p(x)); I(X;Y) = H(X) - H(X|Y)
comp-13	Control theory and changing feedback descriptions	x_(t+1) = A*x_t + B*u_t; u_t = -K_t*x_t; Aclosed(K) = A - B*K
comp-14	Type theory and composition of operator descriptions	MachineState = Sigma_(r in Roles) State_r; Eval_r : Desc_r x State_r x Input_r -> Result_r
comp-15	Digital twins and versioned hybrid simulation	z_(n+1) = Phi_(theta_n)(z_n,u_n,h_n); r_(n+1) = y_measured_(n+1) - Observe_(theta_n)(z_(n+1))
comp-16	Security reproducibility, provenance and forensic replay	s_n = Fold(Step,s_0,E_[0:n)); artifact = Build(source,environment,instructions)
physical-01	Contact-aware robot policy specialization	M(q)*q_ddot + C(q,q_dot)*q_dot + g(q) = tau + J_c(q)^T*lambda; q_next = Flow_mode(q,q_dot,tau,lambda,dt)
physical-02	Transactional FPGA function exchange	q_(n+1) = max(0, q_n - service_n) + arrivals_n; handoff_allowed = (inflight = 0) and compatible(shell,module)
physical-03	Calibrated programmable optical transfer networks	E_out = T_opt(theta,omega)*E_in; T_opt = U_N(theta_N)*...*U_1(theta_1)
physical-04	Discrete RF transfer codebooks	F(direction,f) = sum_j a_j(f)*exp(i*(k*r_j dot direction + phi_j)); r_j,m = (1/T_period)*integral_0^T_period r_j(t)*exp(-i*2*pi*m*t/T_period) dt
physical-05	Quality-aware range and motion event encoding	f_beat approximately 2*S_chirp*R/c_light; f_Doppler = 2*v_radial/lambda

Profile	Domain	Governing relation / organising object
physical-06	Sampled supervisory control over continuous circuitry	L*i_L_dot = d*V_in - v_C; C*v_C_dot = i_L - v_C/R_load
physical-07	Reaction-stage operators with explicit material balance	n_dot = Nu*r(n,T) + f_in - f_out; a_element^T*Nu = 0
physical-08	Geometry-linked thermal history specialization	rho*c_p*T_dot = div(k*grad(T)) + Q_laser; E_line = P_laser/v_scan
physical-09	Model-bound inverter and microgrid supervision	P_i = sum_j V_i*V_j*(G_ij*cos(delta_i-delta_j) + B_ij*sin(delta_i-delta_j)); sum_i P_generation_i - sum_i P_load_i - P_loss = dE_stored/dt
physical-10	Battery policy evolution with charge and heat accounting	E_dot = eta_charge*P_charge - P_discharge/eta_discharge - P_selfloss; C_th*T_dot = Q_generation - h*A_surface*(T-T_ambient)
physical-11	Cooling-plant mode and setpoint specialization	C_zone*T_zone_dot = Q_load - UA*(T_zone-T_water); Q_removed = m_dot*c_p*(T_out-T_in)
physical-12	Mission-state specialization over a flight-control interface	m*r_ddot = F_external; J*omega_dot + omega cross (J*omega) = tau_external
physical-13	Catchment-event encoding and model selection	V_water_dot = A_catchment*(P_rain-E_loss) + Q_in-Q_out; Q_discharge = a_rating*(h-h_zero)^b_rating
physical-14	Classical policy state around sampled quantum jobs	rho_next = E_circuit(rho); Pr(z) = Tr(Pi_z*rho_next)
physical-15	Ensemble-preserving experiment descriptions	pi(x) = exp(-beta*E(x))/Z; beta = 1/(k_B*T); alpha(x,y) = min(1, exp(-beta*(E(y)-E(x)))*q(y,x)/q(x,y))
physical-16	Semantic event graphs for production-stage specialization	marking_next = marking + C_incidence*sigma; enabled(sigma) requires marking >= Pre*sigma
bio-01	Reaction-network simulation and conserved quantities	dx/dt = N * v(x,theta) + b(t); v_j(x) = k_j * product_i(x_i ^ alpha_ij)
bio-02	Enzyme kinetics as a typed model-selection problem	E + S <-> ES -> E + P; KM = (k_minus1 + k_cat) / k_1
bio-03	Cell-free reporter-assay models and observation calibration	dm/dt = k_tx - delta_m*m; dp/dt = k_tl*m - delta_p*p
bio-04	Constraint-based metabolic models and flux envelopes	S*v = 0; lower <= v <= upper
bio-05	Omics workflows, count models and reproducible data lineage	K_ij ~ NegativeBinomial(mu_ij, alpha_i); mu_ij = s_j*q_ij; log2(q_ij) = sum_r(X_jr*beta_ir)
bio-06	Protein conformational trajectories and geometric descriptors	dq_i/dt = p_i/m_i; dp_i/dt = -partial(U(q))/partial(q_i)
bio-07	Microfluidic video analytics and droplet morphometry	A = a_pixel * sum_pixels(mask); d_equiv = 2*sqrt(A/pi)
bio-08	Microscopy morphometrics and spatial image phenotypes	A = a_pixel*sum_pixels(mask); circularity = 4*pi*A/(P^2)
bio-09	Population abundance with imperfect observation	N_i ~ Poisson(lambda_i); y_ij | N_i ~ Binomial(N_i, p_ij)

Profile	Domain	Governing relation / organising object
bio-10	Spiking-neuron models with explicit event semantics	C_m*dV/dt = -g_L*(V-E_L) + I(t); V >= V_threshold => emit_spike and V := V_reset
bio-11	Intentional gaze communication from an off-body camera	B = (X^T*X + lambda*I)^(-1)*X^T*Y; g_hat = phi(frame)*B
bio-12	Rehabilitation movement analysis and inverse-dynamics simulation	M(q)*q_ddot + C(q,q_dot) + G(q) = tau + J(q)^T*F_ext; tau = R(q)*f_muscle
bio-13	Compartment pharmacokinetics and observation models	dA_depot/dt = -k_a*A_depot; dA_central/dt = k_a*A_depot - (CL/V)*A_central
bio-14	Multivariate monitoring of benign process records	Z = (X - mu)/sigma, columnwise; Z = T*P^T + E; P^T*P = I
bio-15	Cardiovascular digital models and conservation across compartments	dV_i/dt = sum_j(Q_ji) - sum_j(Q_ij); Q_ij = (P_i - P_j)/R_ij
bio-16	Reaction-diffusion pattern models and spatial modes	partial_t(u) = f(u,v) + D_u*Laplacian(u); partial_t(v) = g(u,v) + D_v*Laplacian(v)
human-01	Language, parsing and symbolic interfaces	G=(N,Sigma,P,S); L(G)={w : S =>*_G w}; chart[n+1]=ParseStep(chart[n],token[n],G[n])
human-02	Music, audio and adaptive signal processing	X[m,k]=sum_{r=0}^{L-1} x[mH+r]*w[r]*exp(-2*pi*i*k*r/L); y[n]=sum_{j=0}^{J} b[j]*x[n-j]-sum_{j=1}^{K} a[j]*y[n-j]
human-03	Learning systems and adaptive assessment	p_i(theta)=1/(1+exp(-a_i*(theta-b_i))); posterior[n+1](theta) proportional to posterior[n](theta)*P(answer[n]|theta,item[n])
human-04	Logistics, routing and resource allocation	min_x sum_{i,j,k} c[i,j]*x[i,j,k]; sum_{j,k} x[i,j,k]=1 for each serviced customer i
human-05	Economic and social agent models	x_i[n+1]=f_i(x[n],d_i[n],e_i[n]); wealth_total[n]=sum_i wealth_i[n]
human-06	Game theory and strategic interaction	u_i(p)=sum_a u_i(a)*product_j p_j(a_j); p_star is Nash if u_i(p_star)>=u_i(q_i,p_star_-i) for every i,q_i
human-07	Creative geometry and procedural media	v_world = M_parent*M_local*v_homogeneous; mesh[n+1]=ApplyProduction(mesh[n],rule[n])
human-08	Archives, provenance and cultural knowledge	P=(Entities,Activities,Agents,Relations); DerivedFrom(new,old) records an explicit derivation relation

E. The 26 selected role comparisons

This selection contains 23 non-internship roles and three internship routes. It includes direct leads, specialist comparisons and seniority-dependent routes. Each profile preserves the exact role and condition from the earlier research, followed by its proposed lifecycle connection. [S1]

588449 Software Designer (C++) - Image Guided Therapy

Best | Netherlands | Netherlands first lead

Perspective	Retained finding
Lifecycle connection	Product engineering, integration and use
Portfolio evidence	TOMElinedT native C++/Java/JNI; TOM-GROK compiled/GPU runtime; TNOMTOM validation; earlier graphics/game development.
Role conditions	About 5 years relevant development; strong C/C++, Windows/legacy code; model-based design/code generation; TDD/BDD; regulated-industry experience
Application emphasis	Prioritize if you can document professional C/C++ and Windows delivery. Show a readable native-code case study with regression tests.

590735 Data and AI Engineer

Amsterdam | Netherlands | Netherlands conditional lead

Perspective	Retained finding
Lifecycle connection	Enterprise information, commercial operations and renewal
Portfolio evidence	TNOMTOM durable Python/SQLite store and thermal-model fitting on public measurements; TOMK evaluation/provenance; TOMElinedT bounded agent interface.
Role conditions	Databricks/Delta/PySpark/Azure, commercial CRM/ERP/financial analysis, modern production LLM agents, professional years and exact degree remain unverified.
Application emphasis	Use the durable-store and calibration case studies. Add evidence of the named enterprise stack and business outcomes before claiming a direct match.

589928 Senior Methods HW Modelling Engineer

Suzhou | China | International scientific-software lead

Perspective	Retained finding
Lifecycle connection	Physical modelling, component design and verification
Portfolio evidence	XD-TAU v1.5.1 typed physical models and exact reference; TNOMTOM fitting on public measurements; TOM-GROK CUDA; TOMElinedT native integration.
Role conditions	Substantial professional modern C++ product development, Windows, CI/CD, simulation/scientific-software products; bench/scanner/phantom correlation; MR knowledge preferred
Application emphasis	If China is feasible, demonstrate modern C++ product experience and how you validate a physical model against measured data.

584997 GenAI Solutions & Prompt Engineer

Bangalore | India | International AI engineering lead

Perspective	Retained finding
Lifecycle connection	Systems integration, verification and engineering methods
Portfolio evidence	TNOMTOM/TOMK for explicit evaluation, traceability, test design and regression; AI-assisted development provenance; TOMElinedT for interface/tool engineering.
Role conditions	Bangalore office >=3 days/week. Must demonstrate applied LLM workflows and C#/.NET; deterministic agent architecture alone does not meet RAG/LLM experience. No sponsorship statement found in description.
Application emphasis	Potential specialist application if stack experience exists.

589859 Software Technologist C++

Bangalore | India | International software lead

Perspective	Retained finding
Lifecycle connection	Product software engineering and maintenance
Portfolio evidence	TNOMTOM; TOMElinedT; TOM-GROK; TOMK
Role conditions	Clarify degree-specific experience route; substantiate practical C++ design/development/testing.
Application emphasis	Broader international software option

593840 Senior Software Engineer

Shenyang | China | International software lead

Perspective	Retained finding
Lifecycle connection	Product integration, performance and software quality
Portfolio evidence	TOMElinedT, TNOMTOM, TOM-GROK, TOMK: implementation, cross-component integration, performance and test discipline.
Role conditions	Shenyang office >=3 days/week. Verify relocation/work authorization and actual professional software experience. Broad title/specification should be clarified with recruiter. No Chinese-language requirement stated.
Application emphasis	Broader software route, less stack-specific than original role.

593980 Software Technologist I

Bangalore | India | International software lead

Perspective	Retained finding
Lifecycle connection	Application engineering, interaction and maintenance
Portfolio evidence	TOMElinedT/TNOMTOM/TOMK as concrete implementation and verification portfolio.
Role conditions	Bangalore. Need C#/WPF/.NET proof; role has unusual minimum/preferred experience wording, preserve and clarify. No visa language found.
Application emphasis	Potential direct software route.

585168 Design Technologist (Murrysville, PA)

Murrysville, Pennsylvania, United States | United States of America | US work-authorization dependent

Perspective	Retained finding
Lifecycle connection	Need discovery, prototype and user evaluation
Portfolio evidence	TOMElinedT for working Android/UI and coding-agent prototype; XD-TAU/TNOMTOM for systems thinking; older creative-tech work for user-centred prototyping.
Role conditions	US work authorization required; no visa sponsorship now/future; no relocation benefits; commute to Murrysville. Need actual UX research/visual portfolio and proof of front-end proficiency.
Application emphasis	First international design target only if US-authorized.

580470 Senior Software Engineer - Image Reconstruction (C++/CUDA)

Orange (OH), Ohio, United States | United States of America | US work-authorization dependent

Perspective	Retained finding
Lifecycle connection	Scientific computation and imaging performance
Portfolio evidence	XD-TAU v1.5.1; TOM-GROK; TNOMTOM; JTK-Twin-Viper
Role conditions	Bachelor or higher; 5+ years professional C++; GPU or constrained/embedded optimization; US work authorization required, no sponsorship now/future and no relocation benefits
Application emphasis	Strong technical theme; US eligibility gate

586928 Senior Software Engineer - Medical Image Reconstruction (C++/CUDA)

Orange (OH), Ohio, United States; Hamilton, New Jersey, United States | United States of America | US work-authorization dependent

Perspective	Retained finding
Lifecycle connection	Scientific computation and imaging performance
Portfolio evidence	TOM-GROK CUDA/WebGPU and memory/throughput work; XD-TAU numerical kernels; JTK/TK geometry; TOM-MAKTUBIII Fourier/operators; TOMElinedT native code.
Role conditions	US no sponsorship now/future, no relocation; commute Orange OH or Mercerville NJ, hybrid 3 days. Custom integer transition throughput is not CT reconstruction performance. Professional C++ years and image-processing domain need evidence.
Application emphasis	Flag alongside separate similar opening 580470.

588485 Test Development Engineer- fixed term (All genders)

Böblingen | Germany | Germany language/lab dependent

Perspective	Retained finding
Lifecycle connection	Physical instrumentation and test development
Portfolio evidence	TNOMTOM replay/negative testing and TCLab identification; XD-TAU exact reference checks; TOM-GROK backend comparison.
Role conditions	Böblingen onsite, fixed term through September 2028. Portfolio software tests do not establish hands-on electronics/instrumentation. German proficiency unknown.
Application emphasis	Good EU conversation if electronics and German background supports.

586913 Research Scientist - AI in Computed Tomography Image Formation (all genders)

Hamburg | Germany | Research degree/domain dependent

Perspective	Retained finding
Lifecycle connection	Research and image-formation algorithm development
Portfolio evidence	XD-TAU v1.5.1; TOM-GROK; TNOMTOM; JTK-Twin-Viper
Role conditions	Master in AI/data science/informatics; actual CNN training, PyTorch/NumPy expertise; medical-imaging clinical understanding; Python/C++; German optional
Application emphasis	Research stretch; degree and medical-imaging evidence needed

586911 Research Scientist - Physics of Computed Tomography (all genders)

Hamburg | Germany | Research degree/domain dependent

Perspective	Retained finding
Lifecycle connection	Research and physical image-formation modelling
Portfolio evidence	XD-TAU v1.5.1; TOM-GROK; TNOMTOM; JTK-Twin-Viper
Role conditions	Master in physics/electrical engineering/data science-AI; x-ray generation, matter interaction/detection and in-depth CT; Python/C++; German optional
Application emphasis	Physics research stretch

591185 AI Senior Systems Engineer (AI for RAMS & Systems Engineering)

Eindhoven | Netherlands | Netherlands senior stretch

Perspective	Retained finding
Lifecycle connection	Systems methods, reliability and lifecycle verification
Portfolio evidence	TNOMTOM; XD-TAU v1.5.1; TOMK
Role conditions	Master or PhD; extensive Systems Engineering/Architecture/RAMS/Digital Engineering in complex products; engineering transformation; FMEA/HALT/ALT context
Application emphasis	Thematic Netherlands stretch, senior organizational role

580273 Senior Software Architect - Azurion Eye

Best | Netherlands | Netherlands senior stretch

Perspective	Retained finding
Lifecycle connection	Architecture, deployment and product evolution
Portfolio evidence	TNOMTOM; TOMElinedT; TOM-GROK; TOMK
Role conditions	Typically7+ years complex regulated software; Kubernetes/containers/distributed cloud-edge systems; Go/Python; DICOM/FHIR relevant; architecture leadership
Application emphasis	Netherlands architecture stretch

588477 Product Owner/Technical Lead - Image Guided Therapy (Display Software)

Best | Netherlands | Netherlands senior stretch

Perspective	Retained finding
Lifecycle connection	Product delivery, prioritisation and software evolution
Portfolio evidence	TNOMTOM; TOMElinedT; TOM-GROK; TOMK
Role conditions	More than10 years software-development experience and more than3 years Product Owner experience; hands-on C++11/C#/Windows; model-based design, TDD/BDD
Application emphasis	Deprioritize unless lengthy product-owner track record exists

570549 Senior Software Design Quality Engineer - AI

Best | Netherlands | Netherlands senior stretch

Perspective	Retained finding
Lifecycle connection	Design quality and lifecycle management
Portfolio evidence	TNOMTOM; TOMElinedT; TOM-GROK; TOMK
Role conditions	Best Senior Software Design Quality Engineer-AI: BA/MA and15+ years, design-quality NPD/LCM, detailed regulatory standards; no relocation. Strong thematic testing match does not satisfy seniority/regulatory proof.
Application emphasis	Deprioritize unless you can substantiate 15+ years and design-quality ownership under medical software standards.

590246 Assistant Function Development Engineer-预研

Shenzhen | China | Exploratory engineering conditional

Perspective	Retained finding
Lifecycle connection	Physical principles, demonstrators and function development
Portfolio evidence	XD-TAU/TNOMTOM explicit physical modelling and unit/energy contracts; TK-MK-ULTRA/JTK mechanical/geometry models; prior creative-technologist/inventor prototypes potentially more direct.
Role conditions	Shenzhen; need hands-on physical prototype/instrumentation and engineering validation experience. Synthetic/rational simulations do not establish real device function or clinical efficacy. Title says Assistant, description says (Senior), so confirm level. No explicit Chinese-language requirement found.
Application emphasis	Worth an exploratory application if past invention portfolio includes real physical prototypes; senior related role590245 requires5 years.

585855 E-Commerce Data, Automation & AI Manager (m/w/d)

Hamburg | Germany | Commercial-domain dependent

Perspective	Retained finding
Lifecycle connection	Commercial information, digital channels and demand
Portfolio evidence	TNOMTOM empirical data fitting, exact money/inventory logic and workflows; TOMK provenance; original creative-tech consulting.
Role conditions	Hamburg >=3 days/week. Need real Amazon-commerce/KPI/BI experience, predictive/LLM-commerce implementation; theoretical suite does not supply it.
Application emphasis	Potential only if unpublished commercial analytics experience exists.

592685 UI/UX Designer - Advanced Imaging (Bedford, MA)

Bedford, Massachusetts, United States | United States of America | US design qualification dependent

Perspective	Retained finding
Lifecycle connection	User interaction and imaging workflow
Portfolio evidence	TOMElinedT UI/prototype and prior creative technology; HF technical systems alone weaker evidence than visual work.
Role conditions	US authorization/no sponsorship now or future; no relocation; Bedford commute and cath-lab observation. Need shipped UI and Figma/design-system portfolio.
Application emphasis	Conditional creative alternative.

587348 Software Technologist I/II-C# Full Stack

Bangalore | India | Full-stack experience dependent

Perspective	Retained finding
Lifecycle connection	Application and enterprise-service development
Portfolio evidence	TNOMTOM Node/Python/durable SQL and typed APIs; TOMElinedT product delivery; prior game/creative development if C# verified.
Role conditions	Bangalore >=3 days office. Angular and C# professional depth not evidenced by inspected HF files.
Application emphasis	Conditional full-stack alternative.

591151 Software Technologist I - Embedded C++

Pune; Bangalore | India | Embedded/RTOS dependent

Perspective	Retained finding
Lifecycle connection	Embedded product control and hardware interfaces
Portfolio evidence	TOMElinedT JNI/C++ Android ARM64; TOM-GROK finite/native backends; TNOMTOM exact byte kernels and tests.
Role conditions	Pune (listing also multi-location; use detail full locations). Android NDK/ARM64 is not microcontroller ThreadX experience; RTOS/device interfaces/regulated history not evidenced.
Application emphasis	Lower priority unless embedded career experience exists.

593837 Senior/Advanced Algorithm Engineer

Shenyang | China | Senior imaging specialist stretch

Perspective	Retained finding
Lifecycle connection	Advanced imaging algorithms and product computation
Portfolio evidence	XD-TAU v1.5.1; TOM-GROK; TNOMTOM; JTK-Twin-Viper
Role conditions	Master/PhD;10+ years algorithms with computer vision/image processing; actual deep learning; DICOM/ITK/VTK; CT visualization
Application emphasis	Deprioritize unless medical-imaging algorithm career exists

586187 Product Innovation & AI Prototyping Internship - Philips Avent Experience Innovation

Eindhoven | Netherlands | Student route only

Perspective	Retained finding
Lifecycle connection	Early concept, experience innovation and prototyping
Portfolio evidence	TOMElinedT; XD-TAU; TNOMTOM; creative-technology background
Role conditions	Student/training route: confirm its specific eligibility and contract. Avent AI Prototyping Internship Eindhoven: explicitly currently studying industrial/integrated product/mechanical/design engineering or comparable; >=6months32-40hours. Excellent thematic match but student-gated, not ordinary experienced-hire vacancy.
Application emphasis	Confirm the stated specialist experience, degree and location conditions before prioritizing this role.

593359 Internship: Synthetic Data Generation for Dental AI (Master)

Eindhoven | Netherlands | Internship eligibility to confirm

Perspective	Retained finding
Lifecycle connection	Model training data, geometry and visual evaluation
Portfolio evidence	TNOMTOM; TOMK; TOMElinedT
Role conditions	Student/training route: confirm its specific eligibility and contract. Dental synthetic-data internship Eindhoven: MSc-level DataScience/CS/AI/math/physics/biomed; Python+PyTorch/CV,3D rendering. Do not claim mandatory active enrollment from this text: it says MSc and internship, not explicit enrollment. Confirm eligibility with recruiter.
Application emphasis	Confirm the stated specialist experience, degree and location conditions before prioritizing this role.

593360 Internship: Robot-Assisted Toothbrush IMU Validation

Eindhoven | Netherlands | Student route only

Perspective	Retained finding
Lifecycle connection	Physical measurement, motion models and verification
Portfolio evidence	XD-TAU/JTK coordinate and timing models; TNOMTOM reproducible validation.
Role conditions	Student/training route: confirm its specific eligibility and contract. Current student eligibility and hands-on ROS2/robot/sensor integration not verified.
Application emphasis	Only consider if the stated student eligibility applies.

F. Complete Philips vacancy snapshot: 787 requisitions

Snapshot: 2 November 2026, 08:37 CET. Netherlands entries appear first, followed by other primary countries. Select a requisition ID or title to open its official retained URL. Location is the primary listing location; some requisitions also offer other locations. The lifecycle facet is an analytical classification for this report. It identifies a possible organisational connection, rather than asserting a personal qualification. [S1]

Code	Inferred lifecycle facet	Count
A	Cross-lifecycle delivery and support	225
B	Customer adoption and commercial operations	187
C	Deployment, service and renewal	48
D	Enterprise enabling functions	16
E	Manufacturing and industrial delivery	27
F	Materials, supply and industrial delivery	36

Code	Inferred lifecycle facet	Count
G	Product software, data and lifecycle information	33
H	Research, concept and component engineering	81
I	Unresolved	1
J	Use, clinical workflow and adoption	63
K	Verification and lifecycle quality	70

Netherlands

Job ID	Current title at snapshot	Primary location	Facet
591185	AI Senior Systems Engineer (AI for RAMS & Systems Engineering)	Eindhoven	K
588742	Category Manager IT - Software & Cloud	Eindhoven	F
589416	Category Sourcing Manager Ocean Freight	Eindhoven	F
590796	Clinical Application Specialist - Cardiovascular & Radiology Informatics	Eindhoven	J
593286	Clinical Informatics Regulatory Lead	Best	K
590735	Data and AI Engineer	Amsterdam	G
580583	Detection Engineer	Best	G
584975	Director, Process Compliance & Centers of Excellence	Best	K
582700	Graduate Development Program - Factory Best	Best	E
585034	Head of Clinical Affairs Personal Health	Amsterdam	J
588917	Head of IT - Precision Diagnosis	Best	A
594001	Head of SAP Transformation	Amsterdam	A
587911	Implementation Clinical Lead - Central and Eastern Europe	Netherlands - Remote Based	J
591183	Internship: Enabling Execution of 2030 Impact Ambitions	Amsterdam	A
593151	Internship: Enterprise Strategy	Amsterdam	A
570478	Internship: Marketing & Marketing Communication	Amsterdam	B
594772	Internship: Philips Communications & Brand	Amsterdam	A
593360	Internship: Robot-Assisted Toothbrush IMU Validation	Eindhoven	K
593359	Internship: Synthetic Data Generation for Dental AI (Master)	Eindhoven	G
588334	Maintenance Supervisor - Philips - Drachten	Drachten	C
591958	Philips Store For Business Platform Manager - Western Europe Personal Health	Amsterdam	A
593371	Process Operator - Philips - Drachten	Drachten	E
586188	Product Design Internship - Philips Avent Experience Innovation	Amsterdam	H
586187	Product Innovation & AI Prototyping Internship - Philips Avent Experience Innovation	Eindhoven	H
589569	Product Manager IGT-S	Best	B
588477	Product Owner/Technical Lead - Image Guided Therapy (Display Software)	Best	G
590318	Senior Clinical Product Marketing Manager	Best	J
587827	Senior IP Counsel - Audio/Video Technologies	Eindhoven	D
593337	Senior Manager Internal Control over Sustainability Reporting (ESG)	Amsterdam	A
580273	Senior Software Architect - Azurion Eye	Best	G
570549	Senior Software Design Quality Engineer - AI	Best	K
588449	Software Designer (C++) - Image Guided Therapy	Best	H

Job ID	Current title at snapshot	Primary location	Facet
589107	SPS Diagnostic Imaging (DI) Operations Leader	Best	A
593188	Sr. Clinical Product Manager - Neurology	Best	J
587868	Sr. Manager, PR & AI Discovery	Amsterdam	A

Australia

Job ID	Current title at snapshot	Primary location	Facet
592811	Clinical Applications Specialist, Imaging (Australia / New Zealand)	Sydney	J
590364	Digital Marketing & E-Commerce Specialist - Personal Health	Sydney	B
592188	General Manager, Healthcare ANZ	Sydney	A
589825	GI Ultrasound Sales Manager	Brisbane	B
588632	Marketing Operations Manager	Sydney	B

Austria

Job ID	Current title at snapshot	Primary location	Facet
586628	Klagenfurt EHS Officer (all genders)	Klagenfurt	A
591631	Product Industrialization Project Manager - Oral Healthcare (All genders)	Klagenfurt	E
584192	Systems Engineer (m/w/d)	Klagenfurt	K
592175	Werkstudent im Bereich Technische Dokumentation & Digitalisierung (20 h) - befristete Stelle bis Ende 2027 (m/w/d)	Klagenfurt	A

Belgium

Job ID	Current title at snapshot	Primary location	Facet
583908	Clinical Application Specialist IGT - Interventional Cardiology/Radiology (Flanders area)	Brussel (Bruxelles)	J
580711	Clinical Specialist Belux - Image-Guided Therapy Systems	Brussel (Bruxelles)	J
568734	Key Account Manager Health Systems - Flanders	Antwerp	B
586113	Remote Support Engineer Monitoring & Connected Care	Brussel (Bruxelles)	C
580849	Sales & Clinical Specialist BeLux - IGT-Devices	Brussel (Bruxelles)	J
590932	Solution Architect H&AM (Hospital & Ambulatory Monitoring)	Brussel (Bruxelles)	A

Brazil

Job ID	Current title at snapshot	Primary location	Facet
519544	Banco de Talentos | Pessoas com Deficiência	Brazil - Home Based	D
589884	Consultor Vendas Sênior	Barueri	A
592837	Early Career Talent Partner	Varginha	D
592842	Estagiário(a) Sales Operations	Varginha	B
581902	Estágio em Atendimento ao Cliente	Varginha	A
590748	Estágio em Gerencimento de Pedidos	Varginha	A
580990	Medical Devices Maintenance Specialist III	Brazil - Remote Based	C
590154	Product Coordinator	Barueri	A
585906	Programa de Estágio Philips Desenvolve 2026 |Estagiário(a) - Suporte Comercial	Barueri	A
591552	Programa Philips Desenvolve 2025 - Estágio em Configuração de Equipamentos	Varginha	A
590082	Regional LATAM Informatics Delivery Manager	Barueri	A
585478	Remote Service Engineer	Blumenau	C
578802	Safety Risk Engineer (m/f/d)	Varginha	H
589725	Sales Operations Intern	Varginha	B
560844	Senior Support Technician	Blumenau	H
590666	Services BDM US LATAM	Barueri	A
587335	Solution Architect - Interoperability (IBE / Rhapsody)	Blumenau	A

Canada

Job ID	Current title at snapshot	Primary location	Facet
584310	Account Manager, Medical Imaging - Based in British Columbia	Canada - Remote Based, Ontario, Canada - Canada - À distance, Ontario, Canada	B
590384	Account Manager, Patient Monitoring - Atlantic Canada	Halifax, Nova Scotia, Canada - Halifax, Nouvelle-Écosse, Canada	B
571473	Clinical Sales Specialist - CT and DXR (Western Canada)	Canada - Remote Based, Ontario, Canada - Canada - À distance, Ontario, Canada	J
585454	Clinical Sales Specialist- Magnetic Resonance Imaging (ON/QC/Maritimes)	Toronto	J
591906	Field Service Employee - Imaging Systems (Field: London, ON)	London	C
592512	Field Service Employee - Medical Imaging (Field: Sudbury, ON)	Sudbury	C
587817	Integration Consultant, Diagnostic Cardiology Solutions (Remote Based - Canada)	Canada - Remote Based, Ontario, Canada - Canada - À distance, Ontario, Canada	A
591825	Sales, Account Manager - Ultrasound (Southwestern ON, Canada)	Canada - Remote Based, Ontario, Canada - Canada - À distance, Ontario, Canada	B
593285	Sales, Clinical Application Specialist - Cardiovascular Ultrasound (Alberta)	Canada - Home Based, Ontario, Canada - Canada - À domicile, Ontario, Canada	J
594064	Sales, Territory Business Manager - Oral Health/Gestionnaire de territoire - Soins buccodentaires (Ventes professionnelles dentaires, Rive-Sud de Montréal) - Poste à distance, basé sur la Rive-Sud de Montréal (Bilingue)	Montreal, Quebec, Canada - Montréal, Québec, Canada	B
592591	Senior Customer Project Manager (Mississauga, ON)	Mississauga	A

China

Job ID	Current title at snapshot	Primary location	Facet
592945	2027校园招聘 临床应用培训生 Clinical Application Specialist	Beijing	J
592943	2027校园招聘 临床应用培训生 Clinical Application Specialist CT	Shanghai	J
592942	2027校园招聘 制造业卓越运营管培生	Zhuhai	A
592965	2027校园招聘 机械设计培训生 Mechanical Design Engineer	Zhuhai	H
592963	2027校园招聘 生产管理培训生	Zhuhai	A
592941	2027校园招聘 电气设计培训生 Electrical Design Engineer	Zhuhai	H
589885	Advanced Electronics Engineer	Shenzhen	H
589891	Advanced Embedded Software Engineer	Shenzhen	G
573845	Advanced ME (IE)	Suzhou	A
586956	Advanced Mechanical Engineer	Suzhou	H
583704	Advanced Quality Engineer	Suzhou	K
587500	Advanced Quality Engineer	Suzhou	K
586941	Advanced Software Engineer	Suzhou	G
586933	Advanced Software Engineer-C++	Shenyang	G
593750	Advanced Software Engineer-CT	Shenyang	G
591589	Advisory Clinical Scientist	Beijing	J
589226	AI Imaging Data Scientist Intern	Shanghai	H
593760	Ass. Account Manager, Imaging &IGT	Jinan	B
587085	Assistant Digital Platform & Martech Manager	Shanghai	A
591641	Assistant Finance Business Partner	Shanghai	D
590246	Assistant Function Development Engineer-预研	Shenzhen	H
588320	Assistant Innovation Marketing Manager, OHC	Shanghai	B
586170	Assistant Sales Manager, IGTD	Beijing	B
592089	Assistant Service Sales Manager, Imaging & IGT	Beijing	B
593905	Associate Clinical Operations Manager	Shanghai	J
592213	BOM Productivity Program Manager MR	Suzhou	A
592874	Business Operation Manager	Shanghai	A
592881	Category Sourcing Senior Manager - Mainstream Asia	Shanghai	F
592346	Clinical Development Intern	Shanghai	J
589378	Clinical Success Manager, IGTS	Beijing	J
590395	Commercial Policy & Pricing Manager	Shanghai	B
592726	Compliance Investigator	Shanghai	K
590460	Consumer Research Manager	Shenzhen	H
590461	Consumer Research Manager	Shenzhen	H
594392	Consumer Research Team Lead	Shenzhen	H
586952	CT Clinical Scientist	Beijing	J

Job ID	Current title at snapshot	Primary location	Facet
590040	Customer Experience Manager	Shenzhen	A
591702	Design Quality Engineer	Suzhou	K
587476	Digital Architect	Shenzhen	A
594039	E-Marketing Intern	Shanghai	B
591811	FPGA Engineer	Suzhou	H
590631	Head of AI Philips China	Shanghai	G
593784	HR Intern	Suzhou	D
591574	IGT Supervisor	Guangzhou	A
594309	IGT-S Downstream Marketing Leader, G.C.	Beijing	B
590857	IGTS Channel Sales Supervisor	Zhengzhou	B
591067	IGTS Channel Sales Supervisor	Beijing	B
594574	IMC & Social Manager	Shanghai	A
594577	IMC & Social Manager	Shanghai	A
592810	Industrial Design Internship - Shaving Experience Design	Shenzhen	E
585204	Innovation Marketing Manager	Shanghai	B
590442	Integral Project Lead	Shenzhen	A
590490	Integral Project Leader	Shenzhen	A
590505	Integral Project Leader	Shenzhen	A
590229	Integral Project Leader-Oral Healthcare	Shenzhen	A
592626	Integral Project Manager	Shenzhen	A
593976	IT AI Solution Analyst Intern	Shanghai	G
593922	IT PM Intern for PH Business	Shanghai	A
590035	KA Manager	Shanghai	A
587397	Key Account Manager	Shenzhen	B
573167	Lead engineer ESE	Shenzhen	H
591306	Manufacturing Engineering Intern	Suzhou	E
589389	Market Access Intern, IGT-D	Shanghai	B
590681	Marketing Director, Male Grooming	Shanghai	B
594114	Marketing Intern Ventures	Shenzhen	B
593929	Marketing Intern, Hair Care	Shenzhen	B
593930	Marketing Intern, Hair Care	Shenzhen	B
593844	Marketing Intern, Male Grooming	Shanghai	B
591793	Marketing Manager	Shanghai	B
589883	Mechanical Development Engineer	Shenzhen	H
591007	MR Systems Architect	Suzhou	A
591870	MR Systems Engineer	Suzhou	K
591871	MR Systems Engineer	Suzhou	K
591005	MR Systems Engineer/Architect	Suzhou	K

Job ID	Current title at snapshot	Primary location	Facet
583658	Operational Buyer- MR	Suzhou	A
590241	Platform Architect-Oral Healthcare	Shenzhen	A
580109	PQMS Manager	Shenzhen	A
590122	Principal Scientist	Shenzhen	H
591787	Procurement Engineer	Suzhou	F
591573	Procurement Project Manager	Shenzhen	F
586900	Product Designer (Haircare)	Shenzhen	H
592176	Product Industrialization Engineer	Suzhou	E
588379	Product Industrialization Intern	Suzhou	E
580689	Product Industrialization Project Manager	Suzhou	E
591223	Product Industrialization Project Manager OHC	Shenzhen	E
592202	Product Industrialization Project Manager-CT	Suzhou	E
589174	Product Sales	XiAn	B
584687	Project Manager	Shenyang	A
591225	Project Manager	Shenyang	A
590363	Purchase to Pay Coordinator	Suzhou	F
593676	Regulatory Affairs Manager	Shenzhen	K
590274	Remote Service Leader	Shanghai	A
585206	RGM Manager	Shanghai	A
587871	Risk Management Quality Engineer	Shenzhen	K
590036	Sales Enablement Lead	Shanghai	B
590037	Sales Enablement Lead	Shanghai	B
568353	Sales Manager	XiAn	B
592521	Sales Manager, IGT-D	Shenyang	B
592409	Sales Manager, IGTD	Guangzhou	B
576954	Senior Account Sales, Imaging & IGT	Urumqi	B
589488	Senior Account Sales, KA, Imaging &IGT	Chengdu	B
586175	Senior Account Sales, US	Urumqi	B
570227	Senior Account Sales , Imaging&IGT	Nanjing	B
591220	Senior Architect System and Development Engineering	Shenzhen	H
587696	Senior Clinical Development Scientist, Ultrasound &EI	Shanghai	J
585624	Senior Clinical Development Scientist,IGT	Shanghai	J
591643	Senior Consumer Marketing Manager Shaving	Shenzhen	B
592525	Senior Decoration Engineer	Zhuhai	H
588905	Senior Design Quality Manager	Suzhou	K
590848	Senior DevOps Technologist	Shanghai	G
592884	Senior DTC Manager	Shanghai	A
590111	Senior Electrical Engineer	Suzhou	H

Job ID	Current title at snapshot	Primary location	Facet
583566	Senior Electronics Development Engineer	Shenzhen	H
589887	Senior Electronics Engineer	Shenzhen	H
589888	Senior Embedded Software Engineer	Shenzhen	G
590243	Senior Function Development Engineer - Fluid Dynamics	Shenzhen	H
590245	Senior Function Development Engineer-Dental Innovation	Shenzhen	H
593971	Senior Hardware Electrical Engineer	Suzhou	H
585509	Senior Marketing Manager-Oral Healthcare	Shanghai	B
589878	Senior Mechanical Development Engineer	Shenzhen	H
589879	Senior Mechanical Development Engineer	Shenzhen	H
587501	Senior Mechanical Engineer	Suzhou	H
590109	Senior Mechanical Engineer	Suzhou	H
592245	Senior Mechanical Engineer - Ultrasound	Suzhou	H
589928	Senior Methods HW Modelling Engineer	Suzhou	H
575064	Senior Platform Architect	Shenzhen	A
588102	Senior Product Sales, Oncology	Beijing	B
590929	Senior Project Manager (Evergreen)	Suzhou	A
586944	Senior Reliability Design Engineer	Shenyang	K
591731	Senior Service Product Sales, Imaging&IGT	Changsha	B
592949	Senior Services Sales	Lanzhou	B
591222	Senior Software Architect	Shenyang	G
593840	Senior Software Engineer	Shenyang	G
592861	Senior Software Engineer-CT	Suzhou	G
584842	Senior Systems Engineer-SW/IQ	Shenyang	K
589953	Senior Test Development Engineer	Shenzhen	K
593837	Senior/Advanced Algorithm Engineer	Shenyang	H
594036	Senior/Advanced Mechanical Engineer	Suzhou	H
589833	Service Sales, Imaging & IGT	China - Home Based	B
587865	Services Deliver Manager, Ultrasound	Shanghai	A
586792	Site Quality Leader	Zhuhai	K
590740	Software Architect-AI	Suzhou	G
591952	Software Engineer Intern 磁共振软件开发实习生	Suzhou	G
591744	Sr. Account Sales, Imaging & IGT	Beijing	B
584648	Sr. Account Sales,US	Beijing	B
592879	Sr. Manager, Brand Licensing	Shanghai	A
591300	Sr. MR Clinical Scientist	Chengdu	J
592644	SRC Marketing Intern	Shanghai	B
587960	Supply Chain Director	Suzhou	F
594766	System Architect-AI platform	Beijing	H

Job ID	Current title at snapshot	Primary location	Facet
586825	System Engineer	Shenzhen	A
589836	System Engineer	Shenzhen	A
593772	System Engineer-Intern	Suzhou	A
589952	Test and Verification Engineer	Shenzhen	K
591560	Test Engineer	Shenzhen	K
592260	抖音营销经理	Shanghai	A

Colombia

Job ID	Current title at snapshot	Primary location	Facet
570897	Sales Executive - Enterprise Informatics	Bogota	B

Costa Rica

Job ID	Current title at snapshot	Primary location	Facet
593889	Director, Quality Programs-Integrated Supply Chain (ISC)	Alajuela	F
587274	Molding Process Technician I	Alajuela	H
585375	Sr Program Manager	Alajuela	A
592235	Sr. Supplier Quality Engineer	Alajuela	F

Czechia

Job ID	Current title at snapshot	Primary location	Facet
583880	Key Account Manager	Prague	B
591685	Key Account Manager	Prague	B

Egypt

Job ID	Current title at snapshot	Primary location	Facet
585112	Customer Project Manager	Cairo	A
583849	Imaging Modality Sales Specialist (French Speaking)	Cairo	B
580242	Service Manager	Cairo	A

France

Job ID	Current title at snapshot	Primary location	Facet
589092	Chef de Produit OneBlade & Grooming H/F	Suresnes	A
588467	Chef de Projets Imagerie - Région Ouest	Nantes	A
589136	Clinical Application Leader IGT-Systems	Suresnes	J
564122	Cybersecurity Champion	Paris (FR)	A
593080	Intern _Sales Operations & Data Excellence F/H - Suresnes	Suresnes	B
585033	Key Account Manager DPH	Suresnes	B
585587	People Partner & Social Relations (France)	Suresnes	D
567430	Remote Service Engineer - CT	Suresnes	C
563420	Responsable Technique Ultrasons - Ile De France H/F	Suresnes	A
589256	Software Design Assurance Quality Engineer	Paris (FR)	K
593128	Stage Assistant commercial et marketing cardiologie interventionnelle et structurelle (H/F)	Suresnes	B
593121	Stage Assistant Marketing Scanner (Jan/Fév 2027)	Suresnes	B
593099	Stage KAM E-Commerce Amazon France H/F	Suresnes	B
588726	Stage Marketing Manager Assistant One Blade (janvier 2027)	Suresnes	B
592805	Stage Spécialiste Clinique IRM (H/F) (Janvier 2027 | stage 6 mois)	Suresnes	A
592983	Stagiaire HSE (Hygiène, Sécurité, Environnement) _ Janvier 2027	Suresnes	A
591760	Stagiaire Strategic Partnership Analyst (H/F) (janvier 2027)	Suresnes	A

Germany

Job ID	Current title at snapshot	Primary location	Facet
593277	Account Manager (all genders) DI/IGT/EI - Baden Württemberg	Germany - Home Based	B
585205	Account Manager (all genders) Imaging Systems - Baden Württemberg	Germany - Home Based	B
586106	Account Manager Healthcare Informatics (all genders) Region Nordost	Germany - Home Based	B
591746	Channel Leader Online DACH (All genders)	Hamburg	A
588759	Clinical Application Specialist - CT (all genders)	Germany - Home Based	J
588105	Clinical Application Specialist- Ultrasound - Region Bayern (All Genders)	Germany - Home Based	J
590301	Customer Demand Planner (All genders)	Hamburg	F
585855	E-Commerce Data, Automation & AI Manager (m/w/d)	Hamburg	B
589308	Field Specialist (all genders) MRT - Region Sachsen Anhalt/Brandenburg	Germany - Home Based	A
568669	Field Specialist (all genders) DXR/IGT Multimodality - Region Hamburg/Schleswig-Holstein	Germany - Home Based	A
576889	Field Specialist (all genders) für Medizinische Großgeräte - in der Region Thüringen/Sachsen-Anhalt	Germany - Home Based	A
585864	Field Specialist für MRT (all genders) Region Berlin	Germany - Home Based	A
586490	Field Specialist IGT (all genders) Region Thüringen	Germany - Home Based	A
586028	Field Specialist IGT (all genders) - Region Baden Württemberg	Germany - Home Based	A
584149	Field Specialist MRT (all genders) Region NRW-Süd	Germany - Home Based	A
583565	Field Spezialist H&AM (all genders) Region Niedersachsen Süd	Germany - Home Based	A
584673	Industriemechaniker / CNC-Fräser (all genders) 1 Jahr befristet	Hamburg	D
578378	Manufacturing Manager (all genders)	Böblingen	E
584928	PostDoc MR Physiker:in (all genders) an der Uniklinik RWTH Aachen	Aachen	A
588978	PostDoc MRT-Physiker:in (all genders) am Universitätsklinikum Bonn	Germany - Home Based	A
586965	Power Electronics Engineer (all genders)	Hamburg	H
592636	Product Security Architect (All Genders)	Munich	A
586913	Research Scientist - AI in Computed Tomography Image Formation (all genders)	Hamburg	H
586911	Research Scientist - Physics of Computed Tomography (all genders)	Hamburg	H
593858	Sales Trainee Programm Health Systems (18 Monate) Region Ost (all genders)	Germany - Home Based	B
591448	Service Technician H&AM (all genders) im Delivery Center	Germany - Home Based	C
590640	Strategic Account Manager (all genders) Baden Württemberg	Germany - Home Based	B

Job ID	Current title at snapshot	Primary location	Facet
586958	Systems Engineer & Technologist (all genders)	Hamburg	K
588485	Test Development Engineer- fixed term (All genders)	Böblingen	K
586962	Tube Designer Components (all genders)	Hamburg	H
592036	Werkstudent / Praktikum (all genders) Supply Chain Management	Hamburg	F
592263	Werkstudent / Praktikum (all genders) - Künstliche Intelligenz & Datenanalyse	Hamburg	A
591579	Werkstudent / Praktikum (all genders) Project Management Ultrasound	Hamburg	A
587828	Werkstudent / Praktikum (all genders) - SAP Plant Maintenance / Ersatzteilmanagement	Hamburg	C
587826	Werkstudent / Praktikum (all genders) - SAP Plant Maintenance / Instandhaltungsprozesse	Hamburg	C
589673	Werkstudent / Praktikum Projektmanagement - Produktivität (all genders)	Hamburg	A
592784	Werkstudent Business Management & Marketing (all genders)	Hamburg	B
592783	Werkstudent Customer Service Reporting & Data Analytics (all genders)	Hamburg	C
582768	Werkstudent Digitalisierung in der Produktionsplanung und -steuerung (all genders)	Hamburg	A
591200	Werkstudent im Produktionsmanagement (all genders)	Hamburg	A
591199	Werkstudent LEAN Management in der Produktion (all genders)	Hamburg	A
553237	Werkstudent Projektmanagement Medizintechnik & IT (all genders)	Kassel	A
591723	Werkstudent Prozessentwicklung & Digitalisierung (all genders)	Hamburg	A
582340	Werkstudent Prozessentwicklung & Engineering - Qualifizierung / Validierung (all genders)	Hamburg	H
591722	Werkstudent/ Praktikum Qualifizierungsdokumentation Tool Design (all genders)	Hamburg	A
591252	Werkstudent/ Praktikum AI Innovation Marketing (all genders)	Hamburg	B
584671	Zerspanungsmechaniker / CNC-Fräser (all genders) 1 Jahr befristet	Hamburg	D

Hong Kong

Job ID	Current title at snapshot	Primary location	Facet
587097	Field Service Engineer	Hong Kong	C
591645	Finance BP, Hongkong	Hong Kong	D
592950	G&B PIE Project Manager	Hong Kong	A

India

Job ID	Current title at snapshot	Primary location	Facet
588203	Account Manager - Govt, Central	Gurgaon	B
589214	Account Manager - Govt, East	India - Remote Based	B
584565	Account Manager - IS	India - Remote Based	B
592637	Account Manager - MATC	Kolkata (ex Calcutta)	B
585067	Account Manager IS	Gurgaon	B
559394	Account manager- Govt Sales	India - Home Based	B
585430	AI Technical Expert	Bangalore	G
588424	Architect II - DevOps	Bangalore	A
587504	Automation and AI Tech Lead - Personal Health	Bangalore	G
592197	BMM_Cardiology Ultrasound	Gurgaon	A
591177	Clinical Applications Specialist	Bangalore	J
589410	Complaint Handling Specialist	Pune	A
589455	Compliance Learning Operations Specialist	Bangalore	K
587979	Compliance Officer	Gurgaon	K
586582	Consumer Experience Research Lead (Digital)	Bangalore	H
592891	Design Quality Engineer	Pune	K
593975	Director - Product Development	Pune	A
591267	Embedded Software Architect	Pune	G
584997	GenAI Solutions & Prompt Engineer	Bangalore	G
593014	Global Product Manager - Respiratory Accessories	Bangalore	B
593019	Global Product Manager- Accelerators	Bangalore	B
593018	Global Product Manager- IVP & Launch Excellence	Bangalore	B
593826	Integrated Warehouse Distribution (IWD) Solution Lead	Chennai (ex Madras)	F
570606	Lead - E Commerce- Flipkart & Myntra	Bangalore	A
572199	Lead Data Scientist	Bangalore	H
590799	Operational Buyer Planner	Pune	A
589210	Packaging Engineer	Pune	H
593534	Product Industrialization Project Manager	Pune	E
585560	Production Planner	Pune	E
583088	Regional Sales Manager - IS	India - Home Based	B
591074	Resource Planner	Bangalore	A
592257	Senior Software Technologist - C++	Bangalore	G
585968	Senior Global Product Manager (Respiratory Therapy)	Pune	B
592427	Senior Infrastructure & Application Support Engineer	Bangalore	H
593879	Senior IT Architect	Bangalore	A
593674	Senior Manager - Quality & Regulatory	Pune	K
582154	Senior Product Manager	Bangalore	B

Job ID	Current title at snapshot	Primary location	Facet
587144	Senior Security Operations Lead - Digital Networks	Bangalore	A
591495	Senior Software Technologist I - C++	Bangalore	G
593661	Senior. Engineering Manager - PMO	Pune	H
589859	Software Technologist C++	Bangalore	G
593980	Software Technologist I	Bangalore	G
591151	Software Technologist I - Embedded C++	Pune	G
587348	Software Technologist I/II-C# Full Stack	Bangalore	G
593878	Solution Architect	Bangalore	A
593016	Strategist - Sleep & Respiratory Care (S&RC)	Bangalore	A
586408	Supplier Account Manager	Pune	F
586689	System Lead - Electrical	Pune	H
593672	Systems Engineer	Pune	K
593671	Technical Project Manager	Pune	A
593678	Technical Project Manager	Pune	A
589643	Technical Specialist-Technical support	Bangalore	C
591638	Territory Manager- Delhi 1 & Nepal	Gurgaon	A

India (indexed only)

Job ID	Current title at snapshot	Primary location	Facet
591655	Lead Systems Engineer (indexed title; live details unavailable)	Not supplied	I

Indonesia

Job ID	Current title at snapshot	Primary location	Facet
575236	Manufacturing Engineer - Equipment Engineering	Batam Island	E
588470	Modality Sales Specialist IGT, Indonesia	Jakarta (Java)	B
551316	Production Engineering Internship	Batam Island	E

Israel

Job ID	Current title at snapshot	Primary location	Facet
588965	Manual Assembler - יקנ רדחל ה/ביכרמ	Haifa	A
562932	Manual Assembler-יקנ רדחב ביכרמ	Israel - Home Based	A
586919	Physicist	Haifa	H
593609	Product Quality Engineer	Haifa	K
593614	Production Quality Engineer	Haifa	K
587893	Quality Project Manager	Haifa	K
585359	Senior Board Design Engineer	Haifa	H
585361	Senior FPGA Design Engineer	Haifa	H
585347	Senior Mechanical Engineer	Haifa	H
584663	Service & Delivery Quality Process Engineer	Ra'ananna	K
592896	Service Systems Engineer	Haifa	K
591563	Systems Engineer	Haifa	K

Italy

Job ID	Current title at snapshot	Primary location	Facet
583630	Account Manager Healthcare Informatics South Italy	Casoria - Napoli site	B
591022	CMM intern	Milano	A
587651	Strategic Solution Architect Intern	Genoa	A
594146	Technical Consultant	Genoa	A

Japan

Job ID	Current title at snapshot	Primary location	Facet
589830	Account Manager (担当：HPM/CIS、勤務地：松本支店)	Matsumoto	B
592011	Account Manager (担当：PD/IGTS、勤務地：東東京支店)	Tokyo	B
590921	Account Manager (担当：PD/IGTS、勤務地：横浜支店)	Yokohama	B
583193	Field Service Engineer (担当：IGT/DXR/US、勤務地：大阪支店)	Osaka	C
583191	Field Service Engineer (担当：MA/US、勤務地：広島支店)	Hiroshima	C
582033	Field Service Engineer (担当：MA/US、勤務地：東東京支店)	Tokyo	C
583189	Field Service Engineer (担当：MA/US、勤務地：福岡支店)	Fukuoka	C
577219	Field Service Technician (SRC Japan)	Saitama	C
590827	Finance Business Partner - PH	Tokyo	D
593148	HPM Customer Project Manager	Osaka	A
581259	Regulatory Affairs Manager	Tokyo	K
591004	Supply Chain Management Coordinator, SRC	Tokyo	F
586682	【三重】Sales Account Manager (担当製品：在宅医療向けの睡眠・呼吸製品)	Tsu	B
591088	【千葉市】Sales Account Manager (担当製品：在宅医療向け睡眠製品)	Chiba	B
581433	【名古屋】Sales Account Manager (担当製品：在宅医療向けの睡眠および呼吸製品)	Nagoya	B
579935	【名古屋】Sales Account Manager (担当製品：在宅医療向けの睡眠・呼吸製品)	Nagoya	B
582028	【名古屋市】Customer Service Support (担当製品：在宅医療向けの呼吸機製品)	Nagoya	C
582056	【埼玉】Sales Account Manager (担当製品：在宅医療向けの睡眠および呼吸製品)	Saitama	B
585664	【岐阜】Sales Account Manager (担当製品：在宅医療向けの睡眠・呼吸製品)	Gifu	B
591087	【府中市】Sales Account Manager (担当製品：在宅医療向けの呼吸関連製品)	Tokyo	B
585195	【東京】Sales Account Manager (担当製品：在宅医療向け呼吸製品)	Tokyo	B
586539	【東京】Sales Account Manager (担当製品：在宅医療向け睡眠製品)	Tokyo	B
591084	【東京】Sales Account Manager (担当製品：在宅医療向け睡眠製品)	Tokyo	B
591483	【横浜】Sales Account Manager (担当製品：在宅医療向けの睡眠製品)	Yokohama	B
591085	【浅草】Sales Account Manager (担当製品：在宅医療向けの呼吸関連製品)	Tokyo	B
592258	【神戸】Sales Account Manager (担当製品：在宅医療向けの呼吸関連製品)	Kobe	B
591089	【筑波】Sales Account Manager (担当製品：在宅医療向けの睡眠製品)	Tsukuba	B

Job ID	Current title at snapshot	Primary location	Facet
582855	【静岡】 Sales Account Manager	在宅医療機器 スリープ&レスピラトリ ケア (沼津市)	Numazu	B
563247	(障) General Service Support	Tokyo	A

Kazakhstan

Job ID	Current title at snapshot	Primary location	Facet
586232	Key Account Manager Personal Health - Electro	Almaty	B
590815	Partner Account Manager Service Sales	Almaty	B

Korea, Republic of

Job ID	Current title at snapshot	Primary location	Facet
590582	Clinical Application Specialist, IGT-S (Busan or Daegu)	Busan	J
568834	Communications and Brand Senior Manager, Korea and APAC	Seoul	A
583344	Consumer Marketing Lead, Korea	Seoul	B
578484	Consumer Marketing Manager	Seoul	B
587688	Field Service Engineer, Daegu	Daegu	C
593768	Market Procurement Manager	Seoul	F
590204	Sales Account Manager	Seoul	B

Malaysia

Job ID	Current title at snapshot	Primary location	Facet
594058	Business Marketing Manager	Kuala Lumpur - Co-labs	B
588109	Demand Planner	Kuala Lumpur - Co-labs	F
590878	Personal Health Business Zone Leader: JAPAC	Kuala Lumpur - Co-labs	A
590634	Strategic Account Manager	Kuala Lumpur - Co-labs	B

Mexico

Job ID	Current title at snapshot	Primary location	Facet
542862	E-Trade Specialist Mexico & North LATAM	Huixquilucan de Degollado	A
589049	PH Commercial Pricing Policy Lead Mexico & N.Latam	Huixquilucan de Degollado	B

New Zealand

Job ID	Current title at snapshot	Primary location	Facet
590801	Field Service Engineer Medical Imaging - Wellington	New Zealand - Remote Based	C

Panama

Job ID	Current title at snapshot	Primary location	Facet
592232	Collections Specialist	Ciudad de Panama	A
593442	Intern Lead Management	Ciudad de Panama	A
586528	IT Infrastructure Intern	Ciudad de Panama	A
590343	IT Service Support	Ciudad de Panama	A
593274	NAR Integrated Supply Chain Intern - Monitoring	Ciudad de Panama	F
592219	Operational Excellence Intern - HPM NAR	Ciudad de Panama	A
594495	Purchase to Pay Intern- ICA	Ciudad de Panama	F
580540	Talent Sourcing Partner	Ciudad de Panama	F

Philippines

Job ID	Current title at snapshot	Primary location	Facet
591442	Compliance and Quality Specialist (CAPA)	Taguig	K
591417	Modality Sales Specialist - Ultrasound	Taguig	B

Poland

Job ID	Current title at snapshot	Primary location	Facet
593171	593164 Przedstawiciel(ka) Medyczny(a) Sonicare	Poland - Home Based	A
594662	Commercial Excellence Intern (all genders)	Warszawa	B
591505	Consumer Marketing Manager (All genders)	Warszawa	B
592883	Digital Marketing & E-Commerce Specialist (All genders)	Warszawa	B
594581	Field Service Engineer/ Inżynier Serwisu Sprzętu Medycznego, Białystok (All genders)	Poland - Remote Based	C
594793	Field Service Engineer/ Inżynier Serwisu Sprzętu Medycznego, Łódź (All genders)	Poland - Remote Based	C
584867	Implementation Clinical Consultant - Radiology Informatics (All genders)	Poland - Remote Based	J
584868	Implementation Project Manager - Radiology Informatics CEE (All genders)	Poland - Remote Based	A
586331	International Mobility Coordinator - Maternity cover fixed term contract (all genders)	Lodz	A
592831	Key Account Intern (all genders)	Warszawa	B
592045	Key Account Manager - Hospital and Ambulatory Monitoring (Central&Eastern Poland)	Poland - Remote Based	B
587339	Key Account Manager Ultrasound (all genders)	Poland - Remote Based	B
583094	Modality Sales Specialist Healthcare IT - PACS (all genders)	Poland - Remote Based	B
589593	Młodszy Serwisant/Serwisantka Aparatury Medycznej	Warszawa	A
593158	Order Management Specialist (Nordics markets) (all genders)	Lodz	B
593165	Przedstawiciel(ka) Medyczny(a) Sonicare Bydgoszcz	Bydgoszcz	A
593164	Przedstawiciel(ka) Medyczny(a) Sonicare Szczecin	Poland - Remote Based	A
592714	Regional Marketing Intern CEE Mother & Child Care (all genders)	Warszawa	B
588944	Senior Customer Order Specialist (Fixed term) all genders	Lodz	A
591439	Services Intern with German (all genders)	Lodz	A
592559	Services Intern with Italian (all genders)	Lodz	A
588261	Site Planning Intern (all genders)	Lodz	A
594145	Technical Consultant	Warszawa	A
592813	Tender & Contract Operations Intern (all genders)	Warszawa	A
592972	Transport Specialist (all genders)	Lodz	A

Romania

Job ID	Current title at snapshot	Primary location	Facet
591156	Key Account Manager Diagnostic Imaging and Interventional Guided Therapy Romania	Bucharest	B
589602	Key Account Manager Ultrasound	Bucharest	B
589609	Key Account Manager Ultrasound - located in Cluj	Bucharest	B

Russian Federation

Job ID	Current title at snapshot	Primary location	Facet
584931	Market Sales Leader RCA (Russia and Central Asia)	Moskva	B
585333	Quality Specialist	Moskva	K

Singapore

Job ID	Current title at snapshot	Primary location	Facet
588343	Lead Systems Engineer	Singapore	K
588705	Manufacturing engineer plastic molding	Singapore	E
591311	Modality Sales Specialist - EI APAC	Singapore	B

South Africa

Job ID	Current title at snapshot	Primary location	Facet
593585	Key Account Manager	Johannesburg	B
558817	Tender Coordinator	Johannesburg	A

Spain

Job ID	Current title at snapshot	Primary location	Facet
592223	Clinical Application Specialist (Hospital & Ambulatory Monitoring)	Madrid	J
592547	Clinical Application Specialist - Hospital & Ambulatory Monitoring - 6 month fixed term contract	Madrid	J
593429	Clinical Application Specialist Ultrasound Iberia	Madrid	J
591596	Customer Project Manager Health Systems (Imaging & Image Guided Therapy)	Madrid	A
594122	Customer Project Officer (a)	Madrid	A
589185	DSS Sales Manager Spain, IGT-Devices	Madrid	B
589667	Field Service Engineer	Madrid	C
593156	Internship: Customer Care Support Intern	Madrid	A
579308	Modality Sales Specialist CT (medical imaging)	Madrid	B
577463	Professional Marketing Support	Valencia	B
593259	Remote Support Engineer - Digital Computational Pathology	Madrid	C
593489	Supply Chain Demand Support Intern	Madrid	F

Sweden

Job ID	Current title at snapshot	Primary location	Facet
574785	Sales Executive - Enterprise Informatics	Stockholm	B
593196	Sales Leader - Ultrasound Sweden & Scandinavia	Stockholm	B
583026	Shopper Activation Manager, Oral Health Care	Stockholm	A

Switzerland

Job ID	Current title at snapshot	Primary location	Facet
588945	Customer Project Manager (all genders)	Switzerland - Home Based	A
587660	Quote and Tender Specialist (all genders) 80% Teilzeit-Stelle	Horgen	A
578221	Technical Consultant Patient Monitoring (all genders) Alps	Switzerland - Home Based	A

Taiwan

Job ID	Current title at snapshot	Primary location	Facet
586762	ACI (Acute Care Informatics) Product Sales	Taipei	B
586369	Customer Project Engineer	Taipei	A
592953	E-commerce & D2C Manager	Taipei	B
575906	EC Supervisor	Taipei	A
591654	Finance BP Taiwan	Taipei	D
592526	Intern Data Scientist	Taipei	H
586094	Key Account Manager , Imaging and Imaging Guided Therapy	Taipei	B
586077	Senior Product Manager	Taipei	B
587548	Senior Product Manager - OHC	Taipei	B

Thailand

Job ID	Current title at snapshot	Primary location	Facet
579819	3PL Warehouse Manager	Bangkok	F
567738	District Finance Business Partner	Bangkok	D
589278	E-Trade Marketing Manager, Personal Health Thailand (Consumer Electronics)	Bangkok	B
571237	Services Commercial Manager Thailand	Bangkok	B
567885	Services Renewal Specialist Manager for Ultrasound	Bangkok	A
584908	Services Renewal Specialist, Thailand	Bangkok	A
557328	Solution and Key Account Manager (Health Technology)	Bangkok	B

Türkiye

Job ID	Current title at snapshot	Primary location	Facet
584098	Service and Solution Delivery Specialist - Resource Planning	Istanbul	A

United Arab Emirates

Job ID	Current title at snapshot	Primary location	Facet
594205	Business Analytics Graduate - Image Guided Therapy (Emiratization)	Dubai	A

United Kingdom

Job ID	Current title at snapshot	Primary location	Facet
587206	Account Manager - Hospital Patient Monitoring - Northeast England	United Kingdom - Remote Based	B
590104	Advertising Effectiveness Lead	Farnborough	A
593078	Category & Shopper Development Lead	Farnborough	A
591178	Clinical Applications Specialist (Monitoring Analytics) - London & South-Central region	Farnborough	J
582235	Customer Program Manager	United Kingdom - Remote Based	A
566817	Growth Marketing & Acquisition Manager Worry-Free Parenting	Farnborough	B
577810	Implementation Clinical Consulting Lead for North Europe	United Kingdom - Remote Based	J
588868	Modality Sales Specialist - Diagnostic Cardiology Solutions (Midlands & South West UK)	United Kingdom - Remote Based	B
587149	TikTok Shop Account Manager with Italian	Farnborough	B
593084	TikTok Shop Account Manager with Spanish	Farnborough	B

United States of America

Job ID	Current title at snapshot	Primary location	Facet
593801	590569 Field Service Engineer - Medical Imaging (Baltimore and DC area)	Baltimore, Maryland, United States	C
581557	Acoustic Engineer - Medical Device (Cambridge, MA)	Cambridge (US), Massachusetts, United States	H
590804	Administrative Assistant (Cleveland, OH - Orange Village)	Orange (OH), Ohio, United States	A
590303	Advanced Imaging Area Sales Manager - Image Guided Therapy Devices (Southwest)	United States of America - Remote Based, United States	B
590307	Advanced Imaging Area Sales Manager - Image Guided Therapy Devices (Texas - DFW or Houston)	United States of America - Home Based, United States	B
588764	AI Clinical Solutions Engineer, Azurion Eye (Asheville, NC or Nashville, TN)	United States of America - Remote Based, United States	J
593714	Assembler	Latham, New York, United States	A
590884	Assembler - 1st shift	Latham, New York, United States	A
590031	Assembler - 2nd shift	Latham, New York, United States	A
590883	Assembler - 2nd shift	Latham, New York, United States	A
593639	Assembler - 2nd shift	Latham, New York, United States	A
592126	Assembler 1	Reedsville, Pennsylvania, United States	A
593726	Assembler 1	Plymouth, Minnesota, United States	A
593208	Assembler 1 - 2nd Shift	Reedsville, Pennsylvania, United States	A
583046	Assembler 2	Murrysville, Pennsylvania, United States	A
591148	Assembler I - 3rd Shift	Reedsville, Pennsylvania, United States	A
588584	Biomedical Equipment Technician (Columbia, SC)	Columbia, South Carolina, United States	H
591203	Biomedical Equipment Technician (Valhalla, NY)	Valhalla, New York, United States	H
591201	Biomedical Equipment Technician - Lubbock, Texas	Lubbock, Texas, United States	H
592387	Business Excellence Specialist	Plymouth, Minnesota, United States	A
588166	Business Platform Analyst	Plymouth, Minnesota, United States	A
584915	Cardiac Monitoring Technician	Malvern, Pennsylvania, United States	A
586756	Cardiac Monitoring Technician	Pleasanton, California, United States	A
593849	Category Sourcing Manager - Logistics	Nashville, Tennessee, United States	F
590266	Clinical Affairs Director	Orange (OH), Ohio, United States	J
592295	Clinical Application Specialist- IGT- Remote - (IA/IL/WI))	Chicago, Illinois, United States	J

Job ID	Current title at snapshot	Primary location	Facet
589617	Clinical Application Specialist- IGT- Remote, New York, New Jersey, Philadelphia and Delaware .	United States of America - Home Based, United States	J
584583	Clinical Partnership Manager , MR Technology Pediatrics (Cincinnati, OH)	Cincinnati, Ohio, United States	J
583888	Clinical Sales Training Specialist, Structural Heart	United States of America - Home Based, United States	J
594187	Clinical Solutions Consultant - General Imaging Ultrasound (IN)	Indianapolis, Indiana, United States	J
593447	Clinical Solutions Consultant - General Imaging Ultrasound (WI)	Green Bay, Wisconsin, United States	J
591834	Clinical Solutions Delivery Consultant - CV Ultrasound (National-US)	Seattle, Washington, United States	J
591942	Clinical Support Specialist (Manhattan)	Manhattan, New York, United States	J
590387	Co-op - Electrical Engineering - Latham, NY - January-July 2027	Latham, New York, United States	H
590906	Co-op - Manufacturing Engineering - Latham, NY - January-August 2027	Latham, New York, United States	E
580754	Co-op - Manufacturing Engineering - Murrysville, PA - January-June 2027	Murrysville, Pennsylvania, United States	E
592065	Co-op -Embedded Test Engineer - Murrysville, PA-January-June 2027	Murrysville, Pennsylvania, United States	K
591730	Co-op - Data Engineer (Biocompatibility) - Murrysville, PA - January to June 2027	Murrysville, Pennsylvania, United States	G
587486	Co-op-Data Engineering-Reedsville, PA-July-December 2027	Reedsville, Pennsylvania, United States	G
591145	Co-op-Industrial Engineering-Reedsville, PA-(January to June 2027)	Reedsville, Pennsylvania, United States	E
592211	Complaint Handling Investigator	Nashville, Tennessee, United States	A
594190	Complaint Investigator	Bothell, Washington, United States	A
590781	Compliance Engineer, Product Safety & Compliance	New Kensington, Pennsylvania, United States	K
590784	Compliance Technician	New Kensington, Pennsylvania, United States	K
586867	Continuous Improvement Lead	Murrysville, Pennsylvania, United States	A
593046	Critical Care Nurse - Clinical Solution Specialist, Patient Monitoring (Travel: Minnesota/Iowa)	Saint Paul, Minnesota, United States	J
586872	Customer Experience Manager	Bothell, Washington, United States	A
592602	Customer Project Manager - Hospital Patient Monitoring (Field: New York City)	New York, New York, United States	A
592574	Customer Project Manager - Hospital Patient Monitoring (Field: Boston)	Boston, Massachusetts, United States	A
594208	Customer Project Manager, Imaging Systems/Construction (Seattle / Western Washington)	United States of America - Home Based, United States	A
588601	Customer Success Executive (Honolulu, HI)	Honolulu, Hawaii, United States	A

Job ID	Current title at snapshot	Primary location	Facet
592121	Customer Success Executive (NYC)	Manhattan, New York, United States	A
591747	Customer Success Executive (Sioux Falls, SD)	Sioux Falls, South Dakota, United States	H
580822	Delivery and Deployment Lead, Azurion Eye (Cambridge, MA)	Cambridge (US), Massachusetts, United States	A
593877	Demand Planning Leader - North America	Stamford, Connecticut, United States	F
585168	Design Technologist (Murrysville, PA)	Murrysville, Pennsylvania, United States	H
593036	Design Transfer Engineer	Ontario, California, United States	H
593189	Development Engineer	Reedsville, Pennsylvania, United States	H
585578	Development Engineer (Bothell, WA)	Bothell, Washington, United States	H
560036	Digital Signal Processing Ultrasound Engineer - Medical Device (San Diego, CA)	San Diego, California, United States	H
592384	Director, Process Compliance & Centers of Excellence	Plymouth, Minnesota, United States	K
589678	Director, Quality Programs-Integrated Supply Chain (ISC)	Plymouth, Minnesota, United States	F
590792	Disease management Solutions Clinical Support Specialist (90% Travel M-F)	United States of America - Remote Based, United States	J
589701	Distribution Tech 1	Phoenix, Arizona, United States	A
593635	District Clinical Manager- CV Ultrasound - Midwest Zone	United States of America - Home Based, United States	J
592663	District Modality Sales Leader, Diagnostic X-ray (Great Lakes)	Cleveland, Ohio, United States	B
594203	District Modality Sales Leader, Diagnostic X-ray (Mid Atlantic)	Richmond, Virginia, United States	B
590463	District Modality Sales Leader, Diagnostic X-ray (Midwest)	Chicago, Illinois, United States	B
592665	District Modality Sales Leader, Diagnostic X-ray (Pacific Northwest)	Portland, Oregon, United States	B
592664	District Modality Sales Leader, Diagnostic X-ray (Tristate)	New York, New York, United States	B
588160	District Service Manager - Imaging Systems (Field: Bay Area California)	San Francisco Bay Area, California, United States	A
589672	Electrical Design Engineer 3	Gainesville, Florida, United States	H
593554	Electromagnetics R&D Engineer	Latham, New York, United States	H
593902	Environmental Health & Safety Engineer	Ontario, California, United States	H
588597	Executive Legal Assistant (Cambridge, MA)	Cambridge (US), Massachusetts, United States	D
590297	Field Biomedical Equipment Technician (Long Island, NY)	Long Island, New York, United States	H
590702	Field Service Engineer (Houston, Texas)	Houston, Texas, United States	C

Job ID	Current title at snapshot	Primary location	Facet
591733	Field Service Engineer - Albany/Cooperstown, New York	Albany, New York, United States	C
590657	Field Service Engineer - Augusta/Bangor Maine	Augusta, Maine, United States	C
593897	Field Service Engineer - Boston, MA 2nd shift	Boston, Massachusetts, United States	C
590473	Field Service Engineer - CT/MRI (Field: Syracuse, NY)	Syracuse, New York, United States	C
588187	Field Service Engineer - IGT Medical Imaging (Columbia, South Carolina)	Greenville, South Carolina, United States	C
593818	Field Service Engineer - Image Guided Therapy (Field: Charlotte, NC)	United States of America - Remote Based, United States	C
592647	Field Service Engineer - Image Guided Therapy/Diagnostic X-Ray (Field: White Plains, NY)	White Plains, New York, United States	C
593115	Field Service Engineer - Imaging Systems (Field: Bismarck, ND)	Bismarck, North Dakota, United States	C
594016	Field Service Engineer - Imaging Systems (Field: Minneapolis, St Paul MN)	Minneapolis, Minnesota, United States	C
593118	Field Service Engineer - Imaging Systems (Field: Rapid City, SD)	United States of America - Remote Based, United States	C
590569	Field Service Engineer - Medical Imaging (Baltimore and DC area)	Baltimore, Maryland, United States	C
591627	Field Service Engineer - Medical Imaging (Field: Cleveland, OH)	Cleveland, Ohio, United States	C
583510	Field Service Engineer - Medical Imaging (Northwest)	United States of America - Remote Based, United States	C
591854	Field Service Engineer - Medical Imaging (San Antonio, TX)	San Antonio, Texas, United States	C
593950	Field Service Engineer - Ultrasound (Field: Pittsburgh, PA)	Pittsburgh, Pennsylvania, United States	C
590348	Field Service Engineer - Diagnostics and Treatment (Field: Atlanta)	Atlanta, Georgia, United States	C
591282	Field Service Engineer - Diagnostics and Treatment (Field: Houston)	Houston, Texas, United States	C
578571	Field Service Engineer, Multimodality Imaging (Travel: Honolulu, HI)	Honolulu, Hawaii, United States	C
591212	Global CAPA Coach	Cambridge (US), Massachusetts, United States	A
592791	Global Marketing Leader, Claims, Value & Pricing	Cambridge (US), Massachusetts, United States	B
590956	Global Systems Engineering Leader, SRC	Murrysville, Pennsylvania, United States	K
591866	Hardware Component Engineer	Plymouth, Minnesota, United States	H
588915	Head of IT - Connected Care	Cambridge (US), Massachusetts, United States	A
590304	Healthcare Technical Consultant - Field Based in Iowa or Minnesota	Des Moines, Iowa, United States	A
590305	Healthcare Technical Consultant - Field Based in Springfield, Missouri	Springfield, Missouri, United States	A

Job ID	Current title at snapshot	Primary location	Facet
590147	Healthcare Technology Management Specialist (Bothell, Cambridge, Colorado Springs, Nashville or Orlando)	United States of America - Remote Based, United States	A
589334	INR Customer Care Representative	Pleasanton, California, United States	A
590478	INR Enrollment Representative	Malvern, Pennsylvania, United States	A
591021	Inspection Technician	Latham, New York, United States	H
593557	Intern - Demand Generation - Nashville, TN - Summer 2027	Nashville, Tennessee, United States	A
593517	Intern - Service Offer Design - Cambridge, MA - Summer 2027	Cambridge (US), Massachusetts, United States	A
594522	Intern - Advanced Manufacturing Engineer - Plymouth, MN - Summer 2027	Plymouth, Minnesota, United States	E
593012	Intern - AI & Analytics Analyst - Nashville, TN - Spring 2027	Nashville, Tennessee, United States	G
592696	Intern - Finance - Plymouth, MN - Summer 2027	Plymouth, Minnesota, United States	D
593616	Intern - Health Systems Marketing-Cambridge, MA-Summer 2027	Cambridge (US), Massachusetts, United States	B
593133	Intern - Industrial Design - Murrysville, PA - Summer 2027	Murrysville, Pennsylvania, United States	E
590830	Intern - R&D NPI Intern, Disposables - San Diego, CA - 2027	San Diego, California, United States	A
592516	Intern - System Test Automation Engineer - San Diego, CA - Summer 2027	San Diego, California, United States	K
592850	Intern - Systems Engineer - New Kensington, PA - Summer 2027	New Kensington, Pennsylvania, United States	K
586368	Intern - Intellectual Property (IP) - San Diego, CA - Summer 2027	San Diego, California, United States	A
593458	Intern - Marketing Operations - Cambridge, MA - Summer 2027	Cambridge (US), Massachusetts, United States	B
593052	Intern - Systems Engineering (Sustaining & Design Controls) - San Diego, CA - Summer 2027	San Diego, California, United States	K
593459	Intern - North America Brand & Communications Associate - Cambridge, MA - Summer 2027	Cambridge (US), Massachusetts, United States	A
593452	Intern- eCommerce (Health Systems Marketing) - Cambridge, MA - Summer 2027	Cambridge (US), Massachusetts, United States	B
593456	Intern-Customer Experience Analytics (Health Systems Strategy and Growth) Nashville, TN-Summer 2027	Nashville, Tennessee, United States	A
593454	Intern-Insights and Analytics (Health Systems Marketing) Cambridge, MA or Nashville, TN - Summer 2027	Cambridge (US), Massachusetts, United States	B
591053	Intern/Co-op - STET Mechanical Engineer - Latham, NY - Spring/Summer 2027	Latham, New York, United States	H
593028	Intern- Distribution and Reverse Logistics- Nashville, TN - Summer 2027	Nashville, Tennessee, United States	F
592445	Inventory Planning Manager, Ultrasound	Cambridge (US), Massachusetts, United States	A
591209	IT Service Desk Analyst I	Malvern, Pennsylvania, United States	A

Job ID	Current title at snapshot	Primary location	Facet
583315	Manufacturing Engineer (Welding/Brazing)	Latham, New York, United States	E
585253	Manufacturing Supervisor - 2nd Shift	Reedsville, Pennsylvania, United States	E
586748	Manufacturing Test Engineer	Reedsville, Pennsylvania, United States	K
590032	Manufacturing Test Engineer	Murrysville, Pennsylvania, United States	K
593396	Material Handler, 2nd shift (Ontario, CA)	Ontario, California, United States	A
590885	Mechanic - 1st Shift	Latham, New York, United States	A
585686	Mechanic - 2nd Shift	Latham, New York, United States	A
593638	Mechanic - 2nd Shift	Latham, New York, United States	A
588416	National Luminary Manager - Cardiovascular Ultrasound (Northeast)	United States of America - Home Based, United States	A
591391	North America S&OE Leader	Stamford, Connecticut, United States	A
581779	Patient Services Inbound Representative	Malvern, Pennsylvania, United States	A
590092	Patient Services Inbound Representative	Chicago, Illinois, United States	A
590699	Patient Services Outbound Representative	Malvern, Pennsylvania, United States	A
592007	Principal Quality Assurance Engineer	Gainesville, Florida, United States	K
590973	Procurement Engineer	Murrysville, Pennsylvania, United States	F
592378	Procurement Engineer	Chicago, Illinois, United States	F
592925	Product Manager, Upstream Marketing - Advanced Imaging (Plymouth, MN / Bedford, MA)	Plymouth, Minnesota, United States	B
592300	Product Quality Inspection Technician - 1st Shift	Latham, New York, United States	K
587878	Product Safety Engineer	New Kensington, Pennsylvania, United States	H
594367	Production Operations -Lead 1 (2nd Shift)	Ontario, California, United States	E
593394	Production Planner - Integrated Planning (Bedford, MA)	Bedford, Massachusetts, United States	E
593692	Production Worker (1st Shift)	Murrysville, Pennsylvania, United States	E
594186	Production Worker (1st Shift)	Murrysville, Pennsylvania, United States	E
593566	Program Manager, Exploratory Innovation Teams (Cambridge, MA)	Cambridge (US), Massachusetts, United States	A
592067	Project Manager-Ultrasound Imaging	Cambridge (US), Massachusetts, United States	A
588992	QMS Manager	Cambridge (US), Massachusetts, United States	A
591102	QMS Specialist	Latham, New York, United States	A
588287	Quality Control Inspector	Ontario, California, United States	K

Job ID	Current title at snapshot	Primary location	Facet
592604	Quality Control Inspector	Plymouth, Minnesota, United States	K
589515	Quality Engineer 1	Colorado Springs, Colorado, United States	K
593933	Quality Management System/CAPA Specialist	Bothell, Washington, United States	K
593559	R&D Engineer, Mechanical	Latham, New York, United States	H
592216	R&D Project Manager	Bothell, Washington, United States	A
593544	Remote Application Specialist	United States of America - Home Based, United States	J
589759	Remote Service Engineer	United States of America - Remote Based, United States	C
592868	Repair Representative	Murrysville, Pennsylvania, United States	C
586925	Research Scientist - Computed Tomography (Orange, OH)	Orange (OH), Ohio, United States	H
586922	Research Scientist-Computed Tomography	Orange (OH), Ohio, United States	H
593574	Sales Account Manager, General Imaging Ultrasound (South Carolina)	Charleston, South Carolina, United States	B
579807	Sales Support, Clinical Applications Consultant - CV Ultrasound (Orange County/Inland Empire, CA)	San Bernardino, California, United States	J
589711	Sales Support, Clinical Specialist - Coronary - Image Guided Therapy Devices (Seattle WA)	Seattle, Washington, United States	J
593942	Sales Support, Clinical Specialist - Coronary Image Guided Therapy Devices (Indianapolis)	Indianapolis, Indiana, United States	J
580091	Sales Support, Clinical Specialist - Coronary Image Guided Therapy Devices (New York City)	New York, New York, United States	J
593275	Sales Support, Clinical Specialist - Peripheral Image Guided Therapy Devices (Boston South)	Boston, Massachusetts, United States	J
592380	Sales Support, Clinical Specialist - Peripheral Image Guided Therapy Devices (Lafayette, LA)	Lafayette, Louisiana, United States	J
590471	Sales Support, Clinical Specialist - Peripheral Image Guided Therapy Devices (Manhattan)	Manhattan, New York, United States	J
591274	Sales Support, Clinical Specialist - Peripheral Image Guided Therapy Devices (New York City)	New York, New York, United States	J
589925	Sales Support, Clinical Specialist - VeriSight 3D ICE (Intracardiac Echo) (Birmingham, AL)	Birmingham (US), Alabama, United States	J
591061	Sales Support, Clinical Specialist- Image Guided Therapy, Coronary (Los Angeles, CA)	Los Angeles, California, United States	J
592010	Sales, Account Executive - ECG Solutions (Brooklyn/Staten Island)	Brooklyn, New York, United States	B
592148	Sales, Account Executive - ECG Solutions (Jackson, MS)	Jackson, Mississippi, United States	B
592147	Sales, Account Executive - ECG Solutions (Little Rock)	Little Rock, Arkansas, United States	B
592402	Sales, Account Executive - ECG Solutions (Tucson, AZ)	Tucson, Arizona, United States	B

Job ID	Current title at snapshot	Primary location	Facet
591546	Sales, Account Executive - Philips ECG Solutions (CT & RI)	Hartford, Connecticut, United States	B
590845	Sales, Account Executive - Philips ECG Solutions (Maine/New Hampshire/Vermont)	Portland, Maine, United States	B
584047	Sales, Account Executive, Philips ECG Solutions (Miami/Naples, FL) (Bi-Lingual)	Miami, Florida, United States	B
584890	Sales, Account Manager - Diagnostic Imaging (North Dallas, TX)	Dallas, Texas, United States	B
591950	Sales, Account Manager, Cardiovascular Ultrasound (South Carolina)	Columbia, South Carolina, United States	B
583247	Sales, Capital/Clinical Sales Specialist, Image Guided Therapy Fixed Systems (Boston MA, Manchester NH, Burlington VT)	Boston, Massachusetts, United States	J
591258	Sales, Enterprise Client Executive, IDN - Enterprise Strategic Solutions (Field: Texas)	United States of America - Remote Based, United States	B
592372	Sales, Inside Sales Territory Manager - Sleep and Respiratory Care (Murrysville, PA or Nashville, TN)	Murrysville, Pennsylvania, United States	B
588184	Sales, Key Account Manager - Cardiac & Vascular Image Guided Therapy (Pittsburgh)	Pittsburgh, Pennsylvania, United States	B
590961	Sales, Key Account Manager - Mobile Surgery (Northern CA and Nevada)	San Francisco, California, United States	B
590090	Sales, Key Account Manager- Mobile Surgery Solutions (Michigan/North Indiana)	Detroit, Michigan, United States	B
590572	Sales, Regional Sales Manager - Coronary Image Guided Therapy Devices (Gulf Coast)	Birmingham (US), Alabama, United States	B
592733	Sales, Territory Business Manager, Dental Professionals (Northern VA)	Fairfax, Virginia, United States	B
590477	Sales, Territory Manager (Central and Eastern Virginia)	Richmond, Virginia, United States	B
592001	Sales, Territory Manager - Image-Guided Therapy Devices, Coronary (Seattle South)	Seattle, Washington, United States	B
590571	Sales, Territory Manager - Image-Guided Therapy Devices, Peripheral (Twin Cities, North)	Minneapolis, Minnesota, United States	B
594153	Sales, Territory Manager - Peripheral Image Guided Therapy Devices (Houston, TX)	Houston, Texas, United States	B
592849	Sales, Territory Manager - RespirTech (Cleveland/Toledo)	Cleveland, Ohio, United States	B
591193	Sales, Territory Manager - RespirTech (El Paso & New Mexico)	El Paso, Texas, United States	B
591270	Sales, Territory Manager - RespirTech (Houston)	Houston, Texas, United States	B
592698	Sales, Territory Manager - RespirTech (Manhattan/Brooklyn/Bronx)	Manhattan, New York, United States	B
592142	Sales, Territory Manager - Sleep and Respiratory Care (Upstate NY)	Syracuse, New York, United States	B
593815	Sales, Territory Manager - VeriSight 3D/4D ICE (Intracardiac Echo) (Indiana, Michigan, Southwest Ohio)	Indianapolis, Indiana, United States	B
590564	Sales, Territory Manager - Heart Rhythm Management - Image Guided Therapy Devices (Minneapolis MN)	Minneapolis, Minnesota, United States	B
593389	Sales, Territory Manager - Heart Rhythm Management - Image Guided Therapy Devices (St. Louis MO)	St. Louis, Missouri, United States	B

Job ID	Current title at snapshot	Primary location	Facet
589463	Senior Clinical Development Scientist / Medical Safety Manager	Bothell, Washington, United States	J
585582	Senior Clinical Scientist (Cambridge, MA)	Cambridge (US), Massachusetts, United States	J
586163	Senior Clinical Scientist - Computed Tomography (Rochester, MN)	Rochester, Minnesota, United States	J
589909	Senior Cloud Operations & Security Engineer	Nashville, Tennessee, United States	G
592294	Senior Counsel - Labor & Employment Law (Cambridge, MA)	Cambridge (US), Massachusetts, United States	D
592400	Senior Design Quality Engineer	Bothell, Washington, United States	K
592854	Senior Design Quality Engineer	Bedford, Massachusetts, United States	K
593049	Senior Director, HPM Commercial Offerings	Cambridge (US), Massachusetts, United States	B
589633	Senior Electrical Engineer- Medical Device	San Diego, California, United States	H
590807	Senior Forward Deployed UX Designer (Asheville, NC or Nashville, TN)	United States of America - Remote Based, United States	H
592598	Senior Information Security Officer - Ultrasound (Bothell, WA)	Bothell, Washington, United States	A
587156	Senior Legal Counsel, BioTel (Malvern, PA or Cambridge, MA)	Malvern, Pennsylvania, United States	D
589007	Senior Manager Field Reimbursement	United States of America - Remote Based, United States	A
582577	Senior Manager, Field Reimbursement	United States of America - Remote Based, United States	A
591999	Senior Manager, State and Local Government Relations (Western US)	United States of America - Remote Based, United States	A
592395	Senior Manager, Upstream Marketing - Image Guided Therapy Devices (Plymouth, MN or San Diego, CA)	Plymouth, Minnesota, United States	B
589064	Senior Product Manager, Electrophysiology (Cambridge, MA)	Cambridge (US), Massachusetts, United States	B
589054	Senior Quality Operations Manager-Advanced Imaging	Bedford, Massachusetts, United States	K
584896	Senior Quality Operations Manager-Ultrasound	Reedsville, Pennsylvania, United States	K
591555	Senior Quality Systems Engineer	Bedford, Massachusetts, United States	K
593637	Senior Regulatory Affairs Specialist	Orange (OH), Ohio, United States	K
580470	Senior Software Engineer - Image Reconstruction (C++/CUDA)	Orange (OH), Ohio, United States	H
586928	Senior Software Engineer - Medical Image Reconstruction (C++/CUDA)	Orange (OH), Ohio, United States	H
581103	Senior Software Engineer - AM&D ECG Solutions (San Diego, CA)	San Diego, California, United States	G
591542	Senior Supplier Engineer-Ultrasound	Reedsville, Pennsylvania, United States	F

Job ID	Current title at snapshot	Primary location	Facet
592279	Senior System Architect - Ultrasound (Cambridge, MA or Bothell, WA)	Bothell, Washington, United States	A
585187	Senior/Staff Ultrasound System Engineer - Medical Device (San Diego, CA)	San Diego, California, United States	A
593875	Service Parts Planning Lead	Bothell, Washington, United States	A
579884	Software Design Quality Engineer	Bedford, Massachusetts, United States	K
585647	Software Design Quality Engineer	Bedford, Massachusetts, United States	K
580511	Sr Global Product Manager, MCT Clinical Systems	Cambridge (US), Massachusetts, United States	J
594018	Sr Program Manager, EU MDR	Murrysville, Pennsylvania, United States	A
590472	Sr. Data Governance & Quality Systems Lead	Plymouth, Minnesota, United States	K
580324	Sr. Project Manager (Medical Devices)	Cambridge (US), Massachusetts, United States	A
592153	Sr. Technical Writer (New Kensington, PA)	New Kensington, Pennsylvania, United States	A
585188	Staff Electrical Engineer / FPGA - Medical Device (San Diego, CA)	San Diego, California, United States	H
592132	Structural Heart Clinical Adoption Expert - CV Ultrasound (West Zone)	San Francisco, California, United States	J
585656	Supplier Quality Engineer	Bedford, Massachusetts, United States	F
589519	Supplier Quality Engineering Manager	Murrysville, Pennsylvania, United States	F
593899	Supply Planner	Ontario, California, United States	A
592250	Systems Architect- Medical Ultrasound Hardware (Cambridge, MA or Bothell, WA)	Cambridge (US), Massachusetts, United States	A
591251	Systems Engineer	New Kensington, Pennsylvania, United States	K
592311	Systems Engineer - Ultrasound (Cambridge, MA or Bothell, WA)	Bothell, Washington, United States	K
591841	Systems Test Engineer - Medical Device (San Diego, CA)	San Diego, California, United States	K
592055	Technical Consultant - Patient Monitoring (Field: Grand Rapids, Michigan/North Indiana/Toledo, Ohio)	Grand Rapids, Michigan, United States	A
592104	Technical Consultant - Patient Monitoring (Field: Hartford/New Haven, CT)	Hartford, Connecticut, United States	A
588904	Technical Support Engineer (New England)	United States of America - Remote Based, United States	C
593135	Trade Compliance Manager, Master Data & Controls	United States of America - Remote Based, United States	K
592685	UI/UX Designer - Advanced Imaging (Bedford, MA)	Bedford, Massachusetts, United States	H
589262	Ultrasound Systems Engineer	Bothell, Washington, United States	K

Job ID	Current title at snapshot	Primary location	Facet
594029	Warehouse and Production Operator	Latham, New York, United States	F
592822	Warehouse Operator	Mount Pleasant, Pennsylvania, United States	F
593493	Warehouse Operator	Plymouth, Minnesota, United States	F
594141	Warehouse Operator	Mount Pleasant, Pennsylvania, United States	F
594152	Warehouse Operator	Reedsville, Pennsylvania, United States	F
594156	Warehouse Operator (2nd Shift)	Mount Pleasant, Pennsylvania, United States	F
590882	Welder	Latham, New York, United States	A
585685	Welder - 2nd Shift	Latham, New York, United States	A
586445	Welder 1	Latham, New York, United States	A
591517	Welder 1	Latham, New York, United States	A
591905	Zone Installation Specialist - Medical Imaging (Travel: Midwest Zone)	United States of America - Remote Based, United States	A

Vietnam

Job ID	Current title at snapshot	Primary location	Facet
582749	Partner Account Manager	Ho Chi Minh City	B

G. Primary-source register

Source codes in the narrative link to these records. Model URLs are pinned to the reviewed revision. Company pages were researched on 2 November 2026. Recruitment links identify the exact requisition retained in the snapshot; a legacy URL slug can differ from the current job title.

M1 XD-TAU AURUMTOMATON v1.5.1, pinned publication, embedded implementation and reference records.
https://huggingface.co/c0mbatduckzz/XD-TAU_AURUMTOMATON_v1.5.1/tree/da88eaaea164e006c6cf091a1773dfcdbc9cd353

M2 XD-TAU-JKT v1.5, pinned publication.
https://huggingface.co/c0mbatduckzz/XD-TAU-JKT/tree/9bdabc99d277b27c6a891b27b284fa5f3c328e8b

M3 TNOMTOM, pinned implementation and evidence.
https://huggingface.co/c0mbatduckzz/TNOMTOM/tree/8a4a15a36c0e41e4617a4597afd88dfdd173a8cf

M4 TOMElinedT, pinned 0.17 delivery and 0.18 checkpoint.
https://huggingface.co/c0mbatduckzz/TOMElinedT/tree/0235a5bb73f182c78ced6faa514e66ea8ac5cdc8

M5 TOMK, pinned finite learner and compiler/runtime.
https://huggingface.co/c0mbatduckzz/TOMK/tree/0febbaa1ea0df9a80eea6d7e45f75b652d39fa84

M6 TOM-GROK, pinned interpreter, GPU implementation and records.
https://huggingface.co/c0mbatduckzz/TOM-GROK/tree/c9f0959233861726d1d7695fef8206f0d6d949e5

M7 Maktub reference, pinned implementation.
https://huggingface.co/c0mbatduckzz/Maktub-Tom-Klootwijk/tree/902a80294f03f4b5d749c5272bf915be23585932

M8 TOM-MAKTUBIII, pinned implementation.
https://huggingface.co/c0mbatduckzz/TOM-MAKTUBIII/tree/c5c90db368f09aa08ae1cd0d4c7c922f528075b6

M9 JTK-Twin-Viper and CATHIROGLYPHDRALE, pinned composition and representation sources.
https://huggingface.co/c0mbatduckzz/JTK-Twin-Viper/tree/ad10a7a5083e3705118ede04c4b8dd24f665ec08

M10 TK-MK-ULTRA, pinned action, orbit, energy and clock proofs.
https://huggingface.co/c0mbatduckzz/TK-MK-ULTRA/blob/e78ede7d418effa3c912e9925493d55615c6adb1/PROOFS.md

M11 TNOMTOM durable event-store code and tests.
https://huggingface.co/c0mbatduckzz/TNOMTOM/tree/8a4a15a36c0e41e4617a4597afd88dfdd173a8cf/system_of_systems/applications/durable

M12 TNOMTOM thermal identification code and records.
https://huggingface.co/c0mbatduckzz/TNOMTOM/tree/8a4a15a36c0e41e4617a4597afd88dfdd173a8cf/system_of_systems/applications/calibration

M13 TNOMTOM physical-network model and implementation.
https://huggingface.co/c0mbatduckzz/TNOMTOM/blob/8a4a15a36c0e41e4617a4597afd88dfdd173a8cf/language/xd_physical_networks/README.md

M14 TNOMTOM physical SI application profiles.
https://huggingface.co/c0mbatduckzz/TNOMTOM/blob/8a4a15a36c0e41e4617a4597afd88dfdd173a8cf/software/xd_ph_matrix_exact/applications/README.md

M15 TNOMTOM coupled physical plant.
https://huggingface.co/c0mbatduckzz/TNOMTOM/blob/8a4a15a36c0e41e4617a4597afd88dfdd173a8cf/system_of_systems/applications/physics/README.md

M16 TOM-GROK 56-domain catalogue.
https://huggingface.co/c0mbatduckzz/TOM-GROK/blob/c9f0959233861726d1d7695fef8206f0d6d949e5/compendium/TK_SROF_ATLAS_domain_catalogue.json

P1 Philips Annual Report 2025
https://www.results.philips.com/publications/ar25/downloads/files/en/PhilipsFullAnnualReport2025-English.pdf

P2 Philips Environmental Profit & Loss and Material Flow Methodology (2025)
https://www.philips.com/c-dam/corporate/about-philips/sustainability/lives-improved/epl-and-material-flow-methodology.pdf

P3 Philips EcoDesign
https://www.philips.com/a-w/about/sustainability/environmental-impact/ecodesign.html

P4 Philips Supplier sustainability
https://www.philips.com/a-w/about/sustainability/supplier-sustainability.html

P5 Philips MRI system lifecycle management
https://www.usa.philips.com/healthcare/support/philips-mri-system-lifecycle-management

P6 Philips BlueSeal helium-free MRI
https://www.usa.philips.com/healthcare/technology/blueseal-helium-free-mri

P7 Philips Patient monitoring
https://www.usa.philips.com/healthcare/patient-monitoring

P8 Philips Integrated Diagnostics
https://www.usa.philips.com/healthcare/enterprise-solutions/integrated-diagnostics

P9 Philips Diagnostic and clinical informatics
https://www.usa.philips.com/healthcare/diagnostic-and-clinical-informatics

P10 Philips Image-guided therapy portfolio
https://www.usa.philips.com/healthcare/image-guided-therapy

P11 Philips Remote services
https://www.usa.philips.com/healthcare/service/remote-maintenance

P12 Philips Circular offerings
https://www.philips.com/a-w/about/sustainability/environmental-impact/materials/circular-offerings.html

P13 Philips Refurbished medical equipment
https://www.usa.philips.com/healthcare/refurbished-systems

P14 Philips Recycling
https://www.philips.com/a-w/about/sustainability/resources/recycling.html

P15 Philips Fetal and maternal monitors
https://www.usa.philips.com/healthcare/patient-monitoring/patient-monitors/fetal-and-maternal-monitoring

P16 Philips Avent Pregnancy+ Community feature, 30 March 2026
https://www.philips.com/a-w/about/news/archive/standard/news/articles/2026/world-leading-pregnancy-app-philips-avent-pregnancy-brings-moms-together-with-community-feature

P17 Philips Avent Natural Care Wearable Breast Pump launch, 1 September 2026
https://www.philips.com/a-w/about/news/archive/standard/news/press/2026/listening-to-mom-evolving-needs-for-comfort-freedom-and-performance-in-one-pump-philips-avent-launches-new-natural-care-wearable-breast-pump.html

P18 Philips Sonicare on-device AI toothbrush announcement, 7 July 2026
https://www.philips.com/a-w/about/news/archive/standard/news/press/2026/philips-sonicare-sets-new-benchmark-in-oral-healthcare-with-brands-first-electric-toothbrush-powered-by-on-device-ai

P19 Philips Sonicare SenseIQ explanation
https://www.usa.philips.com/c-f/XC000019720/what-is-sonicare-senseiq-technology

P20 Philips Sustainability resources and methodologies
https://www.philips.com/a-w/about/sustainability/resources.html

S1 Philips recruitment catalogue, snapshot 2 November 2026 at 08:37 CET.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers

S2 Supplied professional profile.
https://nl.linkedin.com/in/tom-klootwijk-8747848

S3 MIT Professional Education course credential.
https://www.credential.net/eb8b939c-eaf8-45b2-9b9b-877cd2f3a12d

S4 Public Hugging Face profile and repository inventory.
https://huggingface.co/c0mbatduckzz

J1 588449, Software Designer (C++), Best.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Best/Software-Designer--C------Image-Guided-Therapy_588449-2

J2 590735, Data and AI Engineer, Amsterdam.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Amsterdam/Data-and-AI-Engineer_590735-2

J3 589928, Senior Methods HW Modelling Engineer, Suzhou.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Suzhou/Senior-Methods-HW-Modelling-Engineer_589928

J4 584997, GenAI Solutions & Prompt Engineer, Bangalore.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Bangalore/Integration-test-engineer_584997-1

J5 585168, Design Technologist, Murrysville.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Murrysville-Pennsylvania-United-States/UX-Designer--Murrysville--PA-_585168

J6 589859, Software Technologist C++, Bangalore.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Bangalore/Software-Technologist_589859

J7 593840, Senior Software Engineer, Shenyang.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Shenyang/Senior-Software-Engineer_593840

J8 593980, Software Technologist I, Bangalore.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Bangalore/Software-Technologist-I_593980-1

J9 580470, Image Reconstruction C++/CUDA, Orange.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Orange-OH-Ohio-United-States/R-D-Software-Development-Engineer--Medical_580470

J10 586928, Medical Image Reconstruction C++/CUDA.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Orange-OH-Ohio-United-States/Senior-Software-Engineer---Medical-Image-Reconstruction--C---CUDA-_586928

J11 591185, AI Senior Systems Engineer, Eindhoven.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Eindhoven/AI-Senior-Systems-Engineer--AI-for-RAMS---Systems-Engineering-_591185-2

J12 580273, Senior Software Architect, Azurion Eye.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Best/Senior-Software-Architect-Azurion-Eye_580273-2

J13 588477, Product Owner/Technical Lead, Display Software.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Best/Technical-Product-Owner---Image-Guided-Therapy--Display-Software-_588477-1

J14 570549, Senior Software Design Quality Engineer-AI.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Best/Senior-Software-Design-Quality-Engineer---AI_570549-1

J15 588485, Test Development Engineer, Böblingen.
https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Bblingen/Test-Development-Engineer--fixed-term--All-genders-_588485-1

J16 586913, Research Scientist, AI in CT, Hamburg.

https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Hamburg/Research-Scientist---AI--in-Computed-Tomography-Image-Formation--all-genders-_586913-1

J17 586911, Research Scientist, Physics of CT, Hamburg.

https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Hamburg/Research-Scientist---Physics-of-Computed-Tomography--all-genders-_586911

J18 590246, Assistant Function Development Engineer, Shenzhen.

https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Shenzhen/Assistant-Function-Development-Engineer-Oral-Healthcare_590246

J19 586187, Product Innovation & AI Prototyping Internship.

https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Eindhoven/Product-Innovation---AI-Powered-Video-Creation-Internship---Philips-Avent-Experience-Innovation_586187

J20 593359, Synthetic Data Generation for Dental AI internship.

https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Eindhoven/Internship--Synthetic-Data-Generation-for-Dental-AI--Master-_593359

J21 593360, Robot-Assisted Toothbrush IMU Validation internship.

https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Eindhoven/Internship--Robot-Assisted-Toothbrush-IMU-Validation_593360

J22 585855, E-Commerce Data, Automation & AI Manager, Hamburg.

https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Hamburg/E-Commerce-Data--Automation---AI-Manager--m-w-d-_585855-1

J23 592685, UI/UX Designer - Advanced Imaging, Bedford.

https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Bedford-Massachusetts-United-States/UI-UX-Designer---Advanced-Imaging--Bedford--MA-_592685

J24 587348, Software Technologist I/II-C# Full Stack, Bangalore.

https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Bangalore/Architect-I-II---C-_587348

J25 591151, Software Technologist I - Embedded C++, Pune/Bangalore.

https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Pune/Software-Technologist-I---Embedded-C--_591151-1

J26 593837, Senior/Advanced Algorithm Engineer, Shenyang.

https://philips.wd3.myworkdayjobs.com/jobs-and-careers/job/Shenyang/Senior-Advanced-Algorithm-Engineer_593837